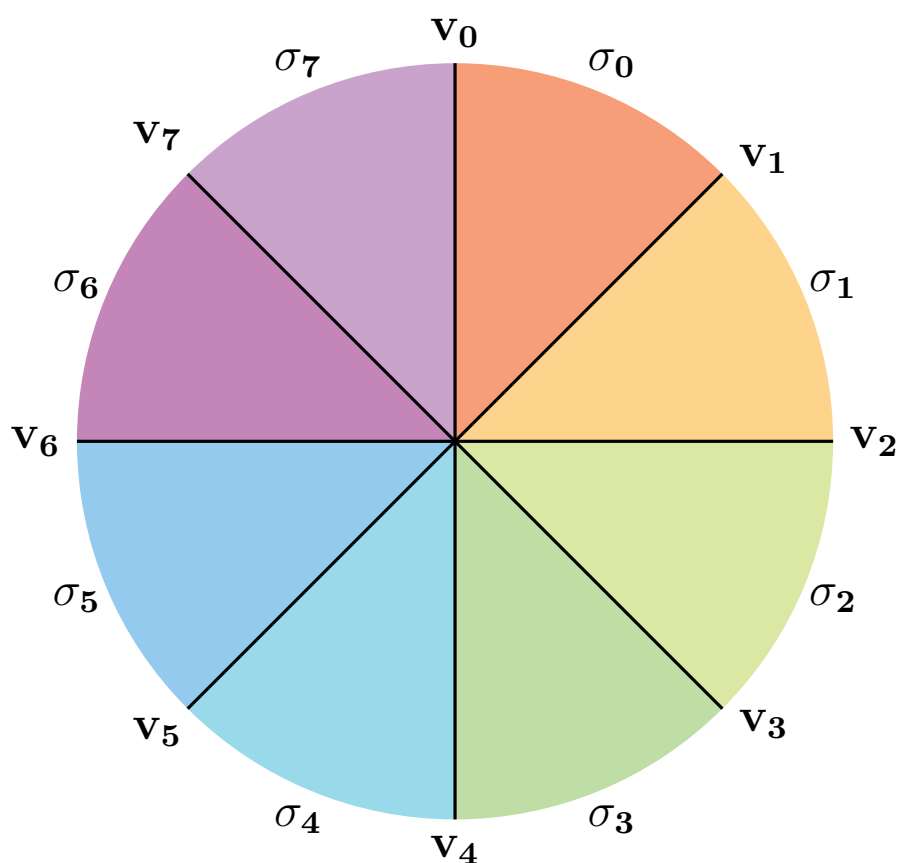


ON THE NON-COERCIVITY OF ENERGY FUNCTIONALS IN KÄHLER GEOMETRY

A COMPUTATIONAL APPROACH ON TORIC VARIETIES

Master's thesis in Mathematics

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Abstract

We study numerical criteria for coercivity of Donaldson’s J -functional, which appears as the energy component of the Mabuchi functional via the Chen–Tian decomposition. Our analysis is carried out in the setting of toric varieties, where intersection-theoretic computations admit a purely combinatorial structure.

Exploiting this combinatorial structure, we observe a relation between the number of distinct fans of smooth complete toric surfaces obtainable by a finite sequence of blow-ups, and the Catalan k -fold convolution numbers.

To investigate the behaviour of the associated numerical invariant, we develop a computational framework implemented in Python for performing calculations on the Kähler cone. This implementation enables the visualization of the induced wall–chamber decomposition determined by optimal destabilizing subvarieties.

We further include the necessary background on toric varieties and their intersection theory, which underpins the framework.

Applying this framework to the constant scalar curvature Kähler problem, we derive results on the coercivity of the J -functional. In particular, we prove that negativity of the numerical invariant implies non-coercivity of Donaldson’s J -functional, and that the invariant is negative for every smooth complete toric surface. We present both a computational approach and a complementary theoretical argument, the latter offering additional structure insight into the wall–chamber decomposition.

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1 Introduction

In Complex Geometry, the search of canonical metrics has long been recognized as a fundamental tool for understanding the geometric and algebraic structure of complex manifolds. Among such metrics, *constant scalar curvature Kähler (cscK) metrics* takes on a central role, as they generalize the notion of Kähler–Einstein metrics and arise naturally as critical points of geometric energy functionals. The study of cscK metrics can be traced back to the work of Calabi, in particular the Calabi conjecture, which concerned the existence of *extremal metrics* (Calabi 1954).

Let X be a compact complex manifold of dimension n , equipped with a Kähler class $\Omega \in H^{1,1}(X, \mathbb{R})$, and denote by $\mathcal{K}_\Omega$ the set of Kähler metrics in Ω .

Definition 1.1. *An extremal metric is a critical point of the Calabi functional*

$$\begin{aligned} \text{Cal}: \mathcal{K}_\Omega &\rightarrow \mathbb{R} \\ \omega &\mapsto \int_X (S(\omega) - \bar{S})^2 \omega^n, \end{aligned}$$

where $S(\omega)$ is the scalar curvature of ω , and $\bar{S}$ is the mean scalar curvature of $S(\omega)$ with respect to the volume form $\frac{\omega^n}{n!}$.

The Calabi conjecture was proven by Yau in 1978 (Yau 1978). Yau proved the existence of Kähler–Einstein metrics when $c_1(X) < 0$, and Aubin–Yau when $c_1(X) = 0$. Then Matsushima–Lichnerowicz proved obstruction to existence of Kähler–Einstein metrics in general when $c_1(X) > 0$, and in more generality, existence of cscK metrics on manifolds with certain types of automorphism groups (Matsushima 1957), (Lichnerowicz 1959).

Let (X, ω) be a compact Kähler manifold of complex dimension n . The scalar curvature is then given as the trace of the Ricci curvature form, defined as $\text{Ric}(\omega) = -dd^c \log \omega^n$, thus,

$$S(\omega) := \text{Tr}_\omega \text{Ric}(\omega) = n \frac{\text{Ric}(\omega) \wedge \omega^{n-1}}{\omega^n}.$$

Furthermore, let α denote the Kähler class $[\omega] \in H^{1,1}(X, \mathbb{R})$. Then the mean scalar curvature is the cohomological constant given by

$$\bar{S} = V^{-1} \int_X S(\omega) \omega^n = n \frac{\int_X c_1(X) \cdot \alpha^{n-1}}{\int_X \alpha^n},$$

where $V := \int_X \omega^n$, is the Kähler volume. The cscK condition then requires that $S(\omega) \equiv \bar{S}$. The corresponding equation is a fourth-order fully nonlinear elliptic partial differential

equation, whose solvability depends on global geometric data. This complexity led to the realization that the existence of such metrics is governed not only by analytic considerations, but also by deep algebro-geometric obstructions. In order to understand these obstructions, decades of research culminated with the study of the algebro-geometric notions of stability.

In 1997 Gang Tian introduced such a notion of stability for the cscK equation, namely *K-stability* (Tian 1997). Later, Donaldson reformulated *K-stability* more algebraically (S. Donaldson 2002). Donaldson provided a concrete reformulation of *K-stability* in terms of test configurations and numerical invariants, that is now known as the Donaldson–Futaki invariant.

K-stability was traditionally only considered for the special case of polarized varieties (X, L) , i.e. X is a complex algebraic variety and L is an ample line bundle over X . Given this setting the following conjecture was introduced by Tian and refined by Donaldson.

Conjecture 1.2. (Yau–Tian–Donaldson conjecture) *A polarized manifold (X, L) admits a constant scalar curvature Kähler metric if and only if it is K -stable.*

The Yau–Tian–Donaldson conjecture was proven in the special case of X being a *Fano* manifold, when cscK metrics are Kähler–Einstein metrics (Chen, Xiuxiong and Donaldson 2014a), (Chen, Xiuxiong and Donaldson 2014b), (Chen, Xiuxiong and Donaldson 2014c). It is worth noting that since the introduction of *K-stability*, there has been a lot of development in the field, and hence many variations of the notions have appeared in the literature. For example uniform *K-stability*, *K-polystability* and filtration *K-stability*.

In general in order to study stability criteria one considers *test configurations* $(\mathcal{X}, \Omega)$ of a compact complex manifold X . A test configuration is an algebro-geometric object encoding $\mathbb{C}^*$ -equivariant degenerations of the initial manifold. The definition of a test configuration vary slightly depending on the setting of stability. One then checks the behaviour of *numerical invariants* with respect to these test configurations, and an associated ‘norm’ of these test configurations in the uniform *K-stability* case. Omitting many details, *K-stability* can then for example be formalised with respect to the Donaldson–Futaki invariant DF as

Definition 1.3. *We say a Kähler manifold $(X, [\omega])$ is*

- *K -semistable if $\text{DF}(\mathcal{X}, \Omega) \geq 0$ for all test configurations $(\mathcal{X}, \Omega)$ for $(X, [\omega])$,*
- *K -stable if $\text{DF}(\mathcal{X}, \Omega) > 0$ for all test configurations $(\mathcal{X}, \Omega)$ for $(X, [\omega])$ with $||(\mathcal{X}, \Omega)|| > 0$,*
- *uniformly K -stable if there exists an $\varepsilon > 0$ such that $\text{DF}(\mathcal{X}, \Omega) \geq \varepsilon ||(\mathcal{X}, \Omega)||$ for all test configurations $(\mathcal{X}, \Omega)$ for $(X, [\omega])$.*

We will further consider certain *energy functionals*, as properness of these functionals are related to *K-stability*. Functionals of interest are Aubin’s *J-functional*, Donaldson’s *J-functional* and the *Mabuchi functional*, which are defined on the space of Kähler potentials defined below. For a fixed Kähler form ω on X , we write

$$\mathcal{H}_\omega := \{\varphi \in C^\infty(X) \mid \omega_\varphi := \omega + dd^c \varphi > 0\}$$

for the space of Kähler potentials on (X, ω) . We consider functionals on the level of potentials, since $\mathcal{H}$ is isomorphic to the space of Kähler metrics cohomologous to ω , up to constants. These functionals play an important role in K -stability, as these functionals are convex along geodesics, so it is sufficient to check coercivity of the Mabuchi functional, as this gives existence of a cscK metric cohomologous to the Kähler form ω . Coercivity of the Mabuchi functional M is defined as; $\exists A, B > 0$ s.t.

$$M(\varphi) \geq AJ(\varphi) - B, \quad \forall \varphi \in \mathcal{H}_\omega,$$

where J is the Aubin's J -functional. As a result of (X. Chen 2000) we further have a decomposition of the Mabuchi functional into a sum of a energy/pluripotential part and an entropy part

$$M_\omega = E_\omega^{-\text{Ric}(\omega)} + H_\omega,$$

The energy part $E^{-\text{Ric}(\omega)}$ is in fact the Donaldson's J -functional. Hence it is not only coercivity of M that is of interest, as coercivity of the energy part is equivalent to solvability of the J -equation, and a sufficient criteria for existence of cscK metrics. Given Kähler metrics ω and χ , the J -equation is defined as $\text{Tr}_{\omega_\varphi} \chi = c$. A special type of test configurations, deformation to the normal cone, relates test configurations to subvarieties, and Lejmi–Székelyhidi conjectured it is enough to consider these test configurations, effectively making the test space that of all, infinitely many, irreducible subvarieties instead (Lejmi and Székelyhidi 2013). Formally they conjecture for a compact Kähler manifold X of dimension n , a solution to the J -equation exists if and only if for all p -dimensional subvarieties $D \subset X$, where for all $p = 1, 2, \dots, n-1$, we have

$$\int_D c_{\chi, \omega} \omega^p - p \omega^{p-1} \wedge \chi > 0.$$

The constant c is given by

$$c_{\chi, \omega} = n \frac{\int_X \chi \wedge \omega^{n-1}}{\int_X \omega^n}. \quad (1.1)$$

It is worth noting that this necessary numerical condition, still ranges over infinitely many subvarieties. Lejmi and Székelyhidi furthermore showed the conjecture for $n = 2$. In general it is now known by Chen and Song (G. Chen 2021), (Song 2020).

In recent work by Khalid and Dyrefelt the criteria of testing against all test configurations, were significantly reduced to testing against a finite number of test configurations (Khalid and Dyrefelt 2024). A consequence of this reduction of the number of test configurations is that it is now possible to numerically check the stability conditions and obtain a complete picture in suitable settings.

Denote the space of all relatively Kähler test configurations for $(X, [\omega])$ by $\mathcal{H}^{\text{NA}}(X, [\omega])$. Given a curve $C \subset X$, one can naturally associate a test configuration $\Phi_{C, \kappa}$ to it via the deformation to the normal cone construction introduced by Ross–Thomas (Ross and Thomas 2005). The test configuration $\Phi_{C, \kappa}$ is relatively Kähler precisely when $\kappa \in (0, \bar{\kappa}_C)$, where

$$\bar{\kappa}_C = \sup \{r > 0 \mid \pi^* \alpha - r[E] > 0\}$$

is the Seshadri constant. The non-archimedean functionals that we wish to consider then take simple polynomial expressions along $\Phi_{C,\kappa}$, varying $\kappa \in (0, \bar{\kappa}_C)$, via the well-known proposition, Proposition 13 (Lejmi and Székelyhidi 2013).

Proposition 1.4. *Suppose X is a smooth projective surface. For slope test configurations $\Phi_{C,\kappa} \in \mathcal{H}^{NA}(X, [\omega])$ we have,*

$$\begin{aligned} \mathbf{E}_{\theta,\omega}^{NA}(\Phi_{C,\kappa}) &= A_1\kappa^2 + B_1\kappa^3, \\ ||\Phi_{C,\kappa}|| &= A_2\kappa^2 + B_2\kappa^3, \end{aligned}$$

where A_1, A_2, B_1, B_2 are real numbers satisfying

$$A_1 = c_{\theta,\omega} \int_C \omega - \int_C \theta, \quad B_1 = - \left(\frac{2[\theta] \cdot [\omega]}{3[\omega]^2} \right) C^2, \quad A_2 = \int_C \omega > 0, \quad B_2 = -\frac{2}{3}C^2.$$

Where $c_{\theta,\omega}$ is the constant from (1.1) for surfaces.

In the polynomial expression of the non-archimedean functional for surfaces,

$$\mathbf{E}_{\theta,\omega}^{NA}(\Phi_{C,\kappa}) = A_1\kappa^2 + B_1\kappa^3,$$

consider the leading term A_1 . Expanding it yields,

$$\begin{aligned} A_1 &= 2 \frac{\int_X \chi \wedge \omega}{\int_X \omega^2} \cdot \int_C \omega - \int_C \chi \\ &= ([\omega] \cdot C) \left(2 \frac{[\chi] \cdot [\omega]}{[\omega]^2} - \frac{[\chi] \cdot C}{[\omega] \cdot C} \right) \\ &= ([\omega] \cdot C) (J([\omega], [\chi])) \end{aligned}$$

Since $[\omega]$ is Kähler, one has $[\omega] \cdot C > 0$ for all curves C , hence the sign of A_1 is entirely governed by $J([\omega], [\chi])$. In particular, if $J([\omega], [\chi]) < 0$, then $\mathbf{E}_{\theta,\omega}^{NA}(\Phi_{C,\kappa}) < 0$ for sufficiently small κ . In this setting, after a suitable normalization, the positivity condition becomes $(2[\omega] - [\chi]) \cdot C > 0$ for all curves C , which is the Nakai–Moishezon criterion for the class $2[\omega] - [\chi]$ to be Kähler. This naturally leads us to consider the quantity,

$$J([\omega], [\chi]) = \min_C \left(2 \frac{[\omega] \cdot [\chi]}{[\omega]^2} - \frac{[\chi] \cdot C}{[\omega] \cdot C} \right).$$

which encodes this positivity in a numerical way. The other direction of the connection between $J([\omega], [\chi])$ and $\mathbf{E}_{\theta,\omega}^{NA}(\Phi_{C,\kappa})$, that $J([\omega], [\chi]) > 0$ implies $\mathbf{E}_{\theta,\omega}^{NA}(\Phi_{C,\kappa}) > 0$, was shown by Chen in the surface case (X. Chen 2000). Let $\alpha := [\omega]$ and $\beta := [\chi]$. We will consider $J(\alpha, \beta)$, alongside the special case $I(\alpha) := J(\alpha, K)$, in part of our main results, in particular a wall-chamber decomposition induced by the intersection numbers. The latter having applications to the cscK problem.

In this thesis we will consider the testing ground of toric varieties, specifically *toric surfaces*. There are two approaches in which one considers toric surfaces, either through the concept of *polytopes* or the concept of *fans*. We will primarily work with fans, however in order to obtain the connection between toric surfaces and Kähler manifolds we consider

the special *Delzant polytopes*, done in **Chapter 3**. Given this connection we can utilize the combinatoric structure of fans, to simplify the study of K -stability of the toric surfaces. In the setting of fans the proper irreducible subvarieties corresponds to the *rays* of the fan, for which the combinatoric structure of the fan makes intersection of divisors into matrix multiplication. In **Chapter 2** we will establish the needed theory of toric varieties, such as the relation of fans to toric varieties and intersection theory on toric varieties. In order to further the understanding of the theory we will provide elaborate examples. We will therefore, through the reduction to finite number of test configuration described in (Khalid and Dyrefelt 2024), consider the quantities,

$$J(\alpha, \beta) = \min_{D_i} \left(2 \frac{\alpha \cdot \beta}{\alpha^2} - \frac{\beta \cdot D_i}{\alpha \cdot D_i} \right),$$

$$I(\alpha) = \min_{D_i} \left(2 \frac{\alpha \cdot K}{\alpha^2} - \frac{K \cdot D_i}{\alpha \cdot D_i} \right).$$

Note that the aforementioned relations of the sign of the quantities, and the energy part $\mathbf{E}_{\theta, \omega}^{\text{NA}}$, relies on α, β being Kähler. For $I(\alpha)$, the canonical divisor K or the anticanonical divisor $-K$, might not be Kähler. A further argument is needed to show this relation.

Main Results

This thesis presents the following results:

- 1 Computation and visualization of wall-chamber decompositions for J -stability.
- 2 A theoretical proof that the energy part of the Mabuchi functional is non-coercive on any toric surface, approached through:
 - a) Computational methods.
 - b) Theoretical methods.
- 3 Observations made throughout the development of the thesis.

Given the reduction of test configurations and the combinatorial structure of fans, and hence toric varieties, computing the above numerical condition becomes feasible. A main goal of this thesis is to produce a program that computes the numerical condition and hence give a broader picture of the behaviour of the numerical criteria. This is done in **Chapter 4**, which will rely on the mathematics software SageMath (The Sage Developers 2025). The aim of the program is to provide data that could lead towards theory regarding the behaviour of the functionals.

Particularly, the program, along with the plotting of the data, provides clear visualizations of the wall-chamber decomposition of the Kähler cone, induced by $J(\alpha, \beta)$ and $I(\alpha)$, with respect to the optimal destabilizing subvarieties. Examples of these visualizations will be shown in **Section 5.1**. Having completed the program, we further found an application to the cscK problem, which is formulated in the following theorem, that we have been unable to find in the literature.

Theorem 1.5. *For every smooth complete toric surface, the intersection number $I(\alpha)$ is negative for every Kähler class α .*

The fact that Theorem 1.5 tells us something about the coercivity of $\mathbf{E}_{\theta,\omega}^{\text{NA}}(\Phi_{C,\kappa})$, relies on the following proposition, that will be proven in **Section 3.3**.

Proposition 1.6. *Negativity of the intersection numbers $I(\alpha)$ implies non-coercivity of $\mathbf{E}_{\theta,\omega}^{\text{NA}}(\Phi_{C,\kappa})$.*

In **Section 5.2** we will provide different approaches to prove Theorem 1.5. The first approach is computational based, being greatly inspired by the development of the program, while the second approach is theoretical based, however only showing a weaker version of Theorem 1.5.

For the computational approach, we observed that the fan Σ carries clear obstructions to the quantity $I(\alpha)$. Specifically, if the fan Σ contains a torus-invariant divisor D_i with self-intersection $D_i \leq -2$, then the *adjunction formula* will immediately result in $I(\alpha) < 0$ for every Kähler class α on X_Σ . This obstruction greatly reduced the toric surfaces to consider, leaving us with 167 toric surfaces.

Based on the computations of the program we found a further obstruction, similar to the first. If the fan Σ contains a torus-invariant divisor D_i with self-intersection $D_i \leq -1$, then $I(\alpha) < 0$ for every Kähler class α on X_Σ . This further obstruction then left us with 15 toric surfaces to consider.

Now that we were left a feasible 15 toric surfaces, we developed an approach, mirroring the program, to determine the behaviour of $I(\alpha)$ in the remaining cases. Combining the obstructions, and the case by case approach, we will then have shown Theorem 1.5.

In the theoretical approach, we show a weaker version of Theorem 1.5, namely that $I(\alpha) \leq 0$ for every Kähler class α on X_Σ . However, this theoretical approach depicts a clearer connection to the wall-chamber decomposition with respect to the optimal destabilizing curves of the numerical criteria, as it utilizes a partial ordering induced by the Kähler cone. It further shows the importance of the program developed, as it guided us towards the proof of Theorem 1.5.

Lastly we round **Chapter 5** off with a combinatorial observation on the number of unique fans consisting of r rays, and the *Catalan k -fold convolution* $C_n^{(k)}$, which to the best of our knowledge, has not yet been documented in the literature.

Having shown Theorem 1.5, a natural question is then how negative $I(\alpha)$ becomes, as it is the energy part of the Chen–Tian decomposition of the Mabuchi functional, and hence carries information about the coercivity of the Mabuchi functional. It is worth emphasising that non-coercivity of the energy part of the Mabuchi functional, in itself does not show non-existence of cscK metrics. For example we have the Fubini–Study metric ω_{FS} on $\mathbb{P}^2$.

However, the energy part does carry deep information relevant for the cscK problem. And while we mainly considered $I(\alpha)$, the program developed can easily be adapted to $J(\alpha, \beta)$, and carries an application to the actively studied geometric PDE Donaldson’s J -equation.

2 Affine Toric Varieties and Fans

In this chapter we introduce and discuss the necessary theory of *affine toric varieties* which we will use in later computations. Our focus will be on how to construct affine toric varieties through the notions of cones and fans. To further understanding, we will, throughout the chapter, construct an example, with the intention that it will clarify the discussed theory. The definitions and results in the following chapter is, unless explicitly stated, based on the book "Toric Varieties" by the authors David Cox, John Little, and Hal Schenck. We will omit the proofs, as they can be seen in (Cox, Little, and Schenck 2011), and focus on elaborating the theory in the example.

2.1 Affine Varieties

First we define affine toric varieties, and thereafter we introduce the notions of cones and fans.

Definition 2.1 (Affine variety). *An irreducible affine algebraic set is called an affine algebraic variety.*

There are some important algebraic notions associated to affine varieties, which we will need to state. We begin with the coordinate ring. Let C denote $\mathbb{C}[x_1, \dots, x_n]$. An affine variety $V \subseteq \mathbb{C}^n$ gives the ideal

$$\mathbf{I}(V) = \{f \in C \mid f(p) = 0, \forall p \in V\}.$$

To the affine variety V , its *coordinate ring* is defined as

$$\mathbb{C}[V] = C/\mathbf{I}(V).$$

We note that a point p of an affine variety V gives the maximal ideal

$$\{f \in \mathbb{C}[V] \mid f(p) = 0\} \subseteq \mathbb{C}[V],$$

and that all maximal ideals of $\mathbb{C}[V]$ arise this way. Closely related to this is the notion of the *spectrum* of a ring. The spectrum of a ring is the set of proper prime ideals. Since a maximal ideal is always a prime ideal, this relates V to $\mathbb{C}[V]$.

An affine variety $V \subseteq \mathbb{C}^n$ has two topologies, the *classical topology* and *Zariski topology*. The classical topology is the induced topology from usual topology on $\mathbb{C}^n$. Zariski

closed sets are subvarieties of V , whose complements are the Zariski open sets. We note that some Zariski open subsets of V are themselves affine varieties. For $f \in \mathbb{C}[V] \setminus \{0\}$ let

$$V_f = \{p \in V \mid f(p) \neq 0\} \subseteq V.$$

V_f is then Zariski open in V , and an affine variety. Let V have the ideal $\mathbf{I} = \langle f_1, \dots, f_s \rangle$ and pick $g \in \mathbb{C}$ representing f . Then $V_f = V \setminus \mathbf{V}(g)$ is Zariski open in V . If we consider a new variable y and let $W = \mathbf{V}(f_1, \dots, f_s, 1 - gy) \subseteq \mathbb{C}^n \times \mathbb{C}$. We can identify V_f with the affine variety W , since the projection $\mathbb{C}^n \times \mathbb{C} \rightarrow \mathbb{C}^n$ maps W bijectively onto V_f .

We have now defined affine varieties and stated some important notions of them, however this thesis focuses on affine toric varieties. Hence we introduce the n -dimensional torus, which is the affine open subset $(\mathbb{C}^*)^n = \mathbb{C}^n \setminus \mathbf{V}(x_1 \dots x_n) \subseteq \mathbb{C}^n$, with coordinate ring $\mathbb{C}[x_1, \dots, x_n]_{x_1 \dots x_n} = \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$. We note that $(\mathbb{C}^*)^n$ is a group under componentwise multiplication. By a *torus* we mean an affine variety T isomorphic to $(\mathbb{C}^*)^n$, where T inherits a group structure from the isomorphism. A *character* of a torus T is a morphism, which is a group homomorphism. For example, $m = (a_1, \dots, a_n) \in \mathbb{Z}^n$ gives a character $\chi^m: (\mathbb{C}^*)^n \rightarrow \mathbb{C}^*$ defined by

$$\chi^m(t_1, \dots, t_n) = t_1^{a_1} \dots t_n^{a_n}.$$

All characters of $(\mathbb{C}^*)^n$ arise this way, thus the characters of $(\mathbb{C}^*)^n$ form a group that is isomorphic to $\mathbb{Z}^n$.

A *one-parameter subgroup* is a morphism $\lambda: \mathbb{C}^* \rightarrow T$, which is a group homomorphism. For example, $v = (b_1, \dots, b_n) \in \mathbb{Z}^n$ gives a one-parameter subgroup $\lambda^v: \mathbb{C}^* \rightarrow (\mathbb{C}^*)^n$ defined by

$$\lambda^v(t) = (t^{b_1}, \dots, t^{b_n}).$$

All one-parameter subgroups of $(\mathbb{C}^*)^n$ arise this way. The group of one-parameter subgroups of $(\mathbb{C}^*)^n$ is naturally isomorphic to $\mathbb{Z}^n$. Hence we have a natural bilinear pairing $\langle \cdot, \cdot \rangle: M \times N \rightarrow \mathbb{Z}$ defined as,

- **Intrinsic:** Given a one-parameter subgroup λ^v and a character χ^m , the composition $\chi^m \circ \lambda^v: \mathbb{C}^* \rightarrow \mathbb{C}^*$ is a character of $\mathbb{C}^*$, given by $t \mapsto t^l$ for some $l \in \mathbb{Z}$. Then $\langle m, v \rangle = l$.
- **Concrete:** If $T = (\mathbb{C}^*)^n$ with $m = (a_1, \dots, a_n) \in \mathbb{Z}^n, v = (b_1, \dots, b_n) \in \mathbb{Z}^n$, then one computes that

$$\langle m, v \rangle = \sum_{i=1}^n a_i b_i,$$

where the pairing is the usual dot product.

We can now define *affine toric varieties*.

Definition 2.2 (Affine toric variety). *An affine toric variety is an irreducible affine variety V containing a torus $T_N \simeq (\mathbb{C}^*)^n$ as a Zariski open subset, such that the action of T_N on itself extends to an algebraic action of T_N on V .*

In order to construct affine toric varieties from cones and fans, we introduce the concepts of *lattices* and *affine semigroups*. A lattice is a free abelian group of finite rank. Hence, a lattice of rank n is isomorphic to $\mathbb{Z}^n$. For example a torus T_N has lattices M of characters and N of one-parameter subgroups. A semigroup is a set S with an associative binary operation and an identity element. Furthermore, affine semigroups must satisfy:

- The binary operation is commutative. The operation will be written as $+$ and the identity element as 0 . A finite set $\mathcal{A} \subseteq S$ gives

$$\mathbb{N}\mathcal{A} = \left\{ \sum_{m \in \mathcal{A}} a_m m \mid a_m \in \mathbb{N} \right\} \subseteq S.$$

- The semigroup is finitely generated, i.e. there is a finite set $\mathcal{A} \subseteq S$ such that $\mathbb{N}\mathcal{A} = S$.
- The semigroup can be embedded in a lattice M .

An affine semigroup $S \subseteq M$, gives the *semigroup algebra* $\mathbb{C}[S]$, which is the vector space over $\mathbb{C}$ with S as basis and multiplication induced by the semigroup structure of S . If we think of M as the character lattice of a torus T_N , then $m \in M$ gives the character χ^m . Then

$$\mathbb{C}[S] = \left\{ \sum_{m \in S} c_m \chi^m \mid c_m \in \mathbb{C} \text{ and } c_m = 0 \text{ for all but finitely many } m \right\},$$

with multiplication induced by

$$\chi^m \cdot \chi^{m'} = \chi^{m+m'}.$$

If $S = \mathbb{N}\mathcal{A}$ for $\mathcal{A} = \{m_1, \dots, m_s\}$, then $\mathbb{C}[S] = \mathbb{C}[\chi^{m_1}, \dots, \chi^{m_s}]$.

Affine semigroups are related to affine toric varieties by the following proposition.

Proposition 2.3. *Let $S \subseteq M$ be an affine semigroup. Then:*

- $\mathbb{C}[S]$ is an integral domain and finitely generated as a $\mathbb{C}$ -algebra.
- $\text{Spec}(\mathbb{C}[S])$ is an affine toric variety whose torus has character lattice $\mathbb{Z}S$.

Lastly the following theorem asserts that this approach to affine toric variety are indeed equivalent.

Theorem 2.4. *Let V be an affine variety. The following are equivalent:*

- (i) V is an affine toric variety according to Definition 2.2.
- (ii) $V = \text{Spec}(\mathbb{C}[S])$ for an affine semigroup S .

2.2 Cones and Fans

We are now ready to discuss the focus points of this chapter, cones and their relation to affine toric varieties. Throughout this discussion, we will construct an example of an affine toric variety, namely $\mathbb{P}^2$. We start by fixing a pair of dual vector spaces $M_{\mathbb{R}}$ and $N_{\mathbb{R}}$.

Definition 2.5 (Convex polyhedral cone). *A convex polyhedral cone in $N_{\mathbb{R}}$ is a set of the form*

$$\sigma = \text{Cone}(S) = \left\{ \sum_{u \in S} \lambda_u u \mid \lambda_u \geq 0 \right\} \subseteq N_{\mathbb{R}},$$

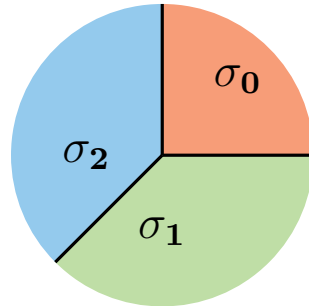
where $S \subseteq N_{\mathbb{R}}$ is finite. For the empty set, we set $\text{Cone}(\emptyset) = \{0\}$.

We quickly note that the naming, convex polyhedral cone, by Definition 2.5 is justified, since $x, y \in \sigma$ implies $\lambda x + (1 - \lambda)y \in \sigma$ for all $0 \leq \lambda \leq 1$. Furthermore, it is in fact a cone, since $x \in \sigma$ implies $\lambda x \in \sigma$ for all $\lambda \geq 0$.

Example 2.6. Let $N \simeq \mathbb{Z}^2$ be a fixed lattice, with dual lattice $M \simeq \mathbb{Z}^2$, such that we have the pair of dual vector spaces $N_{\mathbb{R}} \simeq \mathbb{R}^2$ and $M_{\mathbb{R}} \simeq \mathbb{R}^2$. Let e_1, e_2 and e_1^*, e_2^* be the canonical basis and corresponding dual basis. If we consider the vectors $v_1 = e_1, v_2 = e_2$ and $v_3 = -e_1 - e_2$, we can then create the following cones

$$\sigma_0 = \text{Cone}(v_1, v_2), \quad \sigma_1 = \text{Cone}(v_1, v_3) \quad \text{and} \quad \sigma_2 = \text{Cone}(v_2, v_3).$$

We will elaborate on the importance of these cones in later examples. The cones can be visualised as thus,



◇

Definition 2.7 (Dual Cone). *Given a polyhedral cone $\sigma \subseteq N_{\mathbb{R}}$, its dual cone is defined by*

$$\sigma^\vee = \{m \in M_{\mathbb{R}} \mid \langle m, u \rangle \geq 0 \text{ for all } u \in \sigma\}$$

In the following, the pairing between $M_{\mathbb{R}}$ and $N_{\mathbb{R}}$ is denoted $\langle \cdot, \cdot \rangle$, which is the usual dot product.

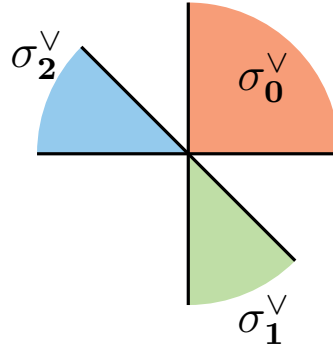
Ongoing Example 2.8. Let $u = (x, y) \in \mathbb{R}^2$, then

$$\langle u, v_1 \rangle = x, \quad \langle u, v_2 \rangle = y \quad \text{and} \quad \langle u, v_3 \rangle = -x - y.$$

The dual cones of σ_0, σ_1 and σ_2 are then respectively

$$\begin{aligned} \sigma_0^\vee &= \{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \text{ and } y \geq 0\} = \text{Cone}(e_1^*, e_2^*), \\ \sigma_1^\vee &= \{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \text{ and } -y \geq x\} = \text{Cone}(-e_2^*, e_1^* - e_2^*), \\ \sigma_2^\vee &= \{(x, y) \in \mathbb{R}^2 \mid y \geq 0 \text{ and } -y \geq x\} = \text{Cone}(-e_1^*, -e_1^* + e_2^*). \end{aligned}$$

The dual cones can be visualised as.



◇

Duality has the following important property:

Proposition 2.9. *Let $\sigma \subseteq N_{\mathbb{R}}$ be a polyhedral cone. Then $\sigma^\vee$ is a polyhedral cone in $M_{\mathbb{R}}$ and $(\sigma^\vee)^\vee = \sigma$.*

Given an $m \in M_{\mathbb{R}}$, we define

$$\begin{aligned} H_m &= \{u \in N_{\mathbb{R}} \mid \langle m, u \rangle = 0\} \subseteq N_{\mathbb{R}}, \\ H_m^+ &= \{u \in N_{\mathbb{R}} \mid \langle m, u \rangle \geq 0\} \subseteq N_{\mathbb{R}}. \end{aligned}$$

If $m \neq 0$ then H_m is a hyperplane and H_m^+ is a closed half-space. If $m_1, \dots, m_s$ generate $\sigma^\vee$, one could show that

$$\sigma = H_{m_1}^+ \cap \dots \cap H_{m_s}^+.$$

We wish to use hyperplanes to define the *faces* of a cone.

Definition 2.10 (Face). *A face of the polyhedral cone σ is $\tau = H_m \cap \sigma$ for some $m \in \sigma^\vee$, denoted by $\tau \preceq \sigma$. Faces where $\tau \neq \sigma$ are called proper faces, denoted $\tau \prec \sigma$.*

Ongoing Example 2.11. Consider $e_1^* \in \sigma_0^\vee$ and let $u = (x, y) \in N_{\mathbb{R}} \simeq \mathbb{R}^2$. Then $\langle e_1^*, u \rangle = x$, hence

$$H_{e_1^*} = \{(x, y) \in \mathbb{R}^2 \mid x = 0\}.$$

We therefore obtain

$$H_{e_1} \cap \sigma_0 = \{(x, y) \in \mathbb{R}^2 \mid x = 0 \text{ and } y \geq 0\} = \text{Cone}(v_2),$$

i.e. the positive y -axis. If we instead consider $e_2^* \in \sigma_0^\vee$, and repeat the computations we obtain

$$H_{e_2} \cap \sigma_0 = \{(x, y) \in \mathbb{R}^2 \mid y = 0 \text{ and } x \geq 0\} = \text{Cone}(v_1),$$

i.e. the positive x -axis. The same computations can be done for σ_1 and σ_2 to obtain

$$H_{-e_2} \cap \sigma_1 = \{(x, y) \in \mathbb{R}^2 \mid -y = 0 \text{ and } x \geq 0\} = \text{Cone}(v_1),$$

$$H_{e_1-e_2} \cap \sigma_1 = \{(x, y) \in \mathbb{R}^2 \mid x = y \text{ and } x \leq 0\} = \text{Cone}(v_3),$$

$$H_{-e_1} \cap \sigma_2 = \{(x, y) \in \mathbb{R}^2 \mid -x = 0 \text{ and } y \geq 0\} = \text{Cone}(v_2),$$

$$H_{-e_1+e_2} \cap \sigma_2 = \{(x, y) \in \mathbb{R}^2 \mid y = x \text{ and } y \leq 0\} = \text{Cone}(v_3).$$

We notice that each of the above faces, actually corresponds to the lines separating the cones in the figure in Example 2.6. It is a more informal way of describing the faces of the cones, however this thesis will stay in cones with maximal dimension 2, where the faces can be identified as the lines separating the cones. $\diamond$

We have some properties for the faces of a polyhedral cone.

Lemma 2.12. *Let $\sigma = \text{Cone}(S)$ be a polyhedral cone. Then:*

- (i) *Every face of σ is a polyhedral cone.*
- (ii) *An intersection of two faces of σ is again a face of σ .*
- (iii) *A face of a face of σ is again a face of σ .*

A *facet* of σ is a face τ of codimension 1, and an *edge* of σ is a face of dimension 1. In the scope of this thesis, a facet and edge coincide. However, the following proposition justifies the discussion of Ongoing Example 2.11.

Proposition 2.13. *Let $\sigma \subseteq N_{\mathbb{R}} \simeq \mathbb{R}^n$ be a polyhedral cone. Then:*

- (i) *If $\sigma = H_{m_1}^+ \cap \dots \cap H_{m_s}^+$ for $m_i \in \sigma^\vee, 1 \leq i \leq s$, then $\sigma^\vee = \text{Cone}(m_1, \dots, m_s)$.*
- (ii) *If $\dim(\sigma) = n$, then in (i) we can assume that the facets of σ are $\tau_i = H_{m_i} \cap \sigma$.*
- (iii) *Every proper face τ of σ is the intersection of the facets of σ containing τ .*

Another property we wish the cones to satisfy is called *strongly convex*. For a cone to be strongly convex, we require that the origin is a face of the cone. Furthermore, we wish the cones to be rational in the following sense.

Definition 2.14 (Rational). *A polyhedral cone $\sigma \subseteq N_{\mathbb{R}}$ is rational if $\sigma = \text{Cone}(S)$ for some finite set $S \subseteq N$.*

We note without proof that faces and duals of rational polyhedral cones are rational.

With these properties we observe that a cone σ has a canonical generating set, constructed in the following way. Let ρ be an edge of σ . We note ρ is a ray, i.e. a half-line, since σ is strongly convex. Furthermore, since ρ is rational, the semigroup $\rho \cap N$ is generated by a unique element $u_\rho \in \rho \cap N$, we call u_ρ the *ray generator* of ρ .

Lemma 2.15. *A strongly convex rational polyhedral cone is generated by the ray generators of its edges.*

It is customary to call the ray generators of the edges the *minimal generators* of a strongly convex rational polyhedral cone.

Definition 2.16. *Let $\sigma \subseteq N_{\mathbb{R}}$ be a strongly convex rational polyhedral cone.*

- (i) σ is smooth if its minimal generators form part of a $\mathbb{Z}$ -basis of N .
- (ii) σ is simplicial if its minimal generators are linearly independent over $\mathbb{R}$.

We are finally ready to bring our cones together, and link the theory to affine toric varieties.

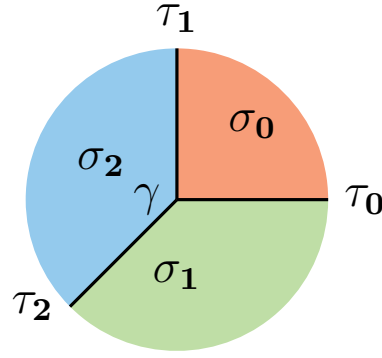
Definition 2.17 (Fan). *A fan Σ in $N_{\mathbb{R}}$ is a finite collection of cones $\sigma \subseteq N_{\mathbb{R}}$ such that:*

- (i) Every $\sigma \in \Sigma$ is a strongly convex rational polyhedral cone.
- (ii) For all $\sigma \in \Sigma$, each face of σ is also in Σ .
- (iii) For all $\sigma_1, \sigma_2 \in \Sigma$, the intersection $\sigma_1 \cap \sigma_2$ is a face of each.

Furthermore, the support of Σ is $|\Sigma| = \bigcup_{\sigma \in \Sigma} \sigma \subseteq N_{\mathbb{R}}$.

Ongoing Example 2.18. Clearly σ_0, σ_1 and σ_2 are strongly convex, rational, smooth and simplicial. Let $\gamma = \{0\}$, $\tau_0 = \text{Cone}(v_1)$, $\tau_1 = \text{Cone}(v_2)$ and $\tau_2 = \text{Cone}(v_3)$ be the origin, and the faces of σ_i . Consider the collection of all of the mentioned cones, and denote it by Σ . We now wish to show that this collection is a fan by Definition 2.17.

- (i) By the note of Definition 2.14 every cone in Σ is rational, or seen by how we described them.
- (ii) Every face of σ_i is described by a τ_j , and γ is a face of every cone in Σ .
- (iii) Consider the figure



Here we can see that (iii) is satisfied. For example take $\sigma_0 \cap \sigma_2$, from the figure, we can see that $\sigma_0 \cap \sigma_2 = \tau_1 \in \Sigma$.

We conclude that Σ is a fan. ◇

For a rational polyhedral cone $\sigma \subseteq N_{\mathbb{R}}$, the lattice points given by

$$S_{\sigma} = \sigma^{\vee} \cap M \subseteq M,$$

form a semigroup.

Lemma 2.19 (Gordan's Lemma). *$S_{\sigma} = \sigma^{\vee} \cap M$ is finitely generated, hence is an affine semigroup.*

We can now combine the theory of cones and fans, with the theory of affine toric varieties.

Theorem 2.20. *Let $\sigma \subseteq N_{\mathbb{R}}$ be a rational polyhedral cone with semigroup $S_{\sigma} = \sigma^{\vee} \cap M$. Then*

$$U_{\sigma} = \text{Spec}(\mathbb{C}[S_{\sigma}]) = \text{Spec}(\mathbb{C}[\sigma^{\vee} \cap M])$$

is an affine toric variety. Furthermore,

$$\dim(U_{\sigma}) = n \iff \text{the torus of } U_{\sigma} \text{ is } T_N = N \otimes_{\mathbb{Z}} \mathbb{C}^* \iff \sigma \text{ is strongly convex.}$$

Proposition 2.21. *Let τ be a face of σ and write $\tau = H_m \cap \sigma$, where $m \in \sigma^{\vee} \cap M$. Then the semigroup algebra $\mathbb{C}[S_{\tau}] = \mathbb{C}[\tau^{\vee} \cap M]$ is the localization of $\mathbb{C}[S_{\sigma}] = \mathbb{C}[\sigma^{\vee} \cap M]$ at $\chi^m \in \mathbb{C}[S_{\sigma}]$. In other words*

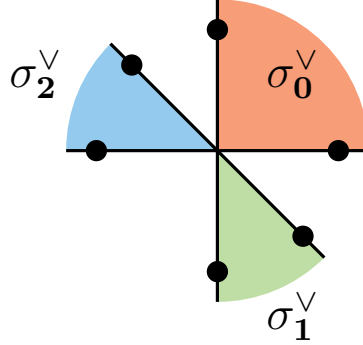
$$\mathbb{C}[S_{\tau}] = \mathbb{C}[S_{\sigma}]_{\chi^m}.$$

Ongoing Example 2.22. Recall that $N \simeq \mathbb{Z}^2$ with dual $M \simeq \mathbb{Z}^2$.

Consider $\sigma_0 = \text{Cone}(v_1, v_2)$, $\sigma_1 = \text{Cone}(v_1, v_3)$ and $\sigma_2 = \text{Cone}(v_2, v_3)$, we then have

$$\begin{aligned} S_{\sigma_0} &= \sigma_0^\vee \cap M = \{(x, y) \in \mathbb{Z}^2 \mid x \geq 0 \text{ and } y \geq 0\}, \\ S_{\sigma_1} &= \sigma_1^\vee \cap M = \{(x, y) \in \mathbb{Z}^2 \mid y \leq 0 \text{ and } -x \geq y\}, \\ S_{\sigma_2} &= \sigma_2^\vee \cap M = \{(x, y) \in \mathbb{Z}^2 \mid y \geq 0 \text{ and } -x \geq y\}. \end{aligned}$$

Considering the below figure, it is easy to describe the generators of the semigroups.



The generators of S_{σ_0} are e_1^* and e_2^* , S_{σ_1} has generators $-e_2^*$ and $e_1^* - e_2^*$, and lastly S_{σ_2} has generators $-e_1^*$ and $-e_1^* + e_2^*$. By Theorem 2.20 the following are affine varieties:

$$\begin{aligned} U_{\sigma_0} &= \text{Spec}(\mathbb{C}[S_{\sigma_0}]), \\ U_{\sigma_1} &= \text{Spec}(\mathbb{C}[S_{\sigma_1}]), \\ U_{\sigma_2} &= \text{Spec}(\mathbb{C}[S_{\sigma_2}]). \end{aligned}$$

Furthermore, the torus of U_{σ_i} is $T_N = (\mathbb{C}^*)^2$. We can elaborate on the semigroups, considering the discussion of affine semigroups in Section 2.1. We have the character lattice $M \simeq \mathbb{Z}^2$ of $(\mathbb{C}^*)^2$, then an $m \in M$ gives a character χ^m . Since $S_{\sigma_0} = \mathbb{N}\mathcal{A}$ for $\mathcal{A} = \{e_1^*, e_2^*\} \subseteq S_{\sigma_0}$, we get that $\mathbb{C}[S_{\sigma_0}] = \mathbb{C}[\chi^{e_1^*}, \chi^{e_2^*}]$. Setting $x_i = \chi^{e_i^*}$ we get the coordinate ring $\mathbb{C}[x_1, x_2]$. Similarly for S_{σ_1} and S_{σ_2} we obtain the coordinate rings $\mathbb{C}[x_1 x_2^{-1}, x_2^{-1}]$ and $\mathbb{C}[x_1^{-1}, x_1^{-1} x_2]$ respectively. $\diamond$

Hence in our ongoing example, every cone of Σ gives an affine toric variety. By (iii) in Definition 2.17 one could imagine there must be a way of gluing these affine varieties together, and indeed there is.

Suppose we have a finite collection $\{V_\alpha\}_\alpha$ of affine varieties, and for all pairs α, β we have Zariski open sets $V_{\beta\alpha} \subseteq V_\alpha$ and isomorphisms $g_{\beta\alpha}: V_{\beta\alpha} \rightarrow V_{\alpha\beta}$ satisfying the following compatibility conditions:

- (i) $g_{\alpha\beta} = g_{\beta\alpha}^{-1}$ for all pairs α, β .
- (ii) $g_{\beta\alpha}(V_{\beta\alpha} \cap V_{\gamma\alpha}) = V_{\alpha\beta} \cap V_{\gamma\beta}$ and $g_{\gamma\alpha} = g_{\gamma\beta} \circ g_{\beta\alpha}$ on $V_{\beta\alpha} \cap V_{\gamma\alpha}$ for all α, β, γ .

Let W be the disjoint union of the V_α , and define the equivalence relation on W by $u \sim v$ if and only if $u \in V_\alpha$ and $v \in V_\beta$ for some α, β with $v = g_{\beta\alpha}(u)$. Compatibility

condition (i) shows that $\sim$ is reflexive and symmetric, and compatibility condition (ii) shows it is transitive. Hence $\sim$ is an equivalence relation. We can then form the quotient space $X = W/\sim$. For each α , let

$$U_\alpha = \{[u] \in X \mid u \in V_\alpha\}.$$

Then $U_\alpha \subseteq X$ is an open set and the map $h_\alpha(u) = [u]$ defines a homeomorphism $h_\alpha: V_\alpha \rightarrow U_\alpha \subseteq X$. Thus X locally looks like an affine variety.

Ongoing Example 2.23. Consider $x_1 \in \mathbb{C}[x_1, x_2]$ and let

$$(U_{\sigma_0})_{x_1} = \{p \in U_{\sigma_0} \mid x_1(p) \neq 0\} \subseteq U_{\sigma_0}.$$

For each of the variables of the coordinate rings in the previous examples, let the following be defined as above

$$(U_{\sigma_0})_{x_2}, (U_{\sigma_1})_{x_1 x_2^{-1}}, (U_{\sigma_1})_{x_2^{-1}}, (U_{\sigma_2})_{x_1^{-1}}, (U_{\sigma_2})_{x_1^{-1} x_2}.$$

By the discussion in the first section of this chapter, these are Zariski open. We will now show how these glue together to form $\mathbb{P}^2$. We consider the finite collection of the affine varieties $\{U_{\sigma_i}\}$ for $i = 0, 1, 2$. For all pairs σ_i, σ_j we have the Zariski open sets as defined above. Consider the maps

$$\begin{aligned} g_{20}: (U_{\sigma_0})_{x_1} &\rightarrow (U_{\sigma_2})_{x_1^{-1}}, \\ g_{10}: (U_{\sigma_0})_{x_2} &\rightarrow (U_{\sigma_1})_{x_2^{-1}}, \\ g_{21}: (U_{\sigma_1})_{x_1 x_2^{-1}} &\rightarrow (U_{\sigma_2})_{x_1^{-1} x_2}. \end{aligned}$$

We note that morphisms between affine varieties corresponds to $\mathbb{C}$ -algebra homomorphisms, namely their coordinate rings. Hence, we can also consider the following maps:

$$\begin{aligned} g_{20}^*: \mathbb{C}[x_1, x_2]_{x_1} &\rightarrow \mathbb{C}[x_1^{-1}, x_1^{-1} x_2]_{x_1^{-1}}, \text{ where } x_1 \mapsto \frac{1}{x_1^{-1}}, x_2 \mapsto \frac{x_1^{-1} x_2}{x_1^{-1}}, \\ g_{10}^*: \mathbb{C}[x_1, x_2]_{x_2} &\rightarrow \mathbb{C}[x_1 x_2^{-1}, x_2^{-1}]_{x_2^{-1}}, \text{ where } x_1 \mapsto \frac{x_1 x_2^{-1}}{x_2^{-1}}, x_2 \mapsto \frac{1}{x_2^{-1}}, \\ g_{21}^*: \mathbb{C}[x_1 x_2^{-1}, x_2^{-1}]_{x_1 x_2^{-1}} &\rightarrow \mathbb{C}[x_1^{-1}, x_1^{-1} x_2]_{x_1^{-1} x_2}, \text{ where } x_1 x_2^{-1} \mapsto \frac{1}{x_1^{-1} x_2}, x_2^{-1} \mapsto \frac{x_1^{-1}}{x_1^{-1} x_2}, \\ g_{02}^*: \mathbb{C}[x_1^{-1}, x_1^{-1} x_2]_{x_1^{-1}} &\rightarrow \mathbb{C}[x_1, x_2]_{x_1}, \text{ where } x_1^{-1} \mapsto \frac{1}{x_1}, x_1^{-1} x_2 \mapsto \frac{x_2}{x_1}, \\ g_{01}^*: \mathbb{C}[x_1 x_2^{-1}, x_2^{-1}]_{x_2^{-1}} &\rightarrow \mathbb{C}[x_1, x_2]_{x_2}, \text{ where } x_1 x_2^{-1} \mapsto \frac{x_1}{x_2}, x_2^{-1} \mapsto \frac{1}{x_2}, \\ g_{12}^*: \mathbb{C}[x_1^{-1}, x_1^{-1} x_2]_{x_1^{-1} x_2} &\rightarrow \mathbb{C}[x_1 x_2^{-1}, x_2^{-1}]_{x_1 x_2^{-1}}, \text{ where } x_1^{-1} \mapsto \frac{x_2^{-1}}{x_1 x_2^{-1}}, x_1^{-1} x_2 \mapsto \frac{1}{x_1 x_2^{-1}}. \end{aligned}$$

We can see that $g_{ij} = g_{ji}^{-1}$ for all pairs i, j .

The second compatibility condition is also satisfied but will be omitted for simplicity. $\diamond$

In general if we have a fan Σ , each cone $\sigma \in \Sigma$ gives the affine toric variety

$$U_\sigma = \text{Spec}(\mathbb{C}[S_\sigma]) = \text{Spec}(\mathbb{C}[\sigma^\vee \cap M]).$$

Recall that a face τ of σ is given by $\tau = H_m \cap \sigma$ for $m \in \sigma^\vee$. In the proof of Proposition 2.21 one would have used the equality

$$S_\tau = S_\sigma + \mathbb{Z}(-m),$$

to show that $\mathbb{C}[S_\tau]$ is the localization $\mathbb{C}[S_\sigma]_{\chi^m}$. Hence, when τ is a face of σ then $U_\tau = (U_\sigma)_{\chi^m}$. We will further need a fact which will be stated without proof, however our example shows it nicely, that if $\tau = \sigma_1 \cap \sigma_2$, then

$$\sigma_1 \cap H_m = \tau = \sigma_2 \cap H_m,$$

for some $m \in \sigma_1^\vee \cap (-\sigma_2)^\vee \cap M$. This shows that

$$U_{\sigma_1} \supseteq (U_{\sigma_1})_{\chi^m} = U_\tau = (U_{\sigma_2})_{\chi^{-m}} \subseteq U_{\sigma_2}.$$

Considering the fan Σ , we obtain the collection of affine toric varieties $U_\sigma = \text{Spec}(\mathbb{C}[S_\sigma])$, where σ iterates over all cones in Σ . Consider any σ_1 and σ_2 in Σ , and let $\tau = \sigma_1 \cap \sigma_2$. By the above equation, we have an isomorphism

$$g_{\sigma_2, \sigma_1}: (U_{\sigma_1})_{\chi^m} \rightarrow (U_{\sigma_2})_{\chi^{-m}},$$

which is the identity on U_τ . The compatibility conditions for gluing the affine varieties U_σ along the subvarieties $(U_\sigma)_{\chi^m}$ are satisfied.

We have then obtained an abstract variety X_Σ associated to the fan Σ . In the case of varieties obtained from the structure of fans, the varieties inherits some properties, namely *normal* and *separated*, defined as follows.

Definition 2.24. A variety X is called *normal* if it is irreducible and the local rings $\mathcal{O}_{X,p}$ are normal for all $p \in X$.

Furthermore, X is called *separated* if the image of the diagonal map $\Delta: X \rightarrow X \times X$ is Zariski closed in $X \times X$.

Theorem 2.25. Let V be an affine toric variety with torus T_N . The following are then equivalent:

- (i) V is normal.
- (ii) $V = \text{Spec}(\mathbb{C}[S_\sigma]) (= U_\sigma)$, where $S_\sigma = \sigma^\vee \cap M$ and $\sigma \subseteq N_{\mathbb{R}}$ is a strongly convex rational polyhedral cone.

Proposition 2.26. Let X be an irreducible variety with a cover of affine open sets V_α . Then X is normal if and only if each V_α is normal.

This leads us to an important theorem, bringing fans and toric varieties together.

Theorem 2.27. *Let Σ be a fan in $N_{\mathbb{R}}$. The variety X_{Σ} is a normal separated toric variety.*

Proof. Since Σ is a fan, every $\sigma \in \Sigma$ is strongly convex, hence $\{0\}$ is a face of every σ . By Theorem 2.20 we have $T_N = \text{Spec}(\mathbb{C}[M]) \simeq (\mathbb{C}^*)^n \subseteq U_{\sigma}$ for all σ . Through the gluing process, all of these tori are identified, hence $T_N \subseteq X_{\Sigma}$, and each U_{σ} has an action of T_N . The gluing isomorphism g_{σ_2, σ_1} is the identity on $\mathbb{C}[S_{\sigma_1 \cap \sigma_2}]$. Hence the actions are compatible on the intersection of every pair of sets in the open affine cover. This patches together to an algebraic action of T_N on X_{Σ} .

X_{Σ} is irreducible, since every U_{σ} is an irreducible affine toric variety containing T_N . By Theorem 2.25 every U_{σ} is a normal affine variety, hence X_{Σ} is normal by Proposition 2.26. Consider the diagonal map

$$\Delta: U_{\tau} \rightarrow U_{\sigma_1} \times U_{\sigma_2}, \quad \tau = \sigma_1 \cap \sigma_2.$$

We wish to show the image of this map is Zariski closed. However, the map Δ comes from the $\mathbb{C}$ -algebra homomorphism

$$\Delta^*: \mathbb{C}[S_{\sigma_1}] \otimes_{\mathbb{C}} \mathbb{C}[S_{\sigma_2}] \rightarrow \mathbb{C}[S_{\tau}]$$

defined by $\chi^m \otimes \chi^n \mapsto \chi^{m+n}$. One can then use the fact that if $\sigma_1, \sigma_2 \in \Sigma$ and $\tau = \sigma_1 \cap \sigma_2$, then $S_{\tau} = S_{\sigma_1} + S_{\sigma_2}$, to show Δ^* is surjective. Hence

$$\mathbb{C}[S_{\tau}] \simeq (\mathbb{C}[S_{\sigma_1}] \otimes_{\mathbb{C}} \mathbb{C}[S_{\sigma_2}]) / \ker(\Delta^*),$$

and the image of Δ is a Zariski closed subset of $U_{\sigma_1} \times U_{\sigma_2}$. □

Some concepts of cones carries over to fans and in turn over to toric varieties, in the following sense.

Definition 2.28. *Let $\Sigma \subseteq N_{\mathbb{R}}$ be a fan.*

- (i) Σ is smooth if every cone σ in Σ is smooth.
- (ii) Σ is simplicial if every cone σ in Σ is simplicial.
- (iii) Σ is complete if its support $|\Sigma| = \bigcup_{\sigma \in \Sigma} \sigma = N_{\mathbb{R}}$.

Theorem 2.29. *Let X_{Σ} be the toric variety defined by a fan $\Sigma \subseteq N_{\mathbb{R}}$. Then:*

- (i) X_{Σ} is a smooth variety if and only if the fan Σ is smooth.
- (ii) X_{Σ} is an orbifold if and only if the fan Σ is simplicial, and we say that X_{Σ} is simplicial.
- (iii) X_{Σ} is compact in the classical topology if and only if Σ is complete.

2.3 The Orbit-Cone Correspondence and Blow-up

In this section we will show the Orbit-Cone Correspondence, that there is a bijective relation of the cones of a fan Σ , and of T_N -orbits in the toric variety X_Σ . Furthermore, we will introduce the notion of blow-up, which will be studied more later.

The Orbit-Cone Correspondence is explicitly given by

Theorem 2.30 (Orbit-Cone Correspondence). *Let X_Σ be the toric variety coming from the fan Σ in $N_{\mathbb{R}}$, where $O(\sigma)$ denotes the torus orbit of a cone $\sigma \in \Sigma$, with a distinguished point. Then:*

(i) *There is a bijective correspondence*

$$\begin{aligned} \{\text{cones } \sigma \text{ in } \Sigma\} &\longleftrightarrow \{T_N\text{-orbits in } X_\Sigma\} \\ \sigma &\longleftrightarrow O(\sigma) \simeq \text{Hom}_{\mathbb{Z}}(\sigma^\perp \cap M, \mathbb{C}^*). \end{aligned}$$

(ii) *Let $n = \dim(N_{\mathbb{R}})$. For each cone $\sigma \in \Sigma$, $\dim(O(\sigma)) = n - \dim(\sigma)$.*

(iii) *The affine open subsets U_σ is the union of orbits*

$$U_\sigma = \bigcup_{\tau \prec \sigma} O(\tau).$$

(iv) *τ is a face of σ if and only if $O(\sigma) \subseteq \overline{O(\tau)}$, and*

$$\overline{O(\tau)} = \bigcup_{\tau \prec \sigma} O(\sigma),$$

where $\overline{O(\tau)}$ denotes the closure both in the classical and Zariski topologies.

In order to make sense of Theorem 2.30, we need to discuss the notion of *actions of groups, orbits and distinguished points* (Conrad 2008).

Definition 2.31. *An action of a group G on a set X is the choice of a permutation $\pi_g: X \rightarrow X$, for each $g \in G$, such that the following conditions hold:*

(i) *π_e is the identity: $\pi_e(x) = x$ for each $x \in X$.*

(ii) *For every g_1 and g_2 in G , $\pi_{g_1} \circ \pi_{g_2} = \pi_{g_1 g_2}$.*

Ongoing Example 2.32. Consider the action of $(t_1, t_2) \in (\mathbb{C}^*)^2$ on $\mathbb{P}^2$ with homogeneous coordinates $[x_0 : x_1 : x_2]$, given by

$$(t_1, t_2) \cdot [x_0 : x_1 : x_2] = [x_0 : t_1 x_1 : t_2 x_2].$$

We wish to describe the fixed points of this action, i.e. the points in $\mathbb{P}^2$ where the action does not change the point.

We can clearly see $[1 : 0 : 0]$ is a fixed point, since the action only works on the remaining factors. One could think that $[0 : 1 : 0]$ and $[0 : 0 : 1]$ are also fixed points, which they indeed are. Let us test it

$$\begin{aligned}(t_1, t_2) \cdot [0 : 1 : 0] &= [0 : t_1 : 0], \\ (t_1, t_2) \cdot [0 : 0 : 1] &= [0 : 0 : t_2].\end{aligned}$$

Since we are working with homogenous coordinates, and $(t_1, t_2) \in (\mathbb{C}^*)^2$, we can rescale by t_1 and t_2 respectively. In other words we have $[0 : t_1 : 0] \sim [0 : 1 : 0]$ and $[0 : 0 : t_2] \sim [0 : 0 : 1]$. Hence we have the fixed points $[1 : 0 : 0]$, $[0 : 1 : 0]$ and $[0 : 0 : 1]$ of the action. $\diamond$

Definition 2.33. *Let a group G act on a set X . For each $x \in X$, its orbit is*

$$O_x = \{g \cdot x \mid g \in G\} \subset X$$

$O(\sigma)$ for a $\sigma \in \Sigma$ in our notation.

Ongoing Example 2.34. Given that we are working with homogenous coordinates $[x_0 : x_1 : x_2] \in \mathbb{P}^2$, one could suspect the orbits would be given by the different combinations of $x_i = 0$ or $x_i \neq 0$. In other words we have the orbits

$$\begin{aligned}O_1 &= \{[x_0 : x_1 : x_2] \in \mathbb{P}^2 \mid x_i \neq 0 \text{ for all } i\} \ni [1 : 1 : 1], \\ O_2 &= \{[x_0 : x_1 : x_2] \in \mathbb{P}^2 \mid x_0 = 0, \text{ and } x_1, x_2 \neq 0\} \ni [0 : 1 : 1], \\ O_3 &= \{[x_0 : x_1 : x_2] \in \mathbb{P}^2 \mid x_1 = 0, \text{ and } x_0, x_2 \neq 0\} \ni [1 : 0 : 1], \\ O_4 &= \{[x_0 : x_1 : x_2] \in \mathbb{P}^2 \mid x_2 = 0, \text{ and } x_0, x_1 \neq 0\} \ni [1 : 1 : 0], \\ O_5 &= \{[x_0 : x_1 : x_2] \in \mathbb{P}^2 \mid x_0 = x_1 = 0 \text{ and } x_2 \neq 0\} = [0 : 0 : 1], \\ O_6 &= \{[x_0 : x_1 : x_2] \in \mathbb{P}^2 \mid x_0 = x_2 = 0 \text{ and } x_1 \neq 0\} = [0 : 1 : 0], \\ O_7 &= \{[x_0 : x_1 : x_2] \in \mathbb{P}^2 \mid x_1 = x_2 = 0 \text{ and } x_0 \neq 0\} = [1 : 0 : 0].\end{aligned}$$

One could show these are all of the orbits of the action on $\mathbb{P}^2$. We highlighted some important points, which we will use later to relate to the fan Σ . We note the last three orbits actually correspond to the fixed points of the action of $(\mathbb{C}^*)^2$ on $\mathbb{P}^2$. $\diamond$

We are almost ready to discuss the Orbit-Cone Correspondence Theorem 2.30. However, we need the last notion of distinguished points of limits of one-parameter subgroups. This will mainly be done through our ongoing example.

Ongoing Example 2.35. We view the torus slightly different from earlier, the torus $(\mathbb{C}^*)^2 \subseteq \mathbb{P}^2$ consists of points with homogenous coordinates $[1 : s : t]$ where $s, t \neq 0$. If we consider a point $u = (a, b) \in N \simeq \mathbb{Z}^2$, we obtain a curve in $\mathbb{P}^2$, truly a curve in $(\mathbb{C}^*)^2$, but we view it as a curve in $\mathbb{P}^2$:

$$\lambda^u(t) = [1 : t^a : t^b].$$

We now look at the limit of $\lambda^u(t)$ as $t \rightarrow 0$. We can already see the limit point depends on $u = (a, b)$. If we consider $u = (a, b) \in \mathbb{Z}^2$ with $a, b > 0$ we obtain

$$\lim_{t \rightarrow 0} [1 : t^a : t^b] = [1 : 0 : 0],$$

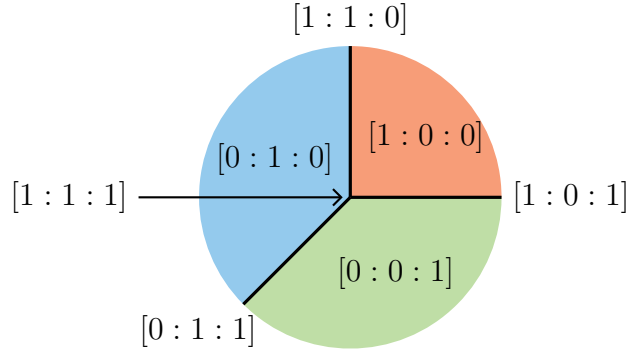
i.e. one of fixed points that correspond to an orbit. Consider instead $u = (a, b)$ with $a < 0$ and $a < b$. Since we are working with homogeneous coordinates, we have $[1 : t^a : t^b] \sim [t^{-a} : 1 : t^{b-a}]$. Hence

$$\lim_{t \rightarrow 0} [1 : t^a : t^b] = \lim_{t \rightarrow 0} [t^{-a} : 1 : t^{b-a}] = [0 : 1 : 0].$$

Using the same arguments and changing the restrictions on a and b and a slight abuse of notation, to further the understanding, we obtain

$$\begin{aligned} b < 0, \quad b < a & : \quad \lim_{t \rightarrow 0} [1 : t^a : t^b] = \lim_{t \rightarrow 0} [t^{-b} : t^{a-b} : 1] = [0 : 0 : 1], \\ b = 0, \quad a > 0 & : \quad \lim_{t \rightarrow 0} [1 : t^a : t^b] = \lim_{t \rightarrow 0} [1 : t^a : 1] = [1 : 0 : 1], \\ a = 0, \quad b > 0 & : \quad \lim_{t \rightarrow 0} [1 : t^a : t^b] = \lim_{t \rightarrow 0} [1 : 1 : t^b] = [1 : 1 : 0], \\ a = b < 0 & : \quad \lim_{t \rightarrow 0} [1 : t^a : t^b] = \lim_{t \rightarrow 0} [1 : t^a : t^a] = \lim_{t \rightarrow 0} [t^{-a} : 1 : 1] = [0 : 1 : 1], \\ a = b = 0 & : \quad \lim_{t \rightarrow 0} [1 : t^a : t^b] = \lim_{t \rightarrow 0} [1 : 1 : 1] = [1 : 1 : 1]. \end{aligned}$$

One could observe that all of the above conditions on $u = (a, b) \in \mathbb{Z}^2$, actually correspond to exactly the cones of the fan Σ .



◇

We have now shown the correspondence between the cones of a fan, and the orbits of the torus action on the variety X_Σ . The next step is to introduce the known concept of *blow-up*, but in the setting of fans. Blow-up introduces the notion of an *exceptional divisor*, which becomes important in the later chapters.

Before we define and compute an example of the blow-up of a fan, we consider the following theorem.

Theorem 2.36. *Let N_1, N_2 be lattices and let Σ_i be a fan in $(N_i)_{\mathbb{R}}, i = 1, 2$.*

(i) *If $\bar{\varphi}: N_1 \rightarrow N_2$ is a $\mathbb{Z}$ -linear map that is compatible with Σ_1 and Σ_2 , then there exists a toric morphism $\varphi: X_{\Sigma_1} \rightarrow X_{\Sigma_2}$, such that $\varphi|_{T_{N_1}}$ is the map*

$$\bar{\varphi} \otimes 1: N_1 \otimes_{\mathbb{Z}} \mathbb{C}^* \rightarrow N_2 \otimes_{\mathbb{Z}} \mathbb{C}^*.$$

(ii) *Contrarily, if $\varphi: X_{\Sigma_1} \rightarrow X_{\Sigma_2}$ is a toric morphism, then φ induces a $\mathbb{Z}$ -linear map $\bar{\varphi}: N_1 \rightarrow N_2$ that is compatible with the fans Σ_1 and Σ_2 .*

This theorem becomes handy, as the blow-up of a toric variety is induced by a morphism between the fans. In the classical sense of blow-up, we have a natural map between the blow-up and the blown-up. We further need the maps to be *compatible*, in the following sense

Definition 2.37. *Let N_1, N_2 be two lattices with fans Σ_1, Σ_2 in $(N_1)_{\mathbb{R}}, (N_2)_{\mathbb{R}}$ respectively. A $\mathbb{Z}$ -linear mapping $\bar{\varphi}: N_1 \rightarrow N_2$ is compatible with the fans Σ_1 and Σ_2 if for every cone $\sigma_1 \in \Sigma_1$, there exists a cone $\sigma_2 \in \Sigma_2$ such that $\bar{\varphi}_{\mathbb{R}}(\sigma_1) \subseteq \sigma_2$.*

In the notion of fans, the blow-up of a fan Σ becomes a *refinement* Σ' . As the name indicates, we wish to take an existing fan, and divide the cones even further, still maintaining the so far discussed properties. We say a fan Σ' *refines* Σ , if every $\sigma \in \Sigma$ is contained in a $\sigma' \in \Sigma'$ and $|\Sigma| = |\Sigma'|$. If we have a refinement Σ' of Σ , then the identity mapping $\bar{\varphi}: N \rightarrow N$ induces a toric morphism $\varphi: X_{\Sigma'} \rightarrow X_{\Sigma}$, since $\bar{\varphi}$ is clearly compatible with Σ' and Σ .

The blow-up requires a specific procedure for the refinement of the fan Σ . That is the procedure of *star subdivision*.

Definition 2.38. *Let Σ be a fan in $N_{\mathbb{R}} \sim \mathbb{R}^n$. Let $\sigma = \text{Cone}(v_1, \dots, v_n) \in \Sigma$ be a smooth cone, so that $v_1, \dots, v_n$ is a basis for N . Let $v_0 = v_1 + \dots + v_n$ and let $\Sigma'(\sigma)$ be the set of all cones generated by subsets of $\{v_0, \dots, v_n\}$ not containing $\{v_1, \dots, v_n\}$. Then*

$$\Sigma^*(\sigma) = (\Sigma \setminus \{\sigma\}) \cup \Sigma'(\sigma)$$

is a fan in $N_{\mathbb{R}}$ called the star subdivision of Σ along σ .

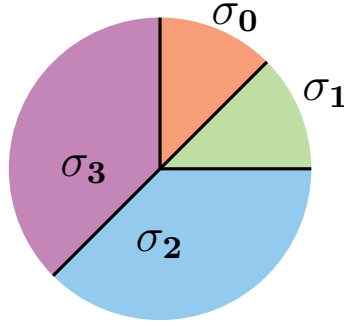
Ongoing Example 2.39. As noted earlier $\sigma_i \in \Sigma$ are smooth, and each correspond to a different distinguished point of $\mathbb{P}^2$. Consider $\sigma_0 = \text{Cone}(v_1, v_2)$, following the procedure of Definition 2.38 we let $v_4 = v_1 + v_2 = e_1 + e_2$. We further let $\Sigma'(\sigma_0)$ denote the set of all cones generated by subsets of $\{v_4, v_1, v_2\}$ not containing $\{v_1, v_2\}$, i.e. it is the set

$$\Sigma'(\sigma_0) = \{\text{Cone}(\emptyset), \text{Cone}(v_4), \text{Cone}(v_1), \text{Cone}(v_2), \text{Cone}(v_4, v_1), \text{Cone}(v_4, v_2)\}.$$

Then consider

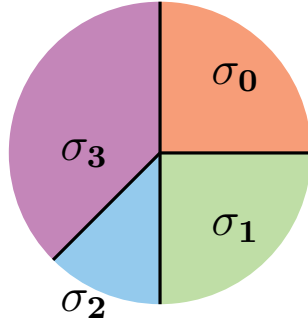
$$\begin{aligned}\Sigma^*(\sigma_0) &= (\Sigma \setminus \{\sigma_0\}) \cup \Sigma'(\sigma_0) \\ &= \{\text{Cone}(\emptyset), \text{Cone}(v_4), \text{Cone}(v_1), \text{Cone}(v_2), \text{Cone}(v_3), \\ &\quad \text{Cone}(v_1, v_3), \text{Cone}(v_2, v_3), \text{Cone}(v_4, v_1), \text{Cone}(v_4, v_2)\}.\end{aligned}$$

This is then the star subdivision of Σ along the distinguished point $[1 : 0 : 0]$, corresponding to the cone σ_0 . The new fan $\Sigma^*(\sigma_0)$ is then,

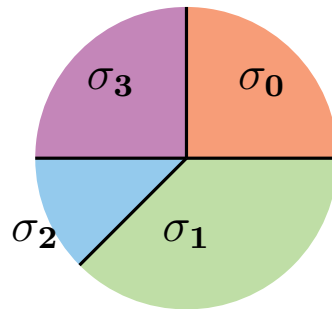


It is worth noting that throughout this thesis we will visualize fans by normalizing their rays to lie in the unit ball. In general, however, the rays may extend beyond the unit ball. While this distinction is important for the computations carried out in this thesis, including it in the visualizations would significantly reduce their readability.

We could also have chosen to consider σ_1 or σ_2 . For example considering the star subdivision of Σ along σ_1 , would yield the fan,



And the star subdivision of Σ along σ_2 , yields the fan,



As we will see in the later chapters, we could continue this process and perform the star subdivision, of the above fans, at the other distinguished points. $\diamond$

Why we consider the star subdivision becomes clear after the following proposition.

Proposition 2.40. $\Sigma^*(\sigma)$ is a refinement of Σ , and the induced toric morphism

$$\varphi: X_{\Sigma^*(\sigma)} \rightarrow X_{\Sigma},$$

makes $X_{\Sigma^*(\sigma)}$ the blow-up of X_{Σ} at the distinguished point γ_{σ} corresponding to the cone σ .

The above fans in our example, are then the blow-ups of $\mathbb{P}^2$ at $[1 : 0 : 0]$, $[0 : 1 : 0]$ and $[0 : 0 : 1]$ respectively. One could apply everything we have done so far in the examples, to the blow-ups and get a slightly different description. Why the description is slightly different, will become clear after the following section.

2.4 Divisors on Affine Toric Varieties

Divisors are defined in terms of irreducible codimension 1 subvarieties. In the case of our example, and later computations, we only consider 2-dimensional fans. Hence one would expect the 1-dimensional cones of Σ , to correspond to a divisor. As we will see they indeed do, in fact a special kind of divisor.

In the Orbit-Cone Correspondence Theorem 2.30 we considered $O(\sigma)$, the torus orbit of a cone $\sigma \in \Sigma$, and especially its closure $\overline{O(\sigma)}$. The closure is in itself a toric variety, which can be seen below, hence if we consider the 1-dimensional cones, which we denote by τ , we obtain a 1-dimensional toric variety. Therefore we would also have obtained a toric variety of codimension 1.

Write $V(\tau) = \overline{O(\tau)}$. By Theorem 2.30 we have $\tau \preceq \sigma$ if and only if $O(\sigma) \subseteq V(\tau)$, and

$$V(\tau) = \bigcup_{\tau \preceq \sigma} O(\sigma).$$

Let N_{τ} be the sublattice of N spanned by the points in $\tau \cap N$, and let $N(\tau) = N/N_{\tau}$. Then the torus $O(\tau) = T_{N(\tau)}$ is an open subset of $V(\tau)$. For each cone $\sigma \in \Sigma$ containing τ , let $\bar{\sigma}$ be the image cone in $N(\tau)_{\mathbb{R}}$ under the quotient map

$$N_{\mathbb{R}} \rightarrow N(\tau)_{\mathbb{R}},$$

coming from the exact sequence

$$0 \rightarrow N_{\tau} \rightarrow N \rightarrow N(\tau) \rightarrow 0.$$

Then the following

$$\text{Star}(\tau) = \{\bar{\sigma} \subseteq N(\tau)_{\mathbb{R}} \mid \tau \text{ is a face of } \sigma \in \Sigma\},$$

is a fan in $N(\tau)_{\mathbb{R}}$.

Proposition 2.41. *For any $\tau \in \Sigma$, the orbit closure $V(\tau) = \overline{O(\tau)}$ is isomorphic to the toric variety $X_{\text{Star}(\tau)}$*

Let X be an irreducible variety. A *prime divisor* $D \subseteq X$ is an irreducible subvariety of codimension 1, i.e. $\dim(D) = \dim(X) - 1$.

Definition 2.42. *Denote by $\text{Div}(X)$ the free abelian group generated by the prime divisors on X . A Weil divisor is then an element of $\text{Div}(X)$.*

A Weil divisor $D \in \text{Div}(X)$, is then a finite sum $D = \sum_i a_i D_i \in \text{Div}(X)$ of prime divisors D_i with $a_i \in \mathbb{Z}$ for all i . We say a divisor D is *effective*, if a_i are all nonnegative, denoted $D \geq 0$.

Definition 2.43. *Let X be a normal variety.*

(i) *The divisor of $f \in \mathbb{C}(X)^*$ is*

$$\text{div}(f) = \sum_D \nu_D(f) D,$$

where the sum is over all prime divisors $D \subseteq X$.

(ii) *$\text{div}(f)$ is called a principal divisor, and the set of all principal divisors is denoted $\text{Div}_0(X)$.*

(iii) *Divisors D and E are linearly equivalent, written $D \sim E$, if their difference is a principal divisor, that is $D - E = \text{div}(f) \in \text{Div}_0(X)$ for some $f \in \mathbb{C}(X)^*$.*

We note that $\text{Div}_0(X)$ is a subgroup of $\text{Div}(X)$. If $D = \sum_i a_i D_i$ is a Weil divisor on X and $U \subseteq X$ is a nonempty open subset, then

$$D|_U = \sum_{U \cap D_i \neq \emptyset} a_i U \cap D_i$$

is a Weil divisor on U called the *restriction* of D on U .

Definition 2.44. *A Weil divisor D on a normal variety X is Cartier if it is locally principal, i.e. X has an open cover $\{U_i\}_{i \in I}$ such that $D|_{U_i}$ is principal in U_i for every $i \in I$.*

We note that Cartier divisors on the normal variety X form a group $\text{CDiv}(X)$ with

$$\text{Div}_0(X) \subseteq \text{CDiv}(X) \subseteq \text{Div}(X).$$

This gives rise to the following important groups, formed by linear equivalence classes, that we will use in the later chapters.

Definition 2.45. *Let X be a normal variety. Its class group is*

$$\mathrm{Cl}(X) = \mathrm{Div}(X)/\mathrm{Div}_0(X),$$

and its Picard group is

$$\mathrm{Pic}(X) = \mathrm{CDiv}(X)/\mathrm{Div}_0(X).$$

Since $\mathrm{CDiv}(X)$ is a subgroup of $\mathrm{Div}(X)$, we have the canonical injection

$$\mathrm{Pic}(X) \hookrightarrow \mathrm{Cl}(X).$$

However, in our case we have a much nicer result, since we only will be considering smooth cones, hence by Definition 2.28 and Theorem 2.29, smooth toric varieties.

Theorem 2.46. *Let X be a normal variety. Then if X is smooth, then every Weil divisor on X is Cartier.*

From the above we can calculate that $\mathrm{Cl}(X) = \mathrm{Pic}(X)$ in our case. Let $\Sigma(1)$ denote the 1-dimensional cones of the fan Σ . We will from now on call a cone $\tau \in \Sigma(1)$ for a *ray*. As discussed earlier, a $\tau \in \Sigma(1)$ gives the codimension 1 orbit $O(\tau)$, whose closure $\overline{O(\tau)}$ is a T_N -invariant prime divisor on X_Σ . To emphasize that $\overline{O(\tau)}$ is a divisor, we will denote it by D_τ , rather than $V(\tau)$ as previously. Since $\tau \in \Sigma(1)$ is 1-dimensional it has a minimal generator $v_\tau \in \tau \cap N$. Recall that for an $m \in M$ we get the character χ^m of the torus T_N .

Proposition 2.47. *For $m \in M$, the character χ^m is a rational function on X_Σ , and its divisor is given by*

$$\mathrm{div}(\chi^m) = \sum_{\tau \in \Sigma(1)} \langle m, v_\tau \rangle D_\tau.$$

Divisors of the form $\sum_{\tau \in \Sigma(1)} a_\tau D_\tau$ are precisely the divisors invariant under the torus action on X_Σ . Hence

$$\mathrm{Div}_{T_N}(X_\Sigma) = \sum_{\tau \in \Sigma(1)} \mathbb{Z} D_\tau \subseteq \mathrm{Div}(X_\Sigma),$$

is the group of T_N -invariant Weil divisors on X_Σ . This brings us to an important theorem.

Theorem 2.48. *Consider the exact sequence*

$$M \rightarrow \operatorname{Div}_{T_N}(X_\Sigma) \rightarrow \operatorname{Cl}(X_\Sigma) \rightarrow 0.$$

The first map is $m \mapsto \operatorname{div}(\chi^m)$ and the second sends a T_N -invariant divisor to its divisor class in $\operatorname{Cl}(X_\Sigma)$. We also have the short exact sequence

$$0 \rightarrow M \rightarrow \operatorname{Div}_{T_N}(X_\Sigma) \rightarrow \operatorname{Cl}(X_\Sigma) \rightarrow 0$$

if and only if $\{v_\tau \mid \tau \in \Sigma(1)\}$ spans $N_\mathbb{R}$.

This gives us an easy way of describing the class group $\operatorname{Cl}(X_\Sigma)$.

Ongoing Example 2.49. We wish to describe the class group $\operatorname{Cl}(X_\Sigma)$ of our example, through the divisors of the characters. Since the first map of the sequence from Theorem 2.48 is given by $m \mapsto \operatorname{div}(\chi^m)$. We further have the short exact sequence of Theorem 2.48

$$0 \rightarrow M \simeq \mathbb{Z}^2 \xrightarrow{f} \operatorname{Div}_{T_N}(X_\Sigma) \xrightarrow{g} \operatorname{Cl}(X_\Sigma) \rightarrow 0,$$

which gives us

$$\operatorname{Cl}(X_\Sigma) \simeq \operatorname{Div}_{T_N}(X_\Sigma) / \operatorname{im}(f) \simeq \operatorname{Div}_{T_N}(X_\Sigma) / \ker(g).$$

Then $\operatorname{Cl}(X_\Sigma)$ is generated by classes of the divisors D_1, D_2, D_3 corresponding to $\tau_0, \tau_1, \tau_2 \in \Sigma(1)$, subject to the relations

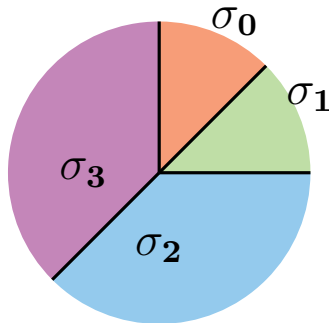
$$\begin{aligned} 0 &\sim \operatorname{div}(\chi^{e_1}) = \langle e_1, v_1 \rangle D_1 + \langle e_1, v_2 \rangle D_2 + \langle e_1, v_3 \rangle D_3 \\ &= D_1 - D_3 \\ 0 &\sim \operatorname{div}(\chi^{e_2}) = \langle e_2, v_1 \rangle D_1 + \langle e_2, v_2 \rangle D_2 + \langle e_2, v_3 \rangle D_3 \\ &= D_2 - D_3. \end{aligned}$$

Thus the class group has generators $[D_1] = [D_2] = [D_3]$, and hence $\operatorname{Cl}(X_\Sigma) \simeq \mathbb{Z}$. As described earlier, since Σ is smooth, X_Σ is smooth, and hence $\operatorname{Cl}(X_\Sigma) = \operatorname{Pic}(X_\Sigma)$.

$\mathbb{P}^2$ is a rather “simple” example. What would happen if we considered one of the blow-ups. Let $\Sigma^*(\sigma_0)$ be the fan constructed in Example 2.39, i.e. the blow-up of $\mathbb{P}^2$ at the distinguished point $[1 : 0 : 0]$. By Theorem 2.48 we again have the short exact sequence,

$$0 \longrightarrow M \simeq \mathbb{Z}^2 \xrightarrow{f} \operatorname{Div}_{T_N}(X_{\Sigma^*(\sigma_0)}) \xrightarrow{g} \operatorname{Cl}(X_{\Sigma^*(\sigma_0)}) \longrightarrow 0,$$

which we will use to describe $\operatorname{Cl}(X_{\Sigma^*(\sigma_0)})$. At this point we recall the fan $\Sigma^*(\sigma_0)$:



And recall $v_1 = e_1, v_2 = e_2, v_3 = -e_1 - e_2$ and $v_4 = e_1 + e_2$.

The short exact sequence gives that $\text{Cl}(X_{\Sigma^*(\sigma_0)})$ is generated by classes of the divisors D_1, D_2, D_3, D_4 corresponding to $\tau_1, \tau_2, \tau_3, \tau_4 \in \Sigma^*(\sigma_0)(1)$, subject to the relations

$$\begin{aligned} 0 \sim \text{div}(\chi^{e_1}) &= \langle e_1, v_1 \rangle D_1 + \langle e_1, v_2 \rangle D_2 + \langle e_1, v_3 \rangle D_3 + \langle e_1, v_4 \rangle D_4 \\ &= D_1 - D_3 + D_4, \\ 0 \sim \text{div}(\chi^{e_2}) &= \langle e_2, v_1 \rangle D_1 + \langle e_2, v_2 \rangle D_2 + \langle e_2, v_3 \rangle D_3 + \langle e_2, v_4 \rangle D_4 \\ &= D_2 - D_3 + D_4. \end{aligned}$$

Thus the class group has generators $[D_1] = [D_2]$ and $[D_3] = [D_1 + D_4]$, and hence $\text{Cl}(X_{\Sigma^*(\sigma_0)}) \simeq \mathbb{Z}^2$. Again, since $\Sigma^*(\sigma_0)$ is smooth, $X_{\Sigma^*(\sigma_0)}$ is smooth and hence

$$\text{Cl}(X_{\Sigma^*(\sigma_0)}) = \text{Pic}(X_{\Sigma^*(\sigma_0)}).$$

We can see the procedure of blow-up has given the class group an extra generator. This will be important for later chapters. $\diamond$

There are other ways to compute the generators of the class group, see Chapter 4, this procedure however nicely relates to the pictures of fans.

2.5 Intersection Numbers on Affine Toric Varieties

Now that we have a description of divisors on a toric variety, we would like to have a description of how these divisors intersect curves and other divisors. In our case the divisors actually correspond to curves or vice versa, since we are working with 2-dimensional fans.

For a Cartier divisor D and complete curve C on a variety X , we would like the number $D \cdot C$ to count the number of points of $D \cap C$, with appropriate multiplicity. We would further wish for the intersection product $D \cdot C$, to posses the following properties:

- (i) $(D + E) \cdot C = D \cdot C + E \cdot C$.
- (ii) $D \cdot C = E \cdot C$ when $D \sim E$.
- (iii) Let D be a prime divisor on X such that $D \cap C$ is finite. Assume each $p \in D \cap C$ is smooth in C, D, X and that the tangent spaces $T_p(C) \subseteq T_p(X)$ and $T_p(D) \subseteq T_p(X)$ meet transversely. Then $D \cdot C = |D \cap C|$.

Explicitly defining the intersection product is beyond the scope of this thesis. We will focus on the properties and notions associated to the intersection product.

We can now define an important class of Cartier divisors, the *numerically effective* Cartier divisors.

Definition 2.50. *Let X be a normal variety. Then a Cartier divisor D on X is numerically effective if*

$$D \cdot C \geq 0$$

for every irreducible complete curve $C \subseteq X$. In shorthand called nef

In the toric case, we have a nicer way of interpreting the nef divisors.

Theorem 2.51. *Let D be a Cartier divisor on a toric variety X_Σ whose fan Σ has convex support of full dimension. The following are then equivalent:*

- (i) D is nef.
- (ii) $D \cdot C \geq 0$ for all torus-invariant irreducible complete curves $C \subseteq X$.

We note that every torus-invariant irreducible complete curve $C \subseteq X$ of a toric variety X , actually correspond precisely to the rays of the fan Σ of X described under Proposition 2.47.

Through the intersection product, we obtain an important equivalence relation on Cartier divisors.

Definition 2.52. *Let X be a normal variety.*

- (i) *A Cartier divisor D on X is numerically equivalent to zero if $D \cdot C = 0$ for all irreducible complete curves $C \subseteq X$.*
- (ii) *Cartier divisors D and E are numerically equivalent, denoted by $D \equiv E$, if $D - E$ is numerically equivalent to zero.*

In the toric case we have the following proposition.

Proposition 2.53. *Let D be a Cartier divisor on a toric variety X_Σ whose fan Σ has convex support of full dimension. Then $D \sim 0$ if and only if $D \equiv 0$.*

By the above equivalence relation we construct the following.

Definition 2.54. *For a normal variety X , define*

$$N^1(X) = (\text{CDiv}(X) / \equiv) \otimes_{\mathbb{Z}} \mathbb{R}.$$

The vector space $N^1(X)$ contain the following cone.

Definition 2.55. $\text{Nef}(X)$ is the cone in $N^1(X)$ generated by classes of nef Cartier divisors. We call $\text{Nef}(X)$ the nef cone.

We can obtain a nice description of the nef cone, by considering a fan Σ of convex support of full dimension, and its associated toric variety X_Σ . We have the inclusion

$$\text{Pic}(X_\Sigma) \subseteq \text{Pic}(X_\Sigma)_\mathbb{R},$$

since $\text{Pic}(X_\Sigma)$ is torsion-free. $\text{Pic}(X_\Sigma)$ is then a lattice in the vector space $\text{Pic}(X_\Sigma)_\mathbb{R}$. We further have

$$\text{Pic}(X_\Sigma)_\mathbb{R} = N^1(X_\Sigma),$$

since numerical equivalence and linear equivalence coincide by Proposition 2.53. We will therefore write $\text{Pic}(X_\Sigma)_\mathbb{R}$ instead of $N^1(X_\Sigma)$. The nef cone will be important in the later chapters.

For the following, we need to list the ray generators of the 2-dimensional cones in Σ as $v_0, \dots, v_{r-1}$ in clockwise order around the origin in $N_\mathbb{R}$. We consider the indices as integers modulo r , i.e. $v_r = v_0$. In order to compute the intersection product in a nice manner, we have the following lemma,

Lemma 2.56. Let $v_0, \dots, v_r$ be the ray generators for a smooth complete fan $\Sigma \in N_\mathbb{R} \simeq \mathbb{R}^2$. There exists $b_i \in \mathbb{Z}$, with $i = 0, \dots, r-1$, such that,

$$v_{i-1} + v_{i+1} = b_i v_i.$$

What would happen in the intersection product $D \cdot C$, if $D = C$? We would obtain the *self-intersection*. The self-intersection, and other intersection products, of divisors on a smooth toric surface, then have the nice description.

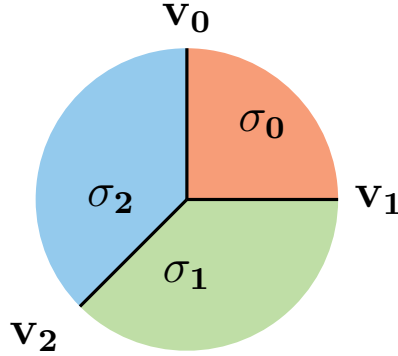
Theorem 2.57. Let D_τ be the divisor on a smooth toric surface X_Σ corresponding to $\tau = \text{Cone}(v)$, which is the intersection of 2-dimensional cones $\text{Cone}(v, v_1)$ and $\text{Cone}(v, v_2)$ in Σ . Then:

- (i) $D_\tau \cdot D_\tau = -b$, where $v_1 + v_2 = bv$ as in Lemma 2.56.
- (ii) For a divisor $D_{\tau'} \neq D_\tau$, we have

$$D_{\tau'} \cdot D_\tau = \begin{cases} 1, & \tau' = \text{Cone}(v_i), \quad i = 1, 2, \\ 0, & \text{otherwise.} \end{cases}$$

Ongoing Example 2.58. Recall that we let Σ denote the fan of the toric variety $\mathbb{P}^2$. Σ consists of the rays $v_1 = e_1, v_2 = e_2$ and $v_3 = -e_1 - e_2$, which we will relabel as $v_0 = e_2, v_1 = e_1, v_2 = -e_1 - e_2$. Note that we always start at the ray e_2 and label clock-

wise from there. Our fan can then be visualised as,



We then have the relations of the ray generators, following the above theorem and lemma

$$\begin{aligned} v_0 + v_2 &= e_2 + (-e_1 - e_2) = (-1) \cdot e_1 = (-1) \cdot v_1, \\ v_1 + v_0 &= e_1 + e_2 = (-1) \cdot (-e_1 - e_2) = (-1) \cdot v_2, \\ v_2 + v_1 &= (-e_1 - e_2) + e_1 = (-1) \cdot e_2 = (-1) \cdot v_0. \end{aligned}$$

We will further relabel the 1-dimensional cones of Σ , i.e. the cones $\tau \in \Sigma(1)$, to $\tau_0 = \text{Cone}(v_0)$, $\tau_1 = \text{Cone}(v_1)$ and $\tau_2 = \text{Cone}(v_2)$, and their corresponding torus-invariant divisors to D_0 , D_1 and D_2 respectively. Then since

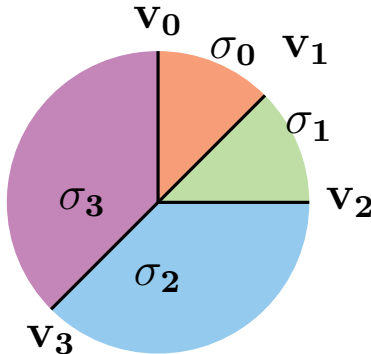
$$\begin{aligned} \tau_0 &= \text{Cone}(v_0) = \text{Cone}(v_0, v_1) \cap \text{Cone}(v_0, v_2) = \sigma_0 \cap \sigma_2, \\ \tau_1 &= \text{Cone}(v_1) = \text{Cone}(v_1, v_0) \cap \text{Cone}(v_1, v_2) = \sigma_0 \cap \sigma_1, \\ \tau_2 &= \text{Cone}(v_2) = \text{Cone}(v_2, v_1) \cap \text{Cone}(v_2, v_0) = \sigma_1 \cap \sigma_2, \end{aligned}$$

we get the following intersection matrix

	D_0	D_1	D_2
D_0	1	1	1
D_1	1	1	1
D_2	1	1	1

This intersection matrix is not that interesting to look at. One could wonder if the blow-up of $\mathbb{P}^2$ at a distinguished point, would yield a more interesting intersection matrix.

Let us recall and relabel the fan of the blow-up of $\mathbb{P}^2$ at $[1 : 0 : 0]$. The rays we use are $v_0 = e_2$, $v_1 = e_1 + e_2$, $v_2 = e_1$ and $v_3 = -e_1 - e_2$. The 1-dimensional cones are then $\tau \in \Sigma^*(\sigma_0)(1)$, which we will denote by $\tau_0 = \text{Cone}(v_0)$, $\tau_1 = \text{Cone}(v_1)$, $\tau_2 = \text{Cone}(v_2)$ and $\tau_3 = \text{Cone}(v_3)$. The fan $\Sigma^*(\sigma_0)$ can then be visualised as,



Following the procedure we did for the fan Σ of $\mathbb{P}^2$, we obtain the following intersection matrix

	D_0	D_1	D_2	D_3
D_0	0	1	0	1
D_1	1	-1	1	0
D_2	0	1	0	1
D_3	1	0	1	1

As we can see a lot more is happening in this intersection matrix. There is suddenly a torus-invariant divisor with self-intersection number -1 , this is the *exceptional divisor*.

◇

To end this section, we have a classification of every smooth complete toric surface.

Theorem 2.59. *Every smooth complete toric surface X_Σ is obtained from either*

$$\mathbb{P}^2, \mathbb{P}^1 \times \mathbb{P}^1, \quad \text{or} \quad \mathcal{H}_r \quad \text{for} \quad r \geq 2$$

by a finite sequence of blow-ups at fixed points of the torus action.

Proposition 2.60. *A complete toric surface X_Σ is projective.*

Throughout the chapter we have computed an ongoing example of the theory in depth, if one wish to see an example of $\mathbb{P}^1 \times \mathbb{P}^1$ we refer to Appendix A. Note that we are missing the example of the Hirzebruch surfaces $\mathcal{H}_r$, $r \geq 2$, for completeness. However, in Chapter 5 we provide a clear obstruction that efficiently rules out the Hirzebruch surfaces with respect to the main quantity of interest $I(\alpha)$.

3 K-Stability on Toric Surfaces

In the previous chapter we discussed the relation between fans and toric varieties. However, we wish to relate fans to Kähler geometry. The question is now, how is this done? To answer this we will discuss the notion of *polytopes*, which are closely related to fans, along with concepts of *symplectic geometry*. We note that polytopes by themselves give rise to toric varieties. The focus is to illustrate the connection between the theories, which essentially is the *Delzant theorem*.

3.1 Affine Toric Varieties Through Polytopes

We are still in the setting where we consider a lattice N together with its dual lattice M . First we discuss the link between cones and polytopes, and later on we link polytopes to Kähler geometry. The definitions in this section originates from (Cox, Little, and Schenck 2011).

Definition 3.1 (Polytope). *A polytope in $N_{\mathbb{R}}$ is a set of the form,*

$$P = \text{Conv}(S) = \left\{ \sum_{v \in S} \lambda_v v \mid \lambda_v \geq 0, \sum_{v \in S} \lambda_v = 1 \right\} \subseteq N_{\mathbb{R}},$$

where $S \subseteq N_{\mathbb{R}}$ is finite, and P is the convex hull of S .

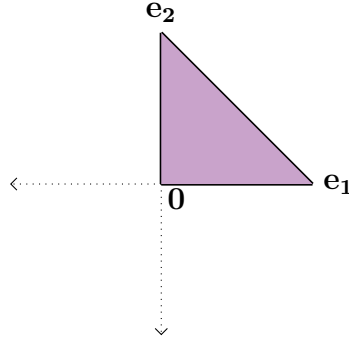
An example of a polytope is a polygon in $\mathbb{R}^2$.

Polytopes are closely related to polyhedral cones, in the sense that a polytope $P \subseteq N_{\mathbb{R}}$ gives a polyhedral cone $C(P) \subseteq N_{\mathbb{R}} \times \mathbb{R}$, called the cone of P . This cone is defined as

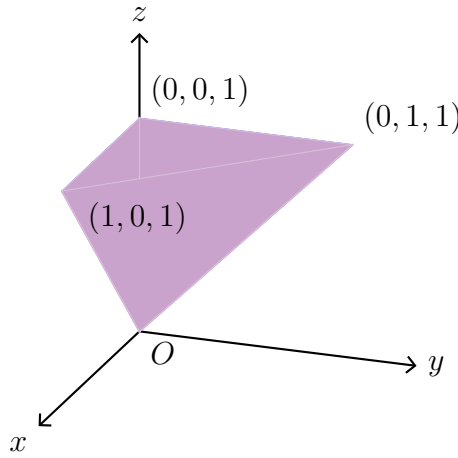
$$C(P) = \{ \lambda \cdot (v, 1) \in N_{\mathbb{R}} \times \mathbb{R} \mid v \in P, \lambda \geq 0 \}.$$

If the polytope has the description $P = \text{Conv}(S)$, then the cone $C(P)$ can be described as $C(P) = \text{Cone}(S \times \{1\})$.

Example 3.2. Consider the set $S = \{0, e_1, e_2\}$. This gives the standard 2-simplex $\Delta^2 = \text{Conv}(S)$



This gives the cone following the above description



However, we note that this cone is 3-dimensional, and in Section 2.2 we only considered 2-dimensional cones. $\diamond$

Example 3.2 shows that we need a different approach to relate cones and fans. The above is still valid, just not what we need. Earlier we discussed some of the theory coming from cones, and it is this we need to relate cones and polytopes. Consider the vector spaces $N_{\mathbb{R}}$ and $M_{\mathbb{R}}$ associated with the lattices N, M respectively. A *lattice polytope* $P \subseteq M_{\mathbb{R}}$ is the convex hull of a finite set $S \subseteq M$. We would like a lattice polytope to have “enough” lattice points, in the following sense.

Definition 3.3. A lattice polytope $P \subseteq M_{\mathbb{R}}$ is *very ample* if for every vertex $m \in P$, the semigroup $S_{P,m} = \mathbb{N}(P \cap M - m)$ generated by the set $P \cap M - m = \{m' - m \mid m' \in P \cap M\}$ is saturated in M .

A semigroup is *saturated* in the following sense,

Definition 3.4. An affine semigroup $S \subseteq M$ is *saturated* if for all $k \in \mathbb{N} \setminus \{0\}$ and $m \in M, km \in S$ implies that $m \in S$.

Let $P \subseteq M_{\mathbb{R}}$ be a very ample polytope of full dimension. If $P \cap M = \{m_1, \dots, m_s\}$, then $X_{P \cap M}$ is the Zariski closure of the image of the map $T_N \rightarrow \mathbb{P}^{s-1}$, given by

$$t \mapsto (\chi^{m_1}(t), \dots, \chi^{m_s}(t)) \in \mathbb{P}^{s-1}.$$

The above then gives the relation to cones, through the following theorem.

Theorem 3.5. *Let $X_{P \cap M}$ be the projective variety of the very ample polytope $P \subseteq M_{\mathbb{R}}$. Assume that P is full dimensional with $\dim(P) = n$. Then:*

(i) *For each vertex $m_i \in P \cap M$, the affine piece $X_{P \cap M} \cap U_i$ is the affine toric variety*

$$X_{P \cap M} \cap U_i = U_{\sigma_i} = \text{Spec}(\mathbb{C}[\sigma_i^{\vee} \cap M]),$$

where $\sigma_i \subseteq N_{\mathbb{R}}$ is the strongly convex rational polyhedral cone dual to the cone $\text{Cone}(P \cap M - m_i) \subseteq M_{\mathbb{R}}$. Furthermore, $\dim(\sigma_i) = n$.

(ii) *The torus of $X_{P \cap M}$ has character lattice M , and hence is the torus T_N .*

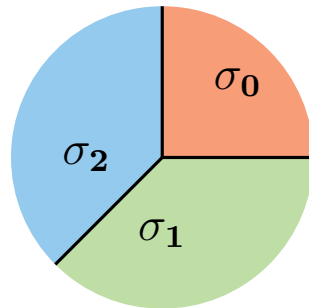
Example 3.6. Consider again the set $S = \{0, e_1, e_2\}$, which gives the standard 2-simplex $\Delta^2 = \text{Conv}(S)$. Clearly S is saturated, since it consists of the basis elements of $M \simeq \mathbb{Z}^2$. Similarly it follows that Δ^2 is very ample, and $\dim(\Delta^2) = 2$. Consider $0 \in \Delta^2 \cap \mathbb{Z}^2$. Following Theorem 3.5, we consider the set $\Delta^2 \cap \mathbb{Z}^2 - 0 = \{e_1, e_2\}$. Which yields the cone

$$\text{Cone}(e_1, e_2) = \sigma_0^{\vee}.$$

If we compare this to the Ongoing Example 2.8, we have obtained the dual cone to $\sigma_0 = \text{Cone}(e_1, e_2)$ of the fan of $\mathbb{P}^2$. We continue by considering the remaining points in S . This yields $\Delta^2 \cap \mathbb{Z}^2 - e_1 = \{-e_1, e_2 - e_1\}$ and $\Delta^2 \cap \mathbb{Z}^2 - e_2 = \{-e_2, e_1 - e_2\}$, hence we have the cones

$$\text{Cone}(-e_1, e_2 - e_1) = \sigma_2^{\vee} \quad \text{and} \quad \text{Cone}(-e_2, e_1 - e_2) = \sigma_1^{\vee}.$$

We note again these are actually the dual cones of the Ongoing Example 2.8. To finish describing the cones, we must take the dual of the above cones. By Proposition 2.9 we obtain the cones from Example 2.6, i.e. the cones



We could continue following Theorem 3.5, and construct the toric variety $\mathbb{P}^2$. That is, however, what we have done throughout the ongoing example of Chapter 2. $\diamond$

Thus we have a relation between polytopes and cones, and hence a relation to affine toric varieties. All that is missing is the link to Kähler geometry.

3.2 Toric Kähler Manifolds

This section heavily refers to (Silva 2006). We start by defining a *Kähler manifold*.

Definition 3.7 (Kähler Manifold). *A Kähler manifold is a symplectic manifold (M, ω) equipped with an integrable compatible almost complex structure. The symplectic form ω is then called a Kähler form. Equivalently a Kähler form ω is a ∂ - and $\bar{\partial}$ -closed form of type $(1, 1)$ which, on a local chart $(\mathcal{U}, z_1, \dots, z_n)$, is given by $\omega = \frac{i}{2} \sum_{j,k=1}^n h_{j\bar{k}} dz_j \wedge d\bar{z}_k$, where, at each $x \in \mathcal{U}$, $(h_{j\bar{k}}(x))$ is a positive definite hermitian matrix.*

By this definition, we do not immediately see a connection to polytopes or cones. That is because we need to consider a *symplectic manifold*, specifically a *symplectic toric manifold*.

Definition 3.8. *A symplectic toric manifold is a compact connected symplectic manifold (M, ω) equipped with an effective hamiltonian action of a torus T of dimension*

$$\dim(T) = \frac{1}{2} \dim(M),$$

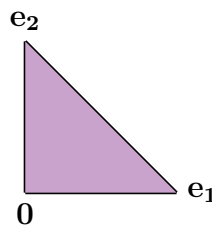
and with a choice of a corresponding moment map μ .

We finally recognize some parts, but again how does this relate to polytopes? In order to decipher this relation, we must consider a *Delzant polytope*, defined as such.

Definition 3.9 (Delzant Polytope). *A Delzant polytope D_Δ in $\mathbb{R}^n$ is a convex polytope satisfying:*

- (i) *It is simple, i.e. there are n edges meeting at each vertex.*
- (ii) *It is rational, i.e. the edges meeting at the vertex p are rational in the sense that each edge is of the form $p + tu_i, t \geq 0$ where $u_i \in \mathbb{Z}^n$.*
- (iii) *It is smooth, i.e. for each vertex, the corresponding $u_1, \dots, u_n$ can be chosen to be a $\mathbb{Z}$ -basis of $\mathbb{Z}^n$.*

Example 3.10. Recall the polytope Δ^2



Clearly Δ^2 is simple. Furthermore, it is also rational. For clarity we show it for one of the vertices. Consider the vertex $e_1 \in \Delta^2$. It is a part of two edges, e_1 to 0 and e_1 to e_2 .

For e_1 to 0, the edge is of the form $e_1 + t(-e_1)$, $t \geq 0$, similarly the edge from e_1 to e_2 is of the form $e_1 + t(e_1 - e_2)$, $t \geq 0$. For the other vertices, the corresponding edges can be described in a similar way. Lastly we see that $\{-e_1, e_1 - e_2\}$ constitutes a $\mathbb{Z}$ -basis of $\mathbb{Z}^2$, the remaining vertices are as easy to show. Hence Δ^2 is smooth, and therefore a Delzant polytope. $\diamond$

We are getting closer, we are just missing the final link.

Theorem 3.11 (Delzant Theorem). *Toric manifolds are classified by Delzant polytopes. More specifically we have the one-to-one correspondence*

$$\begin{aligned} \{\text{toric manifolds}\} &\xrightarrow{1-1} \{\text{Delzant polytopes}\} \\ (M^{2n}, \omega, (\mathbb{C}^*)^n, \mu) &\longmapsto \mu(M) \end{aligned}$$

Example 3.12. Consider the action of $(\mathbb{C}^*)^2$ on $\mathbb{P}^2$, given by

$$(e^{i\theta_1}, e^{i\theta_2}) \cdot [z_0 : z_1 : z_2],$$

with the corresponding moment map

$$\mu[z_0 : z_1 : z_2] = \left(\frac{|z_1|^2}{|z_0|^2 + |z_1|^2 + |z_2|^2}, \frac{|z_2|^2}{|z_0|^2 + |z_1|^2 + |z_2|^2} \right).$$

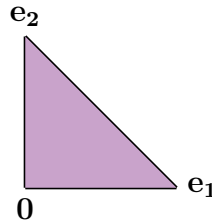
By theorem 27.1 (Silva 2006), the image of μ is the convex hull of the fixed points of the action. As discussed in earlier examples, the fixed points of the torus action are $[1 : 0 : 0]$, $[0 : 1 : 0]$ and $[0 : 0 : 1]$. We then compute

$$\mu[1 : 0 : 0] = (0, 0),$$

$$\mu[0 : 1 : 0] = (1, 0),$$

$$\mu[0 : 0 : 1] = (0, 1).$$

This gives us the polytope we have seen before.



$\diamond$

Hence we now have a guideline for how the theories link together. However, this is merely a rough sketch of the more basic ideas behind the linking. For a more detailed explanation we refer to (Abreu 2003).

The goal of describing the aforementioned theory, is for us to be able to study various stability conditions for toric varieties. We only consider the case of toric varieties coming from a 2-dimensional fan, as Theorem 2.57 gives a rather simple way of computing intersection numbers. Before discussing the stability conditions, we introduce other needed notions, such as the *Kähler cone*.

3.3 Positive Cones, Energy Functionals, and Numerical Invariants

Recall the nef cone $\text{Nef}(X_\Sigma)$, Definition 2.55, is the cone in $N^1(X_\Sigma) = \text{Pic}(X_\Sigma)_\mathbb{R}$ generated by classes of nef Cartier divisors. By Definition 2.50 and Theorem 2.51 we have the following description of $\text{Nef}(X_\Sigma)$. Let $D_{\Sigma(1)}$ be the set of torus-invariant divisors corresponding to the rays $\tau_i \in \Sigma(1)$, then,

$$\text{Nef}(X_\Sigma) = \{D \in \text{Pic}(X_\Sigma) \mid D \cdot D_i \geq 0, D_i \in D_{\Sigma(1)}\}.$$

Closely related we have the Kähler cone.

Definition 3.13. *The Kähler cone $\mathcal{K}(X_\Sigma)$ of the toric variety X_Σ is the interior of $\text{Nef}(X_\Sigma)$, i.e.*

$$\mathcal{K}(X_\Sigma) = \{D \in \text{Pic}(X_\Sigma) \mid D \cdot D_i > 0, D_i \in D_{\Sigma(1)}\}.$$

Example 3.14. Recall from Example 2.49 that the Picard group of $\mathbb{P}^2$ has the generator $[D_1] = [D_2] = [D_3]$. Therefore we will write $H := [D_1]$, which gives the description

$$\text{Pic}(X_\Sigma) \simeq \{aH \mid a \in \mathbb{Z}\} \simeq \mathbb{Z} \cdot H.$$

Hence any divisor D can be written as $D = aH$ for some $a \in \mathbb{Z}$. To describe the nef cone, and in turn the Kähler cone, we need to find the classes of nef Cartier divisors. To this end consider

$$D \cdot C = (aH) \cdot (mH) = amH^2 = am \geq 0 \implies a \geq 0.$$

Recalling that $H^2 = [D_1]^2 = 1$. We can then describe the Nef cone as $\text{Nef}(X_\Sigma) = \mathbb{R}_{\geq 0}H$, which one can visualise as,

$$0 \longrightarrow$$

i.e. the closed half-line from 0 to infinity. The Kähler cone $\mathcal{K}(X_\Sigma)$ is then the open half-line from 0 to infinity.

Now we wish to describe the nef and Kähler cone of the blow-up of $\mathbb{P}^2$ at $[1 : 0 : 0]$. Recall the relabeling done in Example 2.58, which we will use to describe the Picard group. By Theorem 2.48 $\text{Pic}(X_{\Sigma^*(\sigma_0)})$ is generated by classes of the divisors D_0, D_1, D_2, D_3 , corresponding to $\tau_0, \tau_1, \tau_2, \tau_3 \in \Sigma^*(\sigma_0)(1)$, subject to the relations

$$\begin{aligned} 0 \sim \text{div}(\chi^{e_1}) &= \sum_{i=0}^3 \langle e_1, v_i \rangle = D_1 + D_2 - D_3, \\ 0 \sim \text{div}(\chi^{e_2}) &= \sum_{i=0}^3 \langle e_2, v_i \rangle = D_0 + D_1 - D_3. \end{aligned}$$

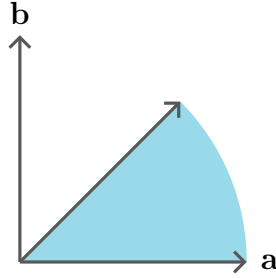
Thus the Picard group has generators $[D_0] = [D_2]$ and $[D_1] = [-D_0 + D_3]$. We recall from Example 2.58 the intersection matrix,

	D_0	D_1	D_2	D_3
D_0	0	1	0	1
D_1	1	-1	1	0
D_2	0	1	0	1
D_3	1	0	1	1

Hence we will write $H := [D_0 + D_1]$ and $E := [D_1]$. As before any divisor D can be written as $D = aH - bE$ for some $a, b \in \mathbb{Z}$. Then we have

$$\begin{aligned} D.H &= (aH - bE).H = aH^2 - bH.E = a \geq 0, \\ D.E &= (aH - bE).E = aH.E - bE^2 = b \geq 0, \\ D.(H - E) &= (aH - bE).(H - E) = aH^2 - aH.E - bE.H + bE^2 = a - b \geq 0. \end{aligned}$$

We can then visualise the nef cone as



hence the Kähler cone is the open lower half of the first quadrant. $\diamond$

In the following subsection we will introduce the theory leading towards our study of the energy part of the *Mabuchi functional*, herein the cscK equation, test configurations, K -stability and more.

The theory introduced in the following subsection are outside the scope of this thesis, but will be included for completeness of the covered theory.

Energy Functionals

In order to study the existence of cscK metrics, we recall that canonical metrics, particularly cscK metrics, can be characterized as critical points of certain energy functionals on the space of Kähler potentials $\mathcal{H}$. Consider a smooth real-valued function ρ on a complex manifold X , ρ is called strictly plurisubharmonic if the real closed $(1, 1)$ -form

$$\omega = \frac{i}{2} \partial \bar{\partial} \rho$$

is positive, i.e. a Kähler form. The function ρ is called a *Kähler potential*. We note by the local $\partial\bar{\partial}$ -lemma, every Kähler metric can locally be described in this manner.

Furthermore, by the $\partial\bar{\partial}$ -lemma we may write

$$\mathcal{H}(X, \omega) := \{\rho \in C^\infty(X) \mid \omega_\rho := \omega + i\partial\bar{\partial}\rho > 0\}$$

for the space of smooth Kähler potentials in $\alpha = [\omega]$. Consider any path $(\rho_t)_{t \geq 0}$ in $\mathcal{H}$. The *Mabuchi functional* (or *K-energy functional*) $M: \mathcal{H} \rightarrow \mathbb{R}$, first introduced in (Mabuchi 1985), is defined by its Euler-Lagrange equation

$$\frac{d}{dt}M(\varphi_t) = -V^{-1} \int_X \dot{\rho}_t (\mathcal{S}(\omega_{\rho_t}) - \bar{\mathcal{S}}) \omega_{\rho_t}^n,$$

where $\mathcal{S}(\omega_{\rho_t})$ is the scalar curvature, which for a Kähler form ω is given by

$$\mathcal{S}(\omega) = \text{Tr}_\omega \text{Ric}(\omega) = n \frac{\text{Ric}(\omega) \wedge \omega^{n-1}}{\omega^n},$$

and $\bar{\mathcal{S}}$ is the mean scalar curvature, which for the Kähler class $[\omega]$ is given by

$$\bar{\mathcal{S}} = V^{-1} \int_X \mathcal{S}(\omega) \omega^n = n \frac{\int_X c_1(X) \cdot [\omega]^{n-1}}{\int_X [\omega]^n}.$$

An *extremal metric* is then defined as

Definition 3.15. A Kähler metric ω is extremal if

$$\bar{\partial} \nabla^{1,0} \mathcal{S}(\omega) = 0$$

Where we are interested in the subclass of extremal Kähler metrics of constant scalar curvature.

Definition 3.16. We say that ω is a constant scalar curvature Kähler (cscK) metric iff the Kähler form ω satisfies the equation

$$\mathcal{S}(\omega) = \bar{\mathcal{S}},$$

referred to as the *cscK equation*.

This defines a fourth-order fully nonlinear elliptic partial differential equation, which solvability depends on global geometric data. Hence with such complexity, algebro-geometric notions of *stability* has been introduced for the cscK equation. Stability conditions link together with the cscK equation through the following guiding conjecture, due to Yau-Tian-Donaldson.

Conjecture 3.17. (Yau-Tian-Donaldson) A polarized manifold (X, L) admits a constant scalar curvature Kähler (cscK) metric if and only if it is *K-stable*.

In order to study stability criteria, one considers *test configurations* $(\mathcal{X}, \Omega)$ of a compact complex manifold X . A test configuration is an algebro-geometric object encoding

$\mathbb{C}^*$ -equivariant degenerations of the initial manifold. The definition of a test configuration vary depending on the setting of stability. In the setting we are considering, we denote the space of all relatively Kähler test configurations for $(X, [\omega])$ by $\mathcal{H}^{\text{NA}}(X, [\omega])$. Then given a curve $C \subset X$, one can naturally associate a test configuration $\Phi_{C,\kappa}$ to it via the *deformation to the normal cone* construction introduced by Ross-Thomas (Ross and Thomas 2005). $\Phi_{C,\kappa}$ is relatively Kähler precisely when $\kappa \in (0, \bar{\kappa}_C)$, where

$$\bar{\kappa}_C = \sup \{r > 0 \mid \pi^* \alpha - r[E] > 0\}$$

is the Seshadri constant. The functional we wish to consider, is then the non-archimedian functional, which take simple polynomial expressions along $\Phi_{C,\kappa}$, varying $\kappa \in (0, \bar{\kappa}_C)$, via the well-known lemma (Lejmi and Székelyhidi 2013).

Lemma 3.18. *Suppose X is a smooth projective surface. For slope test configurations $\Phi_{C,\kappa} \in \mathcal{H}^{\text{NA}}(X, [\omega])$ we have,*

$$\begin{aligned} \mathbf{E}_{\theta,\omega}^{\text{NA}}(\Phi_{C,\kappa}) &= A_1 \kappa^2 + B_1 \kappa^3, \\ ||\Phi_{C,\kappa}|| &= A_2 \kappa^2 + B_2 \kappa^3, \end{aligned}$$

where A_1, A_2, B_1, B_2 are real numbers satisfying

$$A_1 = c_{\theta,\omega} \int_C \omega - \int_C \theta, \quad B_1 = - \left(\frac{2[\theta] \cdot [\omega]}{3[\omega]^2} \right) C^2, \quad A_2 = \int_C \omega > 0, \quad B_2 = -\frac{2}{3} C^2.$$

We say that $(X, [\omega])$ is J-semistable (resp. J-stable) if

$$\mathbf{E}_{\theta,\omega}^{\text{NA}}(\Phi) \geq 0,$$

(resp. > 0) for all normal and relatively Kähler test configurations Φ for $(X, [\omega])$.

Consider the leading term of the polynomial expression of $\mathbf{E}_{\theta,\omega}^{\text{NA}}(\Phi_{C,\kappa})$,

$$\begin{aligned} A_1 &= c_{\theta,\omega} \int_C \omega - \int_C \theta \\ &= 2 \frac{\int_X \chi \wedge \omega}{\int_X \omega^2} \cdot \int_C \omega - \int_C \chi \\ &= [\omega] \cdot C \left(2 \frac{[\chi] \cdot [\omega]}{[\omega]^2} - \frac{[\chi] \cdot C}{[\omega] \cdot C} \right) \end{aligned}$$

Since $[\omega]$ is Kähler, $[\omega] \cdot C > 0$, hence in the surface case the condition becomes, after a normalization, $(2[\omega] - [\chi]) \cdot C > 0$ for all C . This is the Nakai-Moishezon criteria for the class $2[\omega] - [\chi]$ to be Kähler. Let $\alpha := [\omega]$ and $\beta := [\chi]$, through the reduction to finite number of test configurations described in (Khalid and Dyrefelt 2024), we will consider the numerical criteria,

$$J(\alpha, \beta) = \min_{D_i} \left(2 \frac{\alpha \cdot \beta}{\alpha^2} - \frac{\beta \cdot D_i}{\alpha \cdot D_i} \right).$$

Specifically we are interested in the “special” case,

$$\begin{aligned} I(\alpha) &:= J(\alpha, K) = \min_{D_i} \left(2 \frac{\alpha \cdot K}{\alpha^2} - \frac{K \cdot D_i}{\alpha \cdot D_i} \right) \\ &= - \max_{D_i} \left(2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} \right). \end{aligned}$$

For $J(\alpha, \beta)$ we require that α and β are Kähler, where there is a known connection between the sign of $J(\alpha, \beta)$ and coercivity of $\mathbf{E}_{\theta, \omega}^{\text{NA}}(\Phi_{C, \kappa})$ (X. Chen 2000). For $I(\alpha)$, we have to be more careful, as after a number of blow-ups of a fan Σ , the anticanonical divisor $-K$ might not be Kähler. This in fact does not require many blow-ups, as will be seen in Section 5.2.

$I(\alpha)$ is then of interest through the following proposition.

Proposition 3.19. *Let X_Σ be a smooth complex projective surface, then*

$$I(\alpha) < 0 \implies \mathbf{E}_{\theta, \omega}^{\text{NA}}(\Phi_{C, \kappa}) \text{ is not coercive.}$$

Proof. Take a curve $C \subset X_\Sigma$, (omitting details) one can associate a test configuration $\Phi_{C, \kappa}$ to it via the deformation to the normal cone construction described by Ross–Thomas (Ross and Thomas 2005). Then for all $\kappa > 0$ small enough, we have an associated relatively ample line bundle $\Phi_{C, \kappa}$, so that $(\mathcal{X}, \Phi_{C, \kappa})$ forms a test configuration.

For those test configurations, after a normalization and by Lemma 3.18, the non-archimedean J -functional satisfies the polynomial expressions in κ as follows,

$$\begin{aligned} \mathbf{E}_{\theta, \omega}^{\text{NA}}(\Phi_{C, \kappa}) &= I(\alpha)\kappa^2 - \frac{2}{3}C^2\kappa^3 \\ &= \kappa^2 \left(I(\alpha) - \frac{2}{3}C^2\kappa \right), \end{aligned}$$

for $\kappa \in (0, \bar{\kappa}_C)$ (Lejmi and Székelyhidi 2013). Hence if $I(\alpha) < 0$, then for small enough $\kappa \in (0, \bar{\kappa}_C)$, we have $\mathbf{E}_{\theta, \omega}^{\text{NA}}(\Phi_{C, \kappa}) < 0$. Which in turn implies that $(\mathcal{X}, \Phi_{C, \kappa})$ is J-unstable (not J-semistable).

To imply non-coercivity, it is well-known that the destabilizing test configuration has an associated geodesic ray (Phong, Ross, and Sturm 2008), see (Dyrefelt 2017) for the setting we are using. Hence $\mathbf{E}_{\theta, \omega}$ will have negative asymptotic slope along this specific geodesic ray associated to the minimizing C ; thus it cannot be coercive. $\square$

We are interested in $J(\alpha, \beta)$, as $\mathbf{E}_\omega^{-\text{Ric}(\omega)}$ is the energy part of the Mabuchi functional, seen by the Chen-Tian decomposition,

$$M_\omega = \mathbf{E}_\omega^{-\text{Ric}(\omega)} + H_\omega.$$

In the following chapter we will discuss a program developed to compute the quantity on smooth complete toric surfaces.

4 A Computational Framework Based on SageMath

In this project we will use SageMath 10.7, to perform computations on divisors in the Kähler cone (The Sage Developers 2025). SageMath utilizes the programming language Python.

4.1 Setup and Construction of Varieties

As discussed in Chapter 2, we can construct affine toric varieties from the notions of cones and fans. This can also be done in SageMath, following the same procedure. For example, to define the fan of $\mathbb{P}^2$ with lattice $N \simeq \mathbb{Z}^2$, consisting of the cones $\sigma_0 = \text{Cone}(e_1, e_2)$, $\sigma_1 = \text{Cone}(e_1, -e_1 - e_2)$ and $\sigma_2 = \text{Cone}(-e_1 - e_2, e_2)$, where e_1, e_2 is the canonical basis of N , we can write the following.

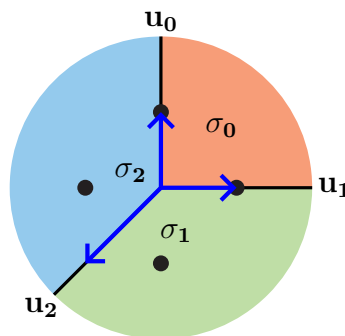
```
P2rays = [(0,1), (1,0), (-1,-1)]
P2cones = [Cone([P2rays[0], P2rays[1]]),
           Cone([P2rays[1], P2rays[2]]),
           Cone([P2rays[2], P2rays[0]])]
P2fan = Fan(cones=P2cones, rays=P2rays)
P2fan
```

Which gives the output.

Rational polyhedral fan in 2-d lattice N

For visualisation and confirmation that it is indeed the correct fan, SageMath can plot the fan.

```
P2fan.plot()
```



To define the toric variety corresponding to the fan, we will call the following.

```
P2X = ToricVariety(P2fan)
P2X
```

2-d toric variety covered by 3 affine patches

In Chapter 2 and Chapter 3 we defined mathematical objects which are of interest, the Picard group Definition 2.45 and the Kähler cone Definition 3.13. In SageMath we can obtain the objects by calling

```
P2pic = P2X.rational_class_group()
P2pic
```

The toric rational divisor class group of a 2-d toric variety covered by 3 affine patches

```
P2Kc = P2X.Kaehler_cone()
P2Kc
```

1-d cone in 1-d lattice

In order to perform computations on the divisors, we further need the cohomology ring of the toric variety. This is done by the following

```
P2coho = P2X.cohomology_ring()
```

Rational cohomology ring of a 2-d toric variety covered by 3 affine patches

Throughout our examples we have used the procedure introduced in Theorem 2.48 to describe the generators of the Picard group. SageMath uses a slightly different approach to describe the generators. The exact sequence

$$0 \longrightarrow M \longrightarrow \operatorname{Div}_{T_N}(X_\Sigma) \longrightarrow \operatorname{Cl}(X_\Sigma) \longrightarrow 0,$$

can be written as

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{A} \mathbb{Z}^r \xrightarrow{B} \mathbb{Z}^{r-2} \longrightarrow 0,$$

where r is the number of rays of Σ , i.e. $\tau \in \Sigma(1)$. As in our example in Chapter 2, the rays are ordered from $(0,1)$ and in clockwise orientation. The matrix A is then a $2 \times r$ matrix whose i 'th column consists of the coefficients of $\tau_i \in \Sigma(1)$. The matrix B is then the *Gale dual* of A , obtained by computing the kernel of A , which is done by solving $A\mathbf{x} = 0$. Consider $\mathbb{P}^2$, as constructed earlier, then

$$A = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & -1 \end{pmatrix}.$$

The reduced row echelon form is then

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}.$$

Next step is to solve the matrix equation

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Taking $x_2 = t$, implies $x_0 = x_1 = t$ and

$$B = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

from whose columns we obtain the relation of the divisors, i.e. $[D_0] = [D_1] = [D_2]$.

Consider a more elaborate example, e.g. the blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along the point $([1 : 0], [1 : 0])$, discussed in the end of Section A.1, with defining vectors

$$v_0 = (0, 1), \quad v_1 = (1, 1), \quad v_2 = (1, 0), \quad v_3 = (0, -1), \quad v_4 = (-1, 0).$$

The matrix A is then

$$\begin{pmatrix} 0 & 1 & 1 & 0 & -1 \\ 1 & 1 & 0 & -1 & 0 \end{pmatrix},$$

with reduced row echelon form

$$\begin{pmatrix} 1 & 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & 0 & -1 \end{pmatrix}.$$

For which we solve the matrix equation

$$\begin{pmatrix} 1 & 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

If we take $x_2 = t, x_3 = s, x_4 = u$, then $x_0 = s + t - u$ and $x_1 = -t + u$. The Gale dual of A is then

$$\begin{pmatrix} 1 & -1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 & 1 \end{pmatrix},$$

from whose columns we obtain the relation of the divisors. In SageMath we define it as

```

rays = [(0,1), (1,1), (1,0), (0,-1), (-1,0)]
cones = [Cone([rays[0], rays[1]]),
          Cone([rays[1], rays[2]]),
          Cone([rays[2], rays[3]]),
          Cone([rays[3], rays[4]]),
          Cone([rays[4], rays[0]])]
fan = Fan(cones=cones, rays=rays)
number_of_rays = len(fan.rays())
X = ToricVariety(fan)
coho = X.cohomology_ring()
pic = X.rational_class_group()
for i in range(number_of_rays):
    print(coho(X.divisor(i)))
print()
for picbasis in pic.gens():
    print(coho(picbasis.lift()))

```

Which yields the output

```

[z2 + z3 - z4]
[-z2 + z4]
[z2]
[z3]
[z4]

```

```

[z2 + z3 - z4]
[-z2 + z4]
[z2]

```

where we can see the relations of the divisors in the cohomology ring. As we will see in later sections, this information is needed in order to compute the intersection number of divisors. However SageMath can also compute the intersection number of divisors, by calling the function 'integrate' on the toric variety. For example as used in the following, to compute the intersection matrix of $\mathbb{P}^2$.

```

import numpy as np

def compute_intersection_matrix(number_of_rays, X):

    intersection_matrix = np.zeros((number_of_rays, number_of_rays))
    for i in range(number_of_rays):
        for j in range(number_of_rays):
            i_coho = coho(X.divisor(i))
            j_coho = coho(X.divisor(j))
            intersection_number = X.integrate(i_coho * j_coho)
            intersection_matrix[i][j] = intersection_number

    return intersection_matrix

```

```
number_of_rays = len(fan.rays())
compute_intersection_matrix(number_of_rays, P2X)
```

Which yields the output

```
array([[1., 1., 1.],
       [1., 1., 1.],
       [1., 1., 1.]])
```

Comparing this to the intersection matrix of Ongoing Example 2.58, we see that the computations agree. However, in the end we aspire to compute variants of the J-functional, which require us to compute $2 + 2n$ intersection numbers, where n is the number of rays of the fan. Furthermore, we wish to compute variants of numerical criterias appearing in K -stability for the many elements of the Kähler cone, aiming for matrices consisting of 10^7 entries. For this many computations, calling the “integrate” function from SageMath becomes rather slow. We therefore introduce another procedure to compute intersection numbers.

This is all the information we need from SageMath in order to perform our computations.

4.2 Convert Information Functions

As seen in the previous section, we require information from SageMath for the given toric variety in order to compute. We will consider divisors in the Picard group in their cohomology class, hence the divisors are defined as a linear combination of the generators of the Picard group. Consider the example of the blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along the point $([1 : 0], [1 : 0])$ from the previous section. We obtained the generators of the Picard group’s cohomology classes to be $g_0 = [z_2 + z_3 - z_4]$, $g_1 = [-z_2 + z_4]$, $g_2 = [z_2]$. A divisor will then be of the form $ag_0 + bg_1 + cg_2$ for $a, b, c \in \mathbb{N}$. However we cannot use the output directly from SageMath, we need to convert it into data that we can use. In order to convert the data we decided to count the occurrences of the rays in each generator, i.e. g_0 would result in $z_2 : 1$, $z_3 : 1$ and $z_4 : -1$. This is done by converting the result from SageMath to a string and then extracting the information using regular expressions. In Python this is simply done by importing the package `re`.

```
import re

def extract_coefficients(s):

    terms = re.findall(r'([+-]?[d]*[*]?z\d+|([+-]?z\d+))', s)

    result = {}
    for term in terms:
        match = re.match(r'([+-]?[d]*)([*]?)z(\d+)', term)
        if match:
            coeff_str, _, z_num = match.groups()
```

```

    if coeff_str in ('', '+'):
        coeff = 1
    elif coeff_str == '-':
        coeff = -1
    else:
        coeff = int(coeff_str)
    result[f'z{z_num}'] = coeff

    return result

```

This function starts by using regular expressions to find all occurrences on the form $\pm j * z_i$, where j is a coefficient and i corresponds to the index of the ray associated to the divisor. Hereafter create a dictionary by splitting the string into its three important parts, the coefficient ' j ', '*' if the coefficient is greater than 1, and lastly the index of ray ' i '. Calling the above function on the generators g_0, g_1 and g_2 respectively yields the following output.

```

{'z2': 1, 'z3': 1, 'z4': -1}
{'z2': -1, 'z4': 1}
{'z2': 1}

```

To end the discussion of the generators of the Picard group, we have the following function.

```

def generate_pic_gen_lists(picard_group, fan, coho):

    gen_list = []
    string_gen_list = []
    for pic_basis in picard_group.basis():
        gen_list.append(pic_basis)
        pic_coho = coho(pic_basis.lift())
        temp = str(pic_coho)[1:-1]
        temp = temp.replace(" ", "")
        string_gen_list.append(temp)

    gen_list_count = [[0 for _ in range(2 * len(fan.rays()))] for _ in
                      range(len(fan.rays()) - 2)]

    coeffs = []

    for i in range(len(string_gen_list)):
        coeffs.append(extract_coefficients(string_gen_list[i]))

    for j in range(len(coeffs)):
        for k in range(len(fan.rays())):
            try:
                if coeffs[j][f"z{k}"] < 0:
                    gen_list_count[j][k + len(fan.rays())] = coeffs[j][f"z{k}"]
            else:
                gen_list_count[j][k] = coeffs[j][f"z{k}"]
        except:
            continue
    return gen_list, gen_list_count

```

This function computes a list of the divisor classes coming from SageMath, `gen_list`. This will only be used to check whether a divisor is represented by the generators or represented as a torus-invariant divisor. The `gen_list_count` is a list of the coefficient of the generators. For example the generators g_0, g_1, g_2 yields the list

```
[0, 0, 1, 1, 0, 0, 0, 0, 0, -1], [0, 0, 0, 0, 1, 0, 0, 0, -1, 0, 0],
[0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]]
```

Where the i 'th entry of the list is the generator g_i , which is split into two parts. Let $n = 5$ be the number of rays, the first n entries count the positive coefficients and the last n counts the negative coefficients. The j 'th entry corresponds to the j 'th torus-invariant divisor, similar the $(n + j)$ 'th entry corresponds to the j 'th invariant divisor.

This information about the generators of the Picard group is then used in the following function to convert our description of the divisors of the form $ag_0 + bg_1 + cg_2$ to its description in the cohomology class.

```
def convert_to_coho(coefficients, gen_list_count, fan):

    coho_divisor = np.zeros(len(fan.rays()))
    for j in range(len(gen_list_count)):
        for i in range(len(fan.rays())):
            coho_divisor[i] += gen_list_count[j][i] * coefficients[j]
            coho_divisor[i] += gen_list_count[j][i + len(fan.rays())] *
                                coefficients[j]

    return coho_divisor
```

The divisor $ag_0 + bg_1 + cg_2$ would then be represented by an array with entries $[0, 0, a - b + c, a, -a + b]$. Consider, as an example, the following,

```
test = np.array([0.5, 0.75, 1])
convert_to_coho(test, gen_list_count, fan)
```

```
array([0. , 0. , 0.75, 0.5 , 0.25])
```

As the variants of the J-functional we are computing require us to find a minimum or maximum ranging over the torus-invariant divisors, we will further need a description of these divisors.

```
def get_coho_C_divisors(number_of_rays):

    C_div_list = [[0 for _ in range(number_of_rays)] for _ in range(
                                                number_of_rays)]

    for i in range(len(C_div_list)):
        C_div_list[i][i] = 1

    return C_div_list
```

This function output is a list of lists, where the i 'th entry of the outer list has a 1 in the inner lists i 'th entry. The output for our ongoing example would be

$[[1, 0, 0], [0, 1, 0], [0, 0, 1]]$

4.3 Computational Functions

Now we have all of the needed information, and converted it to data that we can use. The last part of the program is the actual computation and storing the computed data. As described in Chapter 3 we wish to compute variants of numerical solvability criteria for the J -equation. Since the computations rely heavily on intersection numbers of divisors, we could use the built-in function in SageMath 'integrate'. As discussed earlier this becomes rather slow for our goal of computing towards 10^7 entries. To decrease the computation time we will introduce a different approach of computing the intersection numbers of divisors. We will use the package **numpy** and its matrix multiplication function 'dot'.

```
def intersection_calculator(v1_coeff, v2_coeff, intersection_matrix, fan
                             , gen_list_count, gen_list):

    v1_vec = convert_to_coho(v1_coeff, gen_list_count, fan)
    if len(v2_coeff) == len(gen_list):
        v2_vec = np.array(convert_to_coho(v2_coeff, gen_list_count, fan))
    else:
        v2_vec = np.array(v2_coeff)

    v2_transpose = np.transpose(v2_vec)
    intermediate_result = np.dot(v1_vec, intersection_matrix)

    return np.dot(intermediate_result, v2_transpose)
```

This function takes two divisors, along with the intersection matrix as input. In our case the first divisor will always be a divisor in the Picard group, hence we need to convert it to its cohomology representation. The second divisor can be either a torus-invariant divisor or a divisor in the Picard group, so we need to check which case it is. If the length of the representation list is equal to the number of generators of the Picard group, we know it is described as a divisor in the Picard group.

Consider again the blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along the point $([1 : 0], [1 : 0])$, and two divisors $\gamma_0 = 0.5g_0 + 1g_1 + 0.7g_2$ and $\gamma_1 = 0.6g_0 + 1g_1 + 0.5g_2$. First we convert γ_0 to its cohomology representation

$$\tilde{\gamma}_0 = [0, 0, 0.2, 0.5, 0.5].$$

Since γ_1 is also described as a divisor in the Picard group, we convert it aswell,

$$\tilde{\gamma}_1 = [0, 0, 0.1, 0.6, 0.4].$$

Recall the intersection matrix

$$IM = \begin{array}{c|ccccc} & D_0 & D_1 & D_2 & D_3 & D_4 \\ \hline D_0 & -1 & 1 & 0 & 0 & 1 \\ D_1 & 1 & -1 & 1 & 0 & 0 \\ D_2 & 0 & 1 & -1 & 1 & 0 \\ D_3 & 0 & 0 & 1 & 0 & 1 \\ D_4 & 1 & 0 & 0 & 1 & 0 \end{array}$$

We can then calculate the intersection number of γ_0 and γ_1 as

$$\begin{aligned} \gamma_0 \cdot \gamma_1 &= \tilde{\gamma}_0 \cdot IM \cdot \tilde{\gamma}_1^T \\ &= \begin{pmatrix} 0 & 0 & 0.2 & 0.5 & 0.5 \end{pmatrix} \cdot \begin{pmatrix} -1 & 1 & 0 & 0 & 1 \\ 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 0.1 \\ 0.6 \\ 0.4 \end{pmatrix} \\ &= \begin{pmatrix} 0.5 & 0.2 & 0.3 & 0.7 & 0.5 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 0.1 \\ 0.6 \\ 0.4 \end{pmatrix} \\ &= 0.65. \end{aligned}$$

We are now ready to introduce the main computational functions, that computes the wished expression. For the energy part of the Mabuchi functional,

$$I(\alpha) = -\max_{D_i} \left(2 \cdot \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} \right),$$

we introduce two functions, one that computes the first factor, and one that computes the second factor. As we are considering the maximum ranging over the torus-invariant divisors, we note the first part is only computed once for each α input, while the second factor is computed n times, where n is the number of rays of the fan.

```
def first_factor(alpha, beta, intersection_matrix, fan, gen_list_count,
                 gen_list):

    numerator = intersection_calculator(tuple(alpha), tuple(beta),
                                         intersection_matrix, fan,
                                         gen_list_count, gen_list)
    denominator = intersection_calculator(tuple(alpha), tuple(alpha),
                                           intersection_matrix, fan,
                                           gen_list_count, gen_list)

    if denominator == 0:
        return 0
    else:
        result = 2 * (numerator / denominator)
```



```
return result
```

One observes that, since we are studying the energy part of the Mabuchi functional for the Kähler classes α lying in the Kähler cone, the self-intersection number α^2 is necessarily positive. However, in our computations we employ the Kähler cone defined by SageMath, which returns the closed Kähler cone. Consequently, along the boundary faces of the Kähler cone, it may occur that $\alpha^2 = 0$. In these cases we assign the value zero to the expression. This convention is justified, as it is straightforward to distinguish between the interior points of the Kähler cone and boundary points. The latter does not contribute to the geometric behaviour under consideration.

The second factor is computed similarly.

```
def second_factor(alpha, beta, C, intersection_matrix, fan,
                  gen_list_count, gen_list):

    numerator = intersection_calculator(tuple(beta), tuple(C),
                                         intersection_matrix, fan,
                                         gen_list_count, gen_list)
    denominator = intersection_calculator(tuple(alpha), tuple(C),
                                           intersection_matrix, fan,
                                           gen_list_count, gen_list)

    if denominator == 0:
        return None
    else:
        result = numerator / denominator

    return result
```

We can now compute one iteration of the energy part of the Mabuchi functional, however as mentioned we need to compute the maximum ranging over the torus-invariant divisors. This requires one more function, that iterates over the torus-invariant divisors, and compares their value of $I(\alpha)$.

```
def optim_heat_map_CSCK(alpha, beta, intersection_matrix, C_div_list,
                        fan, gen_list_count, gen_list):

    div_by_zero = []
    value = -float(50000)
    C_div = []
    alpha_beta_part = first_factor(alpha, beta, intersection_matrix, fan,
                                    gen_list_count, gen_list)

    for C in C_div_list:
        C_part = second_factor(alpha, beta, C, intersection_matrix, fan,
                               gen_list_count, gen_list)

        if C_part is None:
            div_by_zero.append(C)
        else:
            C_value = alpha_beta_part - C_part
```

```

    if C_value > value:
        value = C_value
        C_div = C

new_value = -value
return [new_value, C_div]

```

This is the principal computational function, which evaluates the two factors and compares the resulting values of the functional for each torus-invariant divisor. The initial value is set to -50.000 , a choice that is largely arbitrary and motivated solely by empirical observations during the development and testing of the program.

A limitation of this approach, is that it does not account for the possibility that two torus-invariant divisors may be optimizing under the same criteria. Nevertheless, a detailed chamber decomposition of the Kähler cone is not the primary objective of the present implementation. Rather, the purpose of the program is to provide visual and numerical data that highlight regions of interest, thereby guiding further theoretical investigation. The last computational function, simply utilizes the aforementioned functions, explicitly for divisors in the Kähler cone.

```

def process_alpha(alpha_indices, beta, intersection_matrix, Kc,
                  C_div_list, fan, gen_list_count,
                  gen_list, size_of_matrix):

    alpha = np.array(alpha_indices)
    if Kc.contains(tuple(alpha)):
        result = optim_heat_map_CSCK(alpha, beta, intersection_matrix,
                                      C_div_list, fan, gen_list_count,
                                      gen_list)

        value = result[0]
        C_div = convert_C_to_ray_divisor(result[1])
        indices = [fixed_value_to_index(alpha[k], size_of_matrix) for k in
                   range(len(alpha))]

        return (tuple(indices), value, C_div)
    else:
        indices = [fixed_value_to_index(alpha[k], size_of_matrix) for k in
                   range(len(alpha))]

        return (tuple(indices), 50000, None)

```

4.4 Data Storing and Helper Functions

Lastly we need to store the computed data in a usefull manner. We decided to store the data in numpy matrices, however we wished the program to be rather general, such that it does not depend on the dimension of the Kähler cone. In order to achieve this, we introduce helper functions that determines the sizes of the matrices, and creates a mesh-grid of arbitrary size.

```

def find_matrix_size(dim):

```

```
matrix_size = pow(10, 7 / dim)

return math.floor(matrix_size)
```

```
def numpy_cartesian_product(arrays):

    mesh = np.meshgrid(*arrays, indexing='ij')

    return np.column_stack([m.ravel() for m in mesh])
```

The above code is set to determine the size of the matrices, such that for any dimensional Kähler cone, we compute approximately 10^7 entries. Clearly computing more entries would result in a finer plot in the end, but here we are limited by storage space and computing time.

The following function brings the aforementioned functions together, stores the computed data in the matrices, and lastly saves the matrices.

```
def compute_data(beta, basename, output_dir, intersection_matrix, fan,
                 Kc, C_div_list, coho, gen_list_count,
                 gen_list, size_of_matrix):

    alpha_coeffs = [np.linspace(0, 1, size_of_matrix) for _ in range(len(
        fan.rays()) - 2)]

    shape = (size_of_matrix,) * (len(fan.rays()) - 2)
    sign_matrix = np.zeros(shape)
    div_matrix = np.empty(shape, dtype=object)

    for alpha_indices in numpy_cartesian_product(alpha_coeffs):
        indices, value, C_div = process_alpha(tuple(alpha_indices), beta,
                                                intersection_matrix, Kc, C_div_list,
                                                fan, gen_list_count, gen_list,
                                                size_of_matrix)

        sign_matrix[indices] = value
        div_matrix[indices] = C_div

    np.save(os.path.join(output_dir, f'{basename}.SignData.npy'),
            sign_matrix)
    np.save(os.path.join(output_dir, f'{basename}.DivData.npy'),
            div_matrix)
```

The goal of the program is to be as general as possible with as little as possible initial data, for computations in the Kähler cone. Based on everything discussed so far, one observes that the only data the program or SageMath can not compute for us, is the defining rays of the fan. In the case of the energy part of the Mabuchi functional, this also includes the data of the anticanonical divisor $-K$. Hence the following function takes the defining rays and the anticanonical divisor data as input, and then determines everything else that is needed.

```

def compute_with_data(data, output_dir, basename):
    if isinstance(data, np.ndarray):

        rays = data[0]
        beta = data[1]
        cones = define_cones(rays)
        fan = Fan(cones=cones, rays=rays)
        number_of_rays = len(fan.rays())
        X = ToricVariety(fan)
        Kc = X.Kaehler_cone()
        pic = X.rational_class_group()
        coho = X.cohomology_ring()
        size_of_matrix = find_matrix_size(number_of_rays - 2)

        gen_list, gen_list_count = generate_pic_gen_lists(pic, fan, coho)
        C_div_list = get_coho_C_divisors(len(fan.rays()))

        intersection_matrix = compute_intersection_matrix(len(fan.rays()), X
                                                         , coho)

        compute_data(beta, basename, output_dir, intersection_matrix, fan,
                    Kc, C_div_list, coho, gen_list_count
                    , gen_list, size_of_matrix)

    else:
        print("\n\nData type:", type(data))
        print("\n\nData content:", data)

```

As an example we see that the blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along the point $([1 : 0], [1 : 0])$ has initial data

```

[list([(0, 1), (1, 1), (1, 0), (0, -1), (-1, 0)])
list([np.float64(0.6666666666), np.float64(1.0), np.float64(0.6666666666)])]

```

Where the anticanonical divisor is described by the generators of the Picard group, rescaled such that the highest value coefficient is equal to one.

4.5 Plotting Function

Lastly we need a way of visualizing the computed data. Since the dimensions of the Kähler cone can be arbitrary, we wish the plotting function to be able to plot arbitrary dimensions aswell. The two following functions, plots the last two dimensions of the computed data in the Kähler cone, and creates sliders for the remaining dimensions.

```

def plot_heatmap_nd(**kwargs):

    slider_values = {k: v for k, v in kwargs.items() if k.startswith('
                    alpha_')}

```

```

indices = []
for dim in sorted(slider_values.keys(), key=lambda x: int(x.split(
    '_')[1])):
    alpha_val = slider_values[dim]
    dim_index = int(dim.split('_')[1])
    dim_vals = np.linspace(0, 1, sign_data.shape[dim_index])
    index = np.argmin(np.abs(dim_vals - alpha_val))
    indices.append(index)

sign_matrix = sign_data[tuple(indices) + (Ellipsis,)]
div_matrix = div_data[tuple(indices) + (Ellipsis,)]

norm = colors.TwoSlopeNorm(vmin=min_plot_value, vcenter=0, vmax=
    max_plot_value)

plt.imshow(sign_matrix, cmap='RdYlGn', norm=norm, extent=[0, 1, 1, 0])

if annotate_divisors == True:
    rows, cols = sign_matrix.shape
    for m in range(rows):
        for n in range(cols):
            val = div_matrix[m, n]
            for k in range(size_of_matrix + 2):
                if str(k) == str(val):
                    x = (n + 0.5) / (cols)
                    y = (m + 0.5) / (rows)
                    plt.text(x, y, f"{int(val)}", color=ray_color_list[k], ha="
                        center", va="center", fontsize=
                            divisor_size(dimensions))

title_parts = []
for dim in sorted(slider_values.keys(), key=lambda x: int(x.split('_')
    [1])):
    alpha_val = slider_values[dim]
    dim_index = int(dim.split('_')[1])
    if dim_index == 0:
        H_part = f"{round(alpha_val, 4)} * H - "
        title_parts.append(H_part)
    else:
        formatted = format_e_with_subscript(dim_index)
        title_parts.append(f"{round(alpha_val, 4)} * " + formatted + " - "
            )
title_str = f'\alpha = {" ".join(title_parts)}'
plt.title(title_str)
plt.xlabel(f"\alpha_{chr(97 + len(slider_values) + 1)}")
plt.ylabel(f"\alpha_{chr(97 + len(slider_values))}")

fig_color = plt.colorbar(ticks=np.linspace(min_plot_value,
    max_plot_value, 10))

plt.show()

```

```
def plot_func_nd(size_of_matrix, num_dimensions):
    step_counter = 1 / (size_of_matrix - 1)
    sliders = {}

    for dim in range(num_dimensions - 2):
        slider_name = f"alpha_{dim}"
        sliders[slider_name] = FloatSlider(
            value=0.5,
            min=0,
            max=1,
            step=step_counter,
            description=f"\alpha_{chr(97 + dim)}"
        )

    interact(plot_heatmap_nd, **sliders)
```

Remarks

The goal was to develop the program, and therefore we expect that further optimizations are possible. A first improvement would be to remove the dependence on SageMath. At present, SageMath is used to determine whether a divisor is Kähler, although this property is entirely determined by the combinatorial structure of the fan Σ . Implementing this criterion directly could significantly reduce computation time.

Furthermore, SageMath currently constructs auxiliary objects such as the Picard group and the Kähler cone. Since these are one-time constructions, replacing them is unlikely to yield substantial performance gains. However, retaining SageMath for these steps has the advantage of built-in error handling, which ensures that inconsistencies in the initial data file are detected and handled appropriately.

The program, along with useful scripts and precomputed data, can be found at https://github.com/jeppeelb-ctrl/Thesis_plot. Further, at the time of submission, an interactive webpage visualizing precomputed data by the program, was under development. A link to the webpage can be found at <https://sites.google.com/site/zakariassjostromdyrefelt>.

5 Main Theoretical Results and Computational Insights

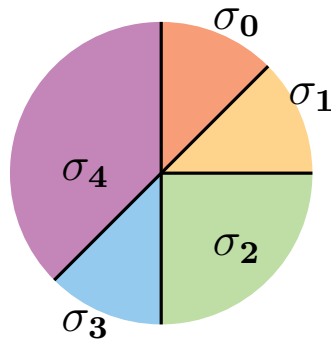
In this chapter we will first use the program to visualize concrete wall-chamber decomposition. Thereafter we will prove Theorem 5.1, providing both a computational and a theoretical approach. The computational approach relying on obstructions arising from the fans Σ , reducing the number of fans of interest. Then treating the remaining cases individually.

5.1 Visualization of Wall-Chamber Decompositions and Computation of Optimal Destabilizing Curves

The code described in Chapter 4 computes the numerical invariants $I(\alpha)$ and $J(\alpha, \beta)$, and plots the computations. However, the invariants relies on determining the optimal destabilizing torus-invariant divisor, which in turn induces a wall-chamber decomposition of the Kähler cone. To make a precise definition of the wall-chamber decomposition, we refer to (Khalid and Dyrefelt 2024), however we visualize it in a broader sense by varying α as well.

Example 1 of $I(\alpha)$

Consider the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 0 : 1]$, i.e. the fan Σ .



Where the generators of the Picard group are

$$g_0 = [D_2 + D_3], \quad g_1 = [-D_2 + D_4] \text{ and } g_2 = [D_2].$$

Hence any divisor $\alpha \in \text{Pic}(X_\Sigma)$ can be described as

$$\alpha = ag_0 + bg_1 + cg_2, \quad a, b, c \in \mathbb{R},$$

and the anticanonical divisor is

$$-K = \sum_{i=0}^4 D_i = [D_2 + 2D_3 + 2D_4] = 2g_0 + 2g_1 + 1g_2.$$

Running the program of Chapter 4 on the initial data

```
[list([(0, 1), (1, 1), (1, 0), (0, -1), (-1, -1)])
list([np.float64(1.0), np.float64(1.0), np.float64(0.5)])]
```

will result in the plot of Figure 5.1 for the fixed value $a = 0.5$, and the plot of Figure 5.2 for the fixed value $a = 1$.

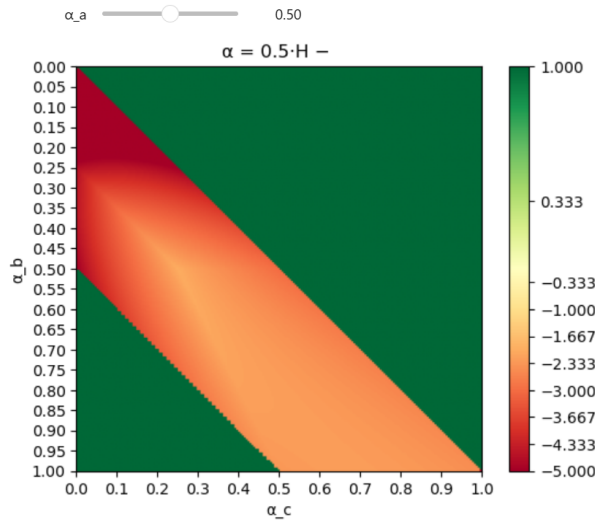


Figure 5.1: Plot of the energy part of the Mabuchi functional $I(\alpha)$ for the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 0 : 1]$, for fixed value $a = 0.5$.

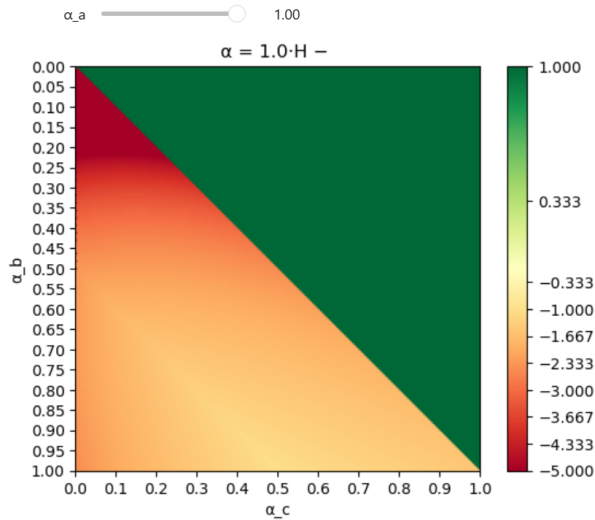
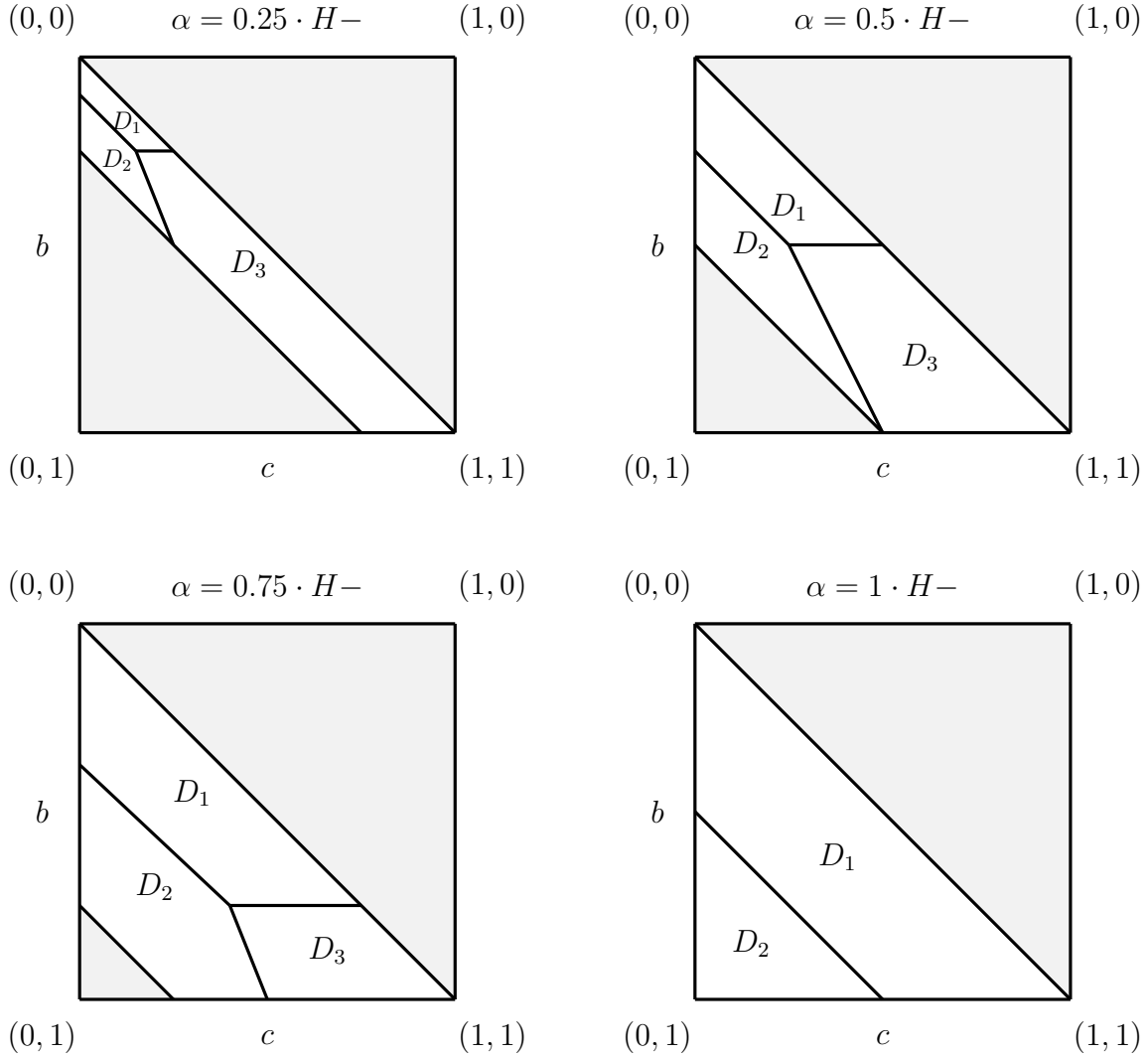


Figure 5.2: Plot of the energy part of the Mabuchi functional $I(\alpha)$ for the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 0 : 1]$, for fixed value $a = 1$.

One observes in the figures that there are brighter lines, which in fact is the wall-chamber decomposition of the Kähler cone with respect to the torus-invariant divisors under $I(\alpha)$. The following figures gives a clearer picture of the wall-chamber decomposition for varying values of a .



These brighter lines are not that visible in the plot, since we are only plotting the explicit values of $I(\alpha)$. Therefore we further implemented to the code a matrix, similar to that of the explicit values, that stores which torus-invariant divisor is optimal destabilizing. Due to the size of our matrices, we have not found a clear way of plotting the wall-chamber decomposition if the Kähler cone is 3-dimensional or lower, however if the Kähler cone is 4-dimensional or higher, we can obtain a clear picture of the wall-chamber decomposition.

Example 2 of $I(\alpha)$

Consider the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$, $[0 : 1 : 0]$ and $[0 : 0 : 1]$, where the generators of the Picard group are

$$g_0 = [D_2 + D_3 - D_5], g_1 = [-D_2 + D_4 + D_5], g_2 = [D_2] \text{ and } g_3 = [D_3].$$

Hence any divisor $\alpha \in \text{Pic}(X_\Sigma)$ can be described as

$$\alpha = ag_0 + bg_1 + cg_2 + dg_3, \quad a, b, c, d \in \mathbb{R},$$

and the anticanonical divisor is

$$-K = \sum_{i=0}^5 D_i = 1g_0 + 2g_1 + 2g_2 + 1g_3.$$

We then obtain the following plot in Figure 5.3 and the plot in Figure 5.4.

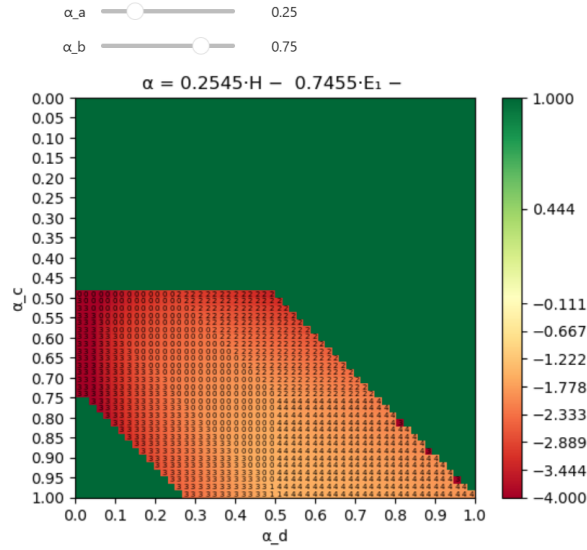


Figure 5.3: Plot of the energy part of the Mabuchi functional $I(\alpha)$ for the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$, $[0 : 1 : 0]$ and $[0 : 0 : 1]$, for fixed values $a = 0.2545$ and $b = 0.7455$.

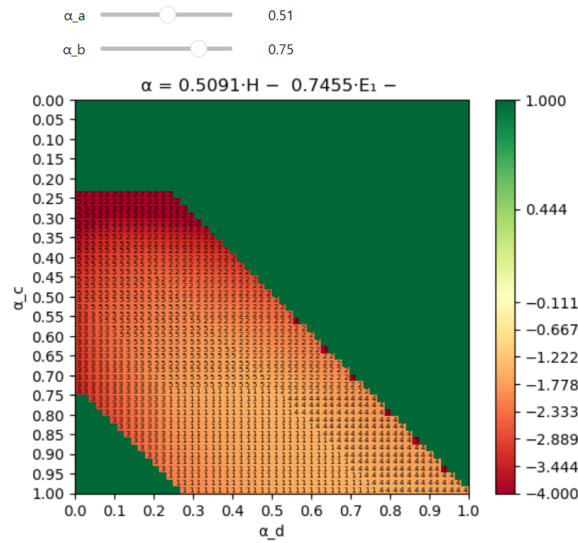
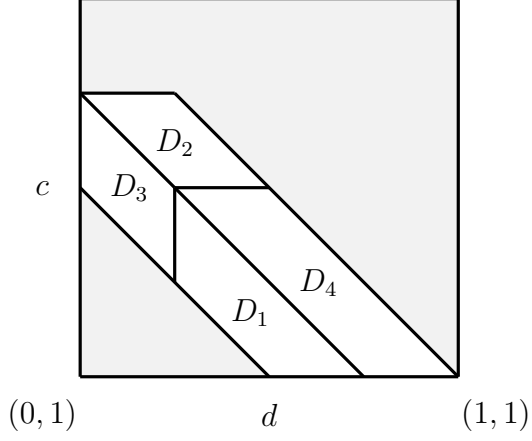


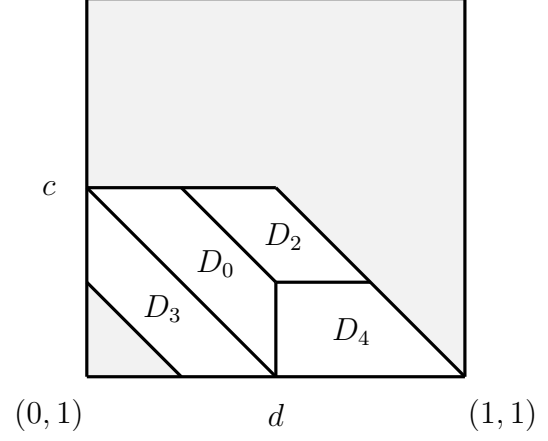
Figure 5.4: Plot of the energy part of the Mabuchi functional $I(\alpha)$ for the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$, $[0 : 1 : 0]$ and $[0 : 0 : 1]$, for fixed values $a = 0.5091$ and $b = 0.7455$.

Again for a clearer picture of the wall-chamber decomposition, consider the following figures.

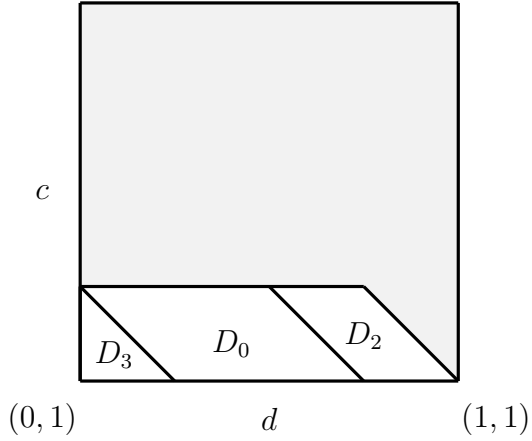
$$(0,0) \quad \alpha = 0.25 \cdot H - 0.5 \cdot E_1 - \quad (1,0)$$



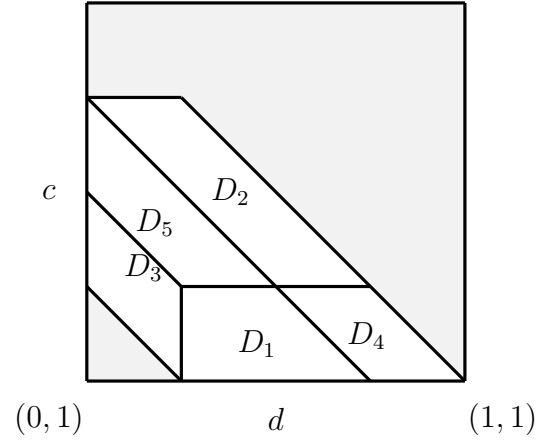
$$(0,0) \quad \alpha = 0.25 \cdot H - 0.75 \cdot E_1 - \quad (1,0)$$



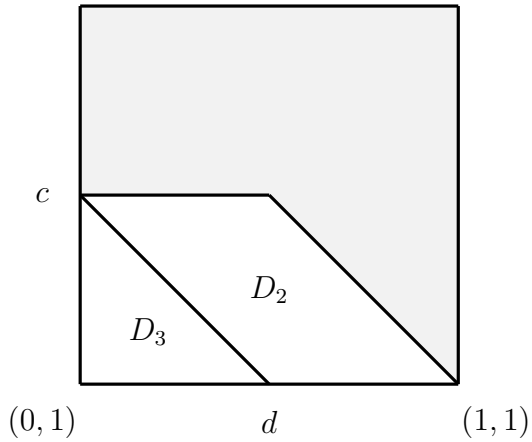
$$(0,0) \quad \alpha = 0.25 \cdot H - 1 \cdot E_1 - \quad (1,0)$$



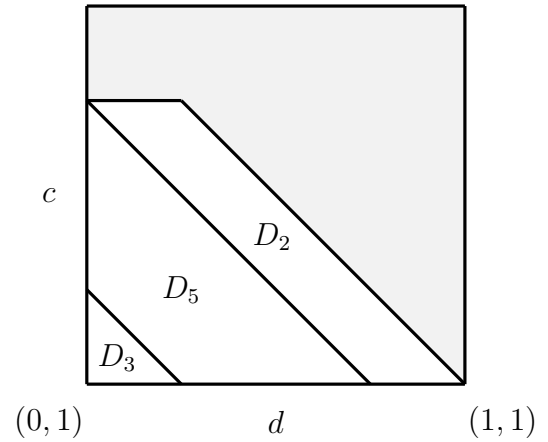
$$(0,0) \quad \alpha = 0.5 \cdot H - 0.75 \cdot E_1 - \quad (1,0)$$



$$(0,0) \quad \alpha = 0.5 \cdot H - 1 \cdot E_1 - \quad (1,0)$$



$$(0,0) \quad \alpha = 0.75 \cdot H - 1 \cdot E_1 - \quad (1,0)$$



We included the description of the anti-canonical divisor $-K$, as we will see in the later sections of this chapter, that the coefficients of $-K$ is the ones determining the decomposition.

Example 1 of $J(\alpha, \beta)$

If we instead considered

$$J(\alpha, \beta) = \min_{D_i} \left(2 \frac{\alpha \cdot \beta}{\alpha^2} - \frac{\beta \cdot D_i}{\alpha \cdot D_i} \right),$$

we will similarly obtain a wall-chamber decomposition with respect to the torus-invariant divisors. Consider again the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 0 : 1]$, our first example of this section, running the code from Chapter 4, adapted to $J(\alpha, \beta)$, with fixed $\beta = -K$, we obtain the plots of Figure 5.5 and Figure 5.6.

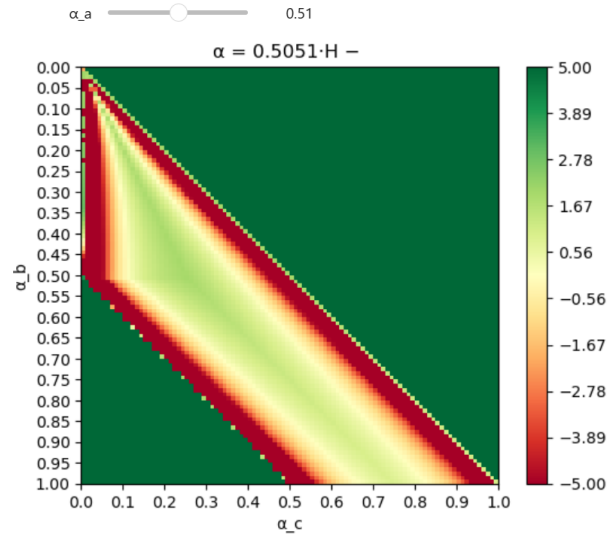


Figure 5.5: Plot of $J(\alpha, -K)$ for the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 0 : 1]$, for fixed value $a = 0.5051$.

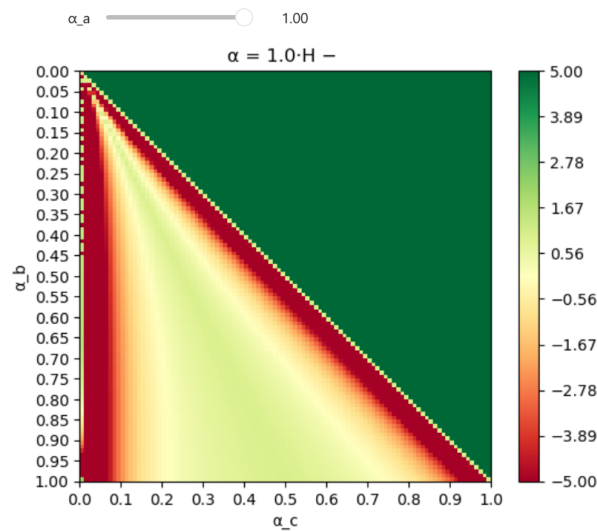
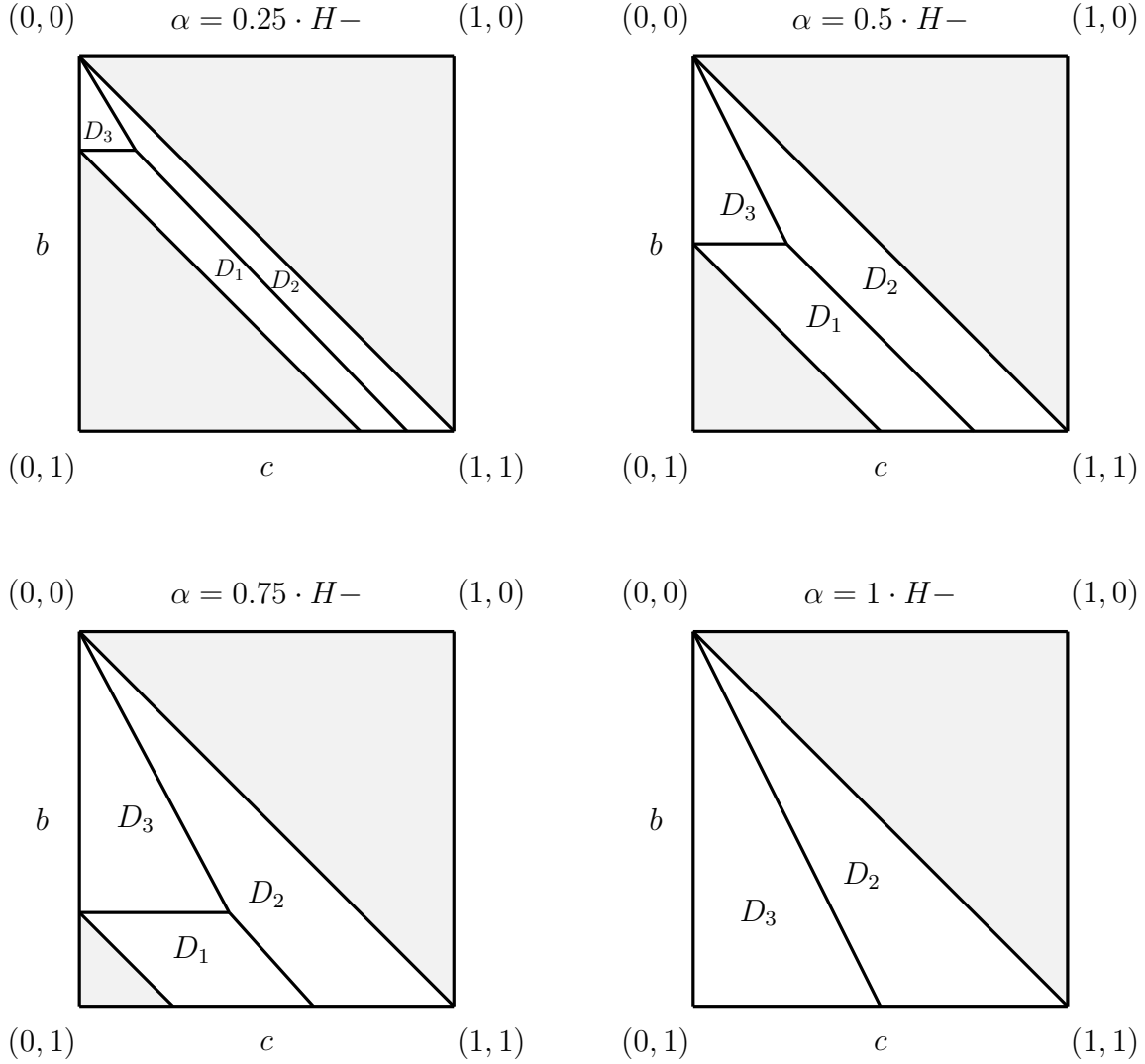


Figure 5.6: Plot of $J(\alpha, -K)$ for the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 0 : 1]$, for fixed value $a = 1$.

Similar to the plots in Figure 5.1 and Figure 5.2, one observes there are lines in the plot, which is in fact the wall-chamber decomposition with respect to the torus-invariant divisors. Consider the following figures for a clearer picture of the decomposition.

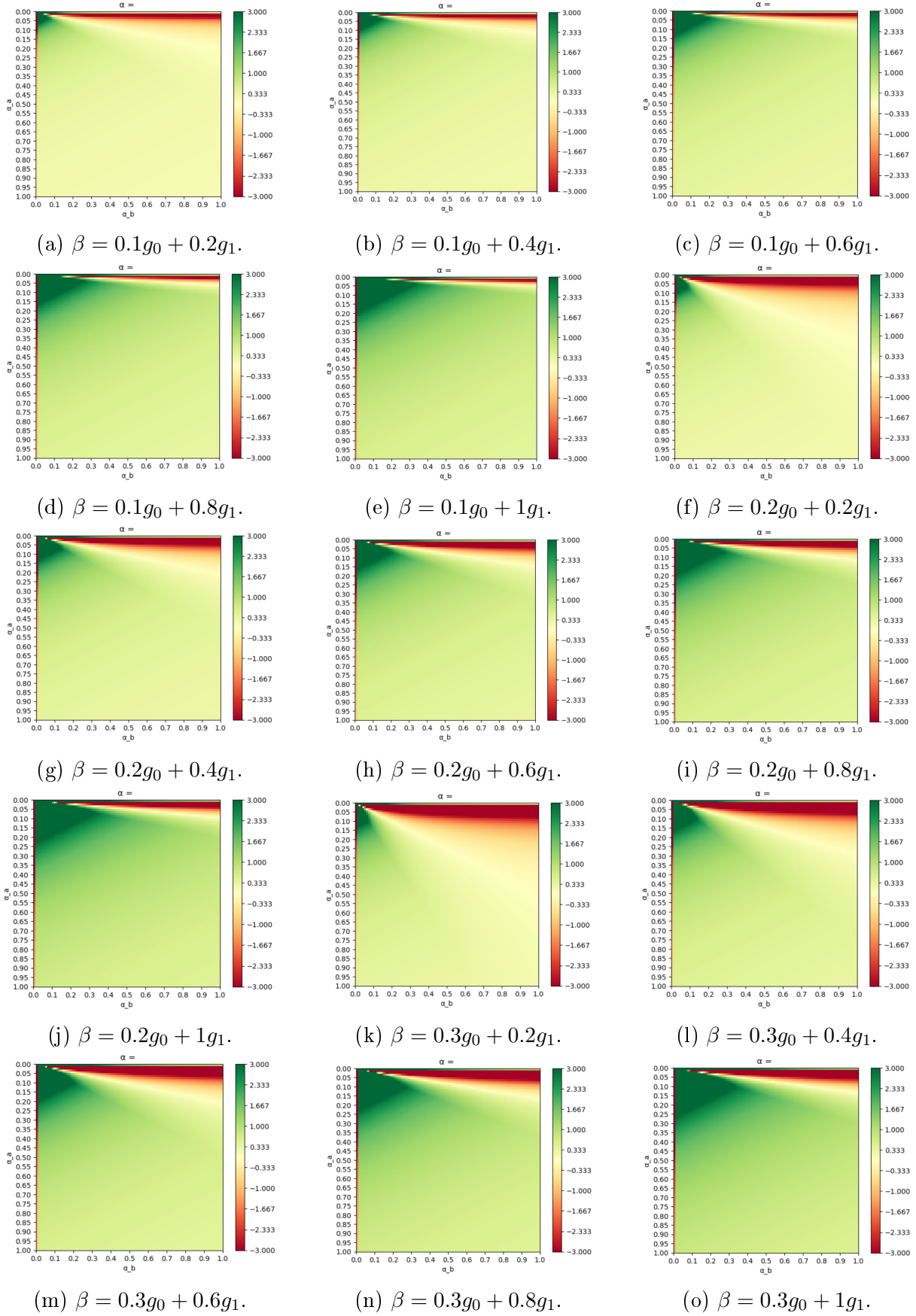


As mentioned for $I(\alpha)$ it is the coefficients of $-K$, which determines the wall-chamber decomposition, for $J(\alpha, \beta)$ we have a similar result. Comparing the above figures of $J(\alpha, -K)$, with the figures for $I(\alpha)$, it is in fact again the coefficients of $\beta = -K$ that determines the decomposition. The notable differences in the decompositions are the fact they are dual to each other. This duality connection, and why the coefficients determines the decomposition becomes clearer in Section 5.2, where we consider a partial ordering induced by the Kähler cone.

$J(\alpha, \beta)$ is not only interesting when $\beta = -K$, in the cases where $-K$ is Kähler, hence to end this section we provide figures of an example, where we vary β .

Example 2 of $J(\alpha, \beta)$

Consider the blow-up of $\mathbb{P}^2$ along the point $[0 : 1 : 0]$, then for varying β we obtain the following figures from the program.

Figure 5.7: Plot of $J(\alpha, \beta)$ for the blow-up of $\mathbb{P}^2$ along the point $[0 : 1 : 0]$, for varying β .

5.2 Proof of Non-Coercivity of The Energy Part of The Mabuchi Functional

We consider the numerical invariant for the the energy part of the Mabuchi functional,

$$I(\alpha) = \inf_D \left(2 \frac{\alpha.K}{\alpha^2} - \frac{K.D}{\alpha.D} \right),$$

where α is a Kähler class on a smooth complex projective surface. $K = K_{X_\Sigma}$ is the canonical class, and the infimum runs over irreducible curves $D \subset X_\Sigma$.

The goal of this section is to prove the following theorem.

Theorem 5.1. *Let X_Σ be a smooth complete toric surface and let α be any Kähler class. Then $I(\alpha) < 0$ for every Kähler class α .*

Then Theorem 5.1 together with Proposition 3.19 shows that the energy part $\mathbf{E}_{\theta,\omega}^{\text{NA}}$ is not coercive. Considering the full Mabuchi functional

$$M = E + H,$$

having shown the energy part is always negative for smooth complex projective surfaces, one wonder how the full functional behaves. We quickly note that non-coercivity of the energy part does not imply non-existence of cscK metrics, as the entropy part H is known to be positive. Natural questions then becomes: When is E so negative that H cannot compensate? When is E negative such that H can compensate? We do not have a clear answer to these questions, but the program developed in Chapter 4 can help guide one in the right directions. From the figures of Section 5.1, one observers that $I(\alpha)$ yields lower values along the rays of the decomposition. Further, we observe that $I(\alpha)$ becomes very negative when one of the coefficients of α tends towards 0.

Computational Approach to Theorem 5.1

In the setting of toric fans one can see that there are infinitely many fans, hence one would aim to limit the number of fans to consider. The following lemma provides criterias for the fan such that $I(\alpha)$ is negative.

Lemma 5.2. *Let X_Σ be a smooth complete toric surface and let α be any Kähler class. Assume:*

- *There exists an irreducible curve $D \subset X_\Sigma$ with $D^2 \leq -3$.*
- *For all Kähler classes α one has $\alpha.K_{X_\Sigma} < 0$ (equivalently $\alpha.(-K) > 0$).*

Then $I(\alpha) < 0$ for every Kähler class α .

Proof. Fix such a curve D . By adjunction on a smooth surface,

$$(K + D).D = 2g(D) - 2,$$

where $g(D)$ is the genus of D , we get

$$K.D = 2g(D) - 2 - D^2.$$

Since $g(D) \geq 0$ and by assumption $D^2 \leq -3$, which implies

$$K.D \geq -2 - (-3) = 1,$$

hence $K.D > 0$.

Consider now

$$I(\alpha) = 2 \frac{\alpha.K}{\alpha^2} - \sup_D \left(\frac{K.D}{\alpha.D} \right).$$

Since α is Kähler $\alpha.D > 0$ and, hence $\frac{K.D}{\alpha.D} > 0$, so the supremum term is strictly positive. By assumption $\alpha.K < 0$ and again α being Kähler yields $\alpha^2 > 0$, implying $2 \frac{\alpha.K}{\alpha^2} < 0$. In total $I(\alpha)$ is a negative term minus a positive term, hence it is negative. We conclude that $I(\alpha) < 0$. $\square$

In our setting of smooth projective toric surfaces, $-K$ is always effective, as $-K = \sum_i D_i$, i.e. sum over torus-invariant divisors. Hence for any Kähler class α we have $\alpha.(-K) > 0$. Therefore, on toric surfaces the second assumption of Lemma 5.2 is automatically satisfied. Thus, if a toric fan contains any torus-invariant divisor D with self-intersection $D^2 \leq -3$, then $I(\alpha) < 0$ for all α .

Recall from Theorem 2.56 that the self-intersection number of the torus-invariant divisor D_i , corresponding to $\tau = \text{Cone}(v_i)$, which is the intersection of 2-dimensional cones $\text{Cone}(v_i, v_j)$ and $\text{Cone}(v_i, v_k)$, comes from the relation of the rays

$$v_j + v_k = b_i v_i, \text{ for some } b_i \in \mathbb{Z},$$

then $D_i^2 = -b_i$. We can then relate the self-intersection numbers of the torus-invariant divisors, to the number of rays of the fan.

Lemma 5.3. *Let X_Σ be a smooth complete toric surface, with associated fan Σ . Further, assume $|\Sigma(1)| = r$ and let a_i be the self-intersection numbers of the torus-invariant divisors D_i associated to the rays $\tau_i \in \Sigma(1)$. Lastly assume that $a_i \geq -n$, $\forall i$ for some $n \in \mathbb{N}$, then the fan Σ consists of maximum*

$$\frac{12}{3-n}$$

rays.

Proof. Consider the invariant sum of self-intersection numbers on toric surfaces,

$$\sum_{i=0}^{r-1} a_i = 12 - 3r.$$

Since we assumed $a_i \geq -n$,

$$\sum_{i=0}^{r-1} a_i \geq -nr.$$

Comparing this with the invariant sum we obtain,

$$12 - 3r \geq -nr \implies \frac{12}{3-n} \geq r.$$

Thus $|\Sigma(1)| \leq \frac{12}{3-n}$. □

From Lemma 5.3 and Lemma 5.2 we observe that for any fan Σ consisting of more than

$$\frac{12}{3-2} = 12 \geq r$$

rays, on the smooth complex projective surface X_Σ associated to Σ , we have $I(\alpha) < 0$ for every Kähler class α .

As we only wish to consider fans having the lowest self-intersection number to be -2 , the above discussion gives an upper bound of the number of fans to consider. However, the above discussion does not take into account Theorem 2.59, and the explicit structures of the fans. In Appendix C we have provided a script which can generate every possible fan based on the number of blow-ups, the initial fan and the self-intersection numbers. Running the script yields that there are only 167 fans with no self-intersection less than -2 .

One observes that it is immensely easier to consider 167 fans compared to infinitely many. However, 167 is still a rather large number, and one would likely want to improve this bound. Based on the computations of the script in Chapter 4, we found a different approach to obtaining such a bound.

Lemma 5.4. *Let X_Σ be a smooth complete toric surface and let α be any Kähler class. Assume:*

- *There exists an irreducible curve $D \subset X_\Sigma$ with $D^2 \leq -2$.*
- *For all Kähler classes α one has $\alpha \cdot K_{X_\Sigma} < 0$ (equivalently $\alpha \cdot (-K) > 0$).*

Then $I(\alpha) < 0$ for every Kähler class α .

Proof. Let Σ be the associated fan to X_Σ . Let r be the number of defining rays of Σ . The intersection matrix is then of the form

$$IM = \begin{pmatrix} D_0^2 & 1 & 0 & 0 & 0 & \cdots & 0 & 0 & 1 \\ 1 & D_1^2 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & D_2^2 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & D_3^2 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & 1 & D_{r-2}^2 & 1 \\ 1 & 0 & 0 & 0 & 0 & \cdots & 0 & 1 & D_{r-1}^2 \end{pmatrix}$$

where D_i^2 is the torus-invariant divisors' self-intersection number.

Following the approach of Chapter 4 to determine the generators of $\text{Pic}(X_\Sigma)$, we can choose the generators to be

$$g_0 = [D_0], g_1 = [D_1], \dots, g_{r-3} = [D_{r-3}],$$

where $[D_0]$ and $[D_1]$ has another description based on the relations in Theorem 2.48.

Let $\alpha = \sum_{k=0}^{r-3} a_k g_k \in \text{Pic}(X_\Sigma)$, with $a_k \in \mathbb{R}$, then

$$[\alpha] = \left[\sum_{k=0}^{r-3} a_k D_k \right].$$

Set $\alpha' = [a_0, a_1, \dots, a_{r-2}, a_{r-1}]$, where $a_{r-2} = a_{r-1} = 0$. Since

$$-K = \sum_{k=0}^{r-1} D_k, \text{ then } [-K] = \left[\sum_{k=0}^{r-1} D_k \right]$$

we set

$$-K' = [1, 1, 1, \dots, 1, 1],$$

with r entries. The i 'th entry of the r long vector $\alpha' \cdot IM$ is then

$$(\alpha' \cdot IM)_i = [a_{i-1} + a_i D_i^2 + a_{i+1}]$$

modulo $r-1$, such that $a_r = a_0$. We also have the i 'th entry of the r long vector $(-K') \cdot IM$ is

$$((-K') \cdot IM)_i = [D_i^2 + 2].$$

For the torus-invariant divisors, we set $D_i = [0, 0, \dots, 1, \dots, 0]$, where the 1 is in the i 'th

entry of the vector. We further compute

$$\begin{aligned}\alpha.(-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= \sum_{k=0}^{r-1} a_{k-1} + a_k D_k^2 + a_{k+1},\end{aligned}$$

$$\begin{aligned}\alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= \sum_{k=0}^{r-1} a_k (a_{k-1} + a_k D_k^2 + a_{k+1}),\end{aligned}$$

$$\begin{aligned}(-K).D_i &= (-K') \cdot IM \cdot D_i^T \\ &= D_i^2 + 2,\end{aligned}$$

$$\begin{aligned}\alpha.D_i &= \alpha' \cdot IM \cdot D_i^T \\ &= a_{i-1} + a_i D_i^2 + a_{i+1}.\end{aligned}$$

Consider

$$\begin{aligned}I(\alpha) &= \inf_D \left(2 \frac{\alpha.K}{\alpha^2} - \frac{K.D}{\alpha.D} \right) \\ &= - \max_{D_i} \left(2 \frac{\alpha.(-K)}{\alpha^2} - \frac{(-K).D_i}{\alpha.D_i} \right).\end{aligned}$$

Rewriting and omitting the negative, yields

$$2 \frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left(\frac{(-K).D_i}{\alpha.D_i} \right).$$

Expanding the terms, we have

$$2 \frac{\alpha.(-K)}{\alpha^2} - \frac{(-K).D_i}{\alpha.D_i} = \frac{2 \sum_{k=0}^r a_{k-1} + a_k D_k^2 + a_{k+1}}{\sum_{k=0}^r a_k (a_{k-1} + a_k D_k^2 + a_{k+1})} - \frac{D_i^2 + 2}{a_{i-1} + a_i D_i^2 + a_{i+1}}.$$

Since α is a Kähler class, we have that

$$\alpha.D_i = a_{i-1} + a_i D_i^2 + a_{i+1} > 0, \quad \forall i,$$

and

$$\alpha^2 = \sum_{k=0}^{r-1} a_k (a_{k-1} + a_k D_k^2 + a_{k+1}) > 0.$$

Since $-K$ is an effective divisor we have that $\alpha.(-K) > 0$. If we then consider the first factor

$$2 \frac{\alpha.(-K)}{\alpha^2} = \frac{2 \sum_{k=0}^r a_{k-1} + a_k D_k^2 + a_{k+1}}{\sum_{k=0}^r a_k (a_{k-1} + a_k D_k^2 + a_{k+1})} > 0.$$

Hence we only need to consider the sign of

$$\frac{(-K).D_i}{\alpha.D_i} = \frac{D_i^2 + 2}{a_{i-1} + a_i D_i^2 + a_{i+1}},$$

where we note that it only depends on the sign of $D_i^2 + 2$.

Hence if $D_i^2 \leq -2$, then $\frac{(-K).D_i}{\alpha.D_i} \leq 0$, and we get

$$-\max_{D_i} \left(2 \frac{\alpha.(-K)}{\alpha^2} - \frac{(-K).D_i}{\alpha.D_i} \right) < 0.$$

We conclude that if the associated fan Σ to the smooth complex projective surface X_Σ has an irreducible curve with self-intersection number less than -2 , then $I(\alpha) < 0$. $\square$

From the above proof it can be observed that

$$(-K) \in \mathcal{K}(X_\Sigma) \iff (-K).D_i = D_i^2 + 2 > 0 \iff D_i^2 \geq -1, \quad \forall i.$$

By Lemma 5.3 and Lemma 5.4 we obtain,

$$\frac{12}{3-n} = \frac{12}{3-1} = 6 \geq r$$

Thus a fan consisting of more than 6 rays will have a torus-invariant divisor with self-intersection less than -1 .

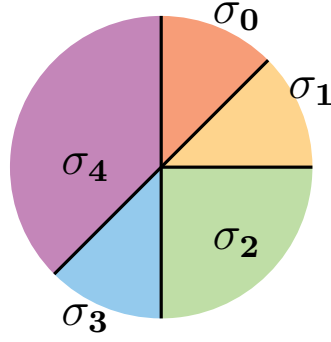
Again using the script from Appendix C, we obtain that there are only 15 cases with no self-intersection less than -1 .

It is worth noting by Lemma 5.4 and Theorem 2.59, that we have effeciently ruled out the Hirzebruch surfaces $\mathcal{H}_r$, as they are forced to have a divisor with self-intersection less than -1 . It is also worth noting that for the remaining 15 cases, 3 cases coincides in the data of the Picard group and the Kähler cone. These 3 cases are the blow-up of $\mathbb{P}^2$ along the three distinguished points $[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]$, and the blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along the points $([1 : 0], [1 : 0]), ([0 : 1], [0 : 1])$ or along the points $([0 : 1], [1 : 0]), ([1 : 0], [0 : 1])$. This can be seen by the fact they share the same reduced row echelon form in the approach of Chapter 4 to determine the generators of the Picard group, and everything else is determined exactly by this. In order to study the behaviour of the numerical invariant $I(\alpha)$ in the remaining 13 cases, we developed an approach similar to that of Lemma 5.4. We will now give an example of this approach for the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 0 : 1]$, note this is exactly the example at the end of Chapter 4.

Blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 0 : 1]$

Proof. Fan

The fan of the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 0 : 1]$, can be visualised as



Intersection Matrix.

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|ccccc} & D_0 & D_1 & D_2 & D_3 & D_4 \\ \hline D_0 & 0 & 1 & 0 & 0 & 1 \\ D_1 & 1 & -1 & 1 & 0 & 0 \\ D_2 & 0 & 1 & -1 & 1 & 0 \\ D_3 & 0 & 0 & 1 & -1 & 1 \\ D_4 & 1 & 0 & 0 & 1 & 0 \end{array}$$

Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2 + D_3], g_1 = [-D_2 + D_4] \text{ and } g_2 = [D_2].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_1 + g_2, \tilde{g}_1 = g_0 \text{ and } \tilde{g}_2 = g_0 + g_1.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 + x_2 \tilde{g}_2 \\ &= (x_1 + x_2)g_0 + (x_0 + x_2)g_1 + x_0 g_2, \end{aligned}$$

$$\begin{aligned} g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 + y_2 \tilde{g}_2 \\ &= (y_1 + y_2)g_0 + (y_0 + y_2)g_1 + y_0 g_2, \end{aligned}$$

$$\begin{aligned} g_2 &= z_0 \tilde{g}_0 + z_1 \tilde{g}_1 + z_2 \tilde{g}_2 \\ &= (z_1 + z_2)g_0 + (z_0 + z_2)g_1 + z_0 g_2, \end{aligned}$$

gives

$$\begin{array}{lll} x_0 = 0, & x_1 = 1, & x_2 = 0 \\ y_0 = 0, & y_1 = -1, & y_2 = 1 \\ z_0 = 1, & z_1 = 1, & z_2 = -1 \end{array}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 + cg_2 \in \text{Pic}(X_\Sigma), \quad a, b, c \in \mathbb{R},$$

can be written as

$$\alpha = c\tilde{g}_0 + (a - b + c)\tilde{g}_1 + (b - c)\tilde{g}_2.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b, c > 0, \quad b > c, \quad a + c > b.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [(a - b + c)D_2 + aD_3 + bD_4].$$

Following the approach of Chapter 4 we set

$$\alpha' = [0, 0, a - b + c, a, b].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^4 D_i = [D_2 + 2D_3 + 2D_4] = 2g_0 + 2g_1 + 1g_2.$$

After rescaling, we set

$$-K' = \left[0, 0, \frac{1}{2}, 1, 1\right].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [b, a - b + c, b - c, c, a] \cdot (\alpha')^T \\ &= 2ab + 2bc - b^2 - c^2, \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= \frac{2a + b + c}{2}. \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{2a + b + c}{2ab + 2bc - b^2 - c^2} > 0.$$

Intersections with Torus Divisors.

We further compute,

$$(-K) \cdot D_i = (-K') \cdot IM \cdot D_i^T = \begin{cases} 1, & \text{if } i = 0, \\ \frac{1}{2}, & \text{if } i = 1, 2, 3, \\ 1, & \text{if } i = 4. \end{cases}$$

$$\alpha.D_i = \alpha' \cdot IM \cdot D_i^T = \begin{cases} b, & \text{if } i = 0, \\ a - b + c, & \text{if } i = 1, \\ b - c, & \text{if } i = 2, \\ c, & \text{if } i = 3, \\ a, & \text{if } i = 4. \end{cases}$$

hence

$$\frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{1}{b}, & \text{if } i = 0, \\ \frac{1}{2(a-b+c)}, & \text{if } i = 1, \\ \frac{1}{2(b-c)}, & \text{if } i = 2, \\ \frac{1}{2c}, & \text{if } i = 3, \\ \frac{1}{a}, & \text{if } i = 4. \end{cases}$$

We note that for each i the above is positive.

Optimization Functional.

We consider the intersection numbers

$$2 \frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\}.$$

Assume $i = 0$ minimizes, hence

$$\begin{aligned} \frac{1}{b} < \frac{1}{2(a-b+c)} &\implies 2(a+c) < 3b, \\ \frac{1}{b} < \frac{1}{2(b-c)} &\implies b < 2c, \\ \frac{1}{b} < \frac{1}{2c} &\implies 2c < b, \\ \frac{1}{b} < \frac{1}{a} &\implies a < b, \end{aligned}$$

which is a contradiction, hence D_0 will never optimize.

Assume $i = 1$ minimizes, hence

$$\begin{aligned} \frac{1}{2(a-b+c)} < \frac{1}{b} &\implies 3b < 2(a+c), \\ \frac{1}{2(a-b+c)} < \frac{1}{2(b-c)} &\implies 2b < a+2c, \\ \frac{1}{2(a-b+c)} < \frac{1}{2c} &\implies b < a, \\ \frac{1}{2(a-b+c)} < \frac{1}{a} &\implies 2b < a+2c, \end{aligned}$$

hence D_1 is optimizing under the above restrictions.

Assume $i = 2$ minimizes, hence

$$\begin{aligned}\frac{1}{2(b-c)} &< \frac{1}{b} \implies 2c < b, \\ \frac{1}{2(b-c)} &< \frac{1}{2(a-b+c)} \implies a+2c < 2b, \\ \frac{1}{2(b-c)} &< \frac{1}{2c} \implies 2c < b, \\ \frac{1}{2(b-c)} &< \frac{1}{a} \implies a+2c < 2b,\end{aligned}$$

hence D_2 is optimizing under the above restrictions.

Assume $i = 3$ minimizes, hence

$$\begin{aligned}\frac{1}{2c} &< \frac{1}{b} \implies b < 2c, \\ \frac{1}{2c} &< \frac{1}{2(a-b+c)} \implies a < b, \\ \frac{1}{2c} &< \frac{1}{2(b-c)} \implies b < 2c, \\ \frac{1}{2c} &< \frac{1}{a} \implies a < 2c,\end{aligned}$$

hence D_3 is optimizing under the above restrictions.

Assume $i = 4$ minimizes, hence

$$\begin{aligned}\frac{1}{a} &< \frac{1}{b} \implies b < a, \\ \frac{1}{a} &< \frac{1}{2(a-b+c)} \implies a+2c < 2b, \\ \frac{1}{a} &< \frac{1}{2(b-c)} \implies 2b < a+2c, \\ \frac{1}{a} &< \frac{1}{2c} \implies 2c < a,\end{aligned}$$

which is a contradiction, hence D_4 is never optimizing.

We now have 2 areas within $\mathcal{K}(X_\Sigma)$ with optimizing divisors D_1, D_2 or D_3 . Consider now the entire expression. Recall

$$2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} = \begin{cases} \frac{2a+b+c}{2ab+2bc-b^2-c^2} - \frac{1}{2(a-b+c)}, & \text{if } i = 1, \\ \frac{2a+b+c}{2ab+2bc-b^2-c^2} - \frac{1}{2(b-c)}, & \text{if } i = 2, \\ \frac{2a+b+c}{2ab+2bc-b^2-c^2} - \frac{1}{2c}, & \text{if } i = 3. \end{cases}$$

Using the fraction rule $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$, we can conclude that the denominator is positive in each case.

Then for $i = 1$, we get

$$\frac{2a+b+c}{2ab+2bc-b^2-c^2} - \frac{1}{2(a-b+c)} = \frac{4a^2+6ac+3c^2-4ab-2bc-b^2}{2(a-b+c)(2ab+2bc-b^2-c^2)},$$

Consider the numerator.

$$\begin{aligned}
4a^2 + 6ac + 3c^2 - 4ab - 2bc - b^2 &= 4a^2 - 4ab - b^2 + c(6a - 2b + 3c) \\
&> 4a^2 - 4ab - b^2 + c(6a - 2b + c + 2b - a) \\
&= 4a^2 - 4ab - b^2 + c(5a + c) \\
&= a(4a - 4b + 5c) - b^2 + c^2 \\
&> a(4a - 4b + c + 4b - 2a) - b^2 + c^2 \\
&= a(2a + c) - b^2 + c^2 \\
&= 2a^2 + ac - b^2 + c^2 \\
&> a^2 + ac + c^2 > 0.
\end{aligned}$$

Using the restrictions $a, b, c > 0$ and $a + 2c > 2b$.

For $i = 2$, we get

$$\frac{2a + b + c}{2ab + 2bc - b^2 - c^2} - \frac{1}{2(b - c)} = \frac{2ab + 3b^2 - 4ac - 2bc - c^2}{2(b - c)(2ab + 2bc - b^2 - c^2)}.$$

Consider the numerator

$$2ab + 3b^2 - 4ac - 2bc - c^2 = 2a(b - 2c) + (3b + c)(b - c),$$

where we recall the restrictions $b > c, b > 2c$ and $a, b, c > 0$, hence we conclude the numerator is positive.

For $i = 3$, we get

$$\frac{2a + b + c}{2ab + 2bc - b^2 - c^2} - \frac{1}{2c} = \frac{4ac + 3c^2 + b^2 - 2ab}{2c(2ab + 2bc - b^2 - c^2)}.$$

Consider the numerator

$$4ac + 3c^2 + b^2 - 2ab = 2a(2c - b) + 3c^2 + b^2,$$

where we recall the restrictions $2c > b$ and $a, b, c > 0$, hence we conclude the numerator is positive.

Conclusion.

Thus, in every region where D_i is optimizing, we obtain

$$-\left(2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i}\right) < 0, \quad i = 1, 2, 3.$$

□

A beautiful consequence of this approach to determine the behaviour of $I(\alpha)$, is we get the restrictions that determine the chamber decomposition mentioned in Section 5.1. If we compare the intermediate conclusions of the approach, namely that of the restrictions for a divisor to lie in the Kähler cone to the plot in the following figure, we can see that the restrictions $a, b, c > 0$, $b > c$ and $a + c > b$ exactly determines the boundary of the Kähler cone.

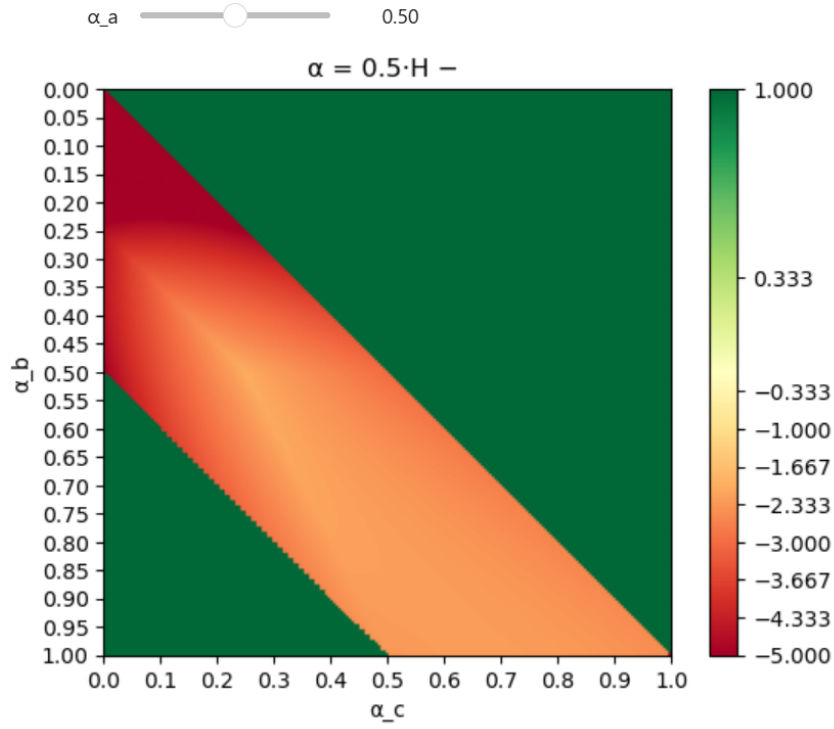
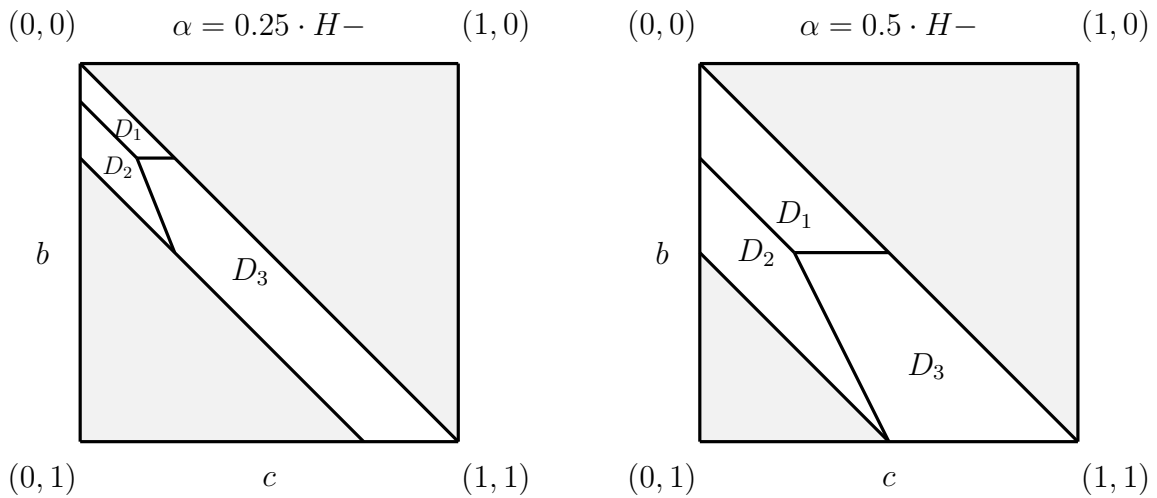


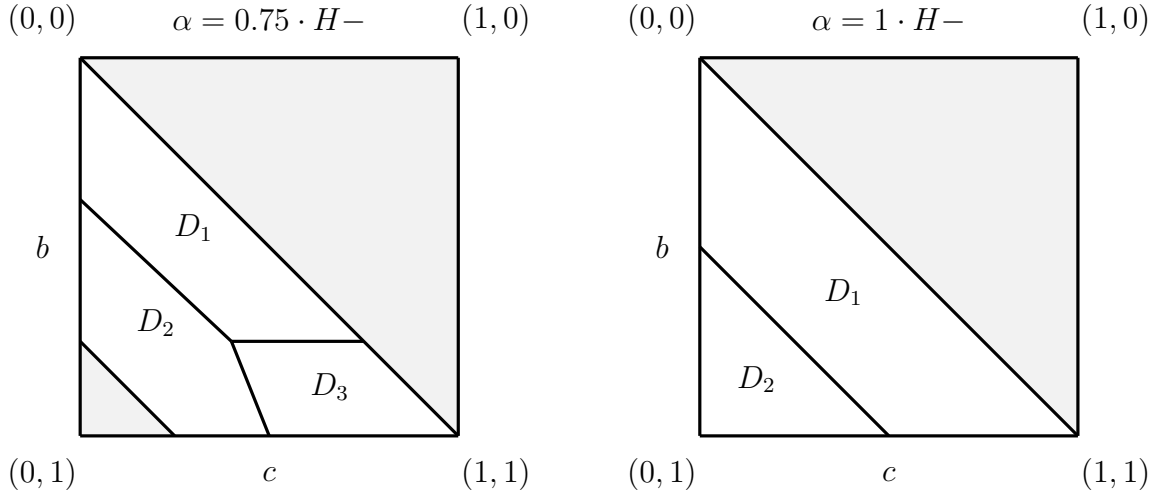
Figure 5.8: Plot of the energy part of the Mabuchi functional for the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 0 : 1]$, for fixed value $a = 0.5$.

Furthermore, we concluded in the approach that the minimizing invariant divisors are D_1, D_2 and D_3 , under their respective restrictions

$$\begin{aligned} a &> b \text{ and } a + 2c > 2b, \text{ for } i = 1, \\ b &> 2c \text{ and } 2b > a + 2c, \text{ for } i = 2, \\ 2c &> b \text{ and } b > a, \text{ for } i = 3. \end{aligned}$$

Which is precisely the decomposition seen in the figures, for varying values of a .





We are now left with 12 cases, which can be seen in Appendix B.

Direct Approach to Weaker Version of Theorem 5.1

The above approach is rather cumbersome, as it requires treating the remaining cases individually. We now present an alternative argument for a weaker version of Theorem 5.1.

Theorem 5.5. *Let X_Σ be a smooth complete toric surface and let α be any Kähler class. Then $I(\alpha) \leq 0$ for every Kähler class α .*

We first set up a linear-algebraic description of intersection theory on X_Σ .

Let X_Σ be a smooth complete toric surface with associated fan Σ , further suppose $\Sigma(1) = \{\tau_1, \dots, \tau_r\}$. By Theorem 2.46 and Theorem 2.48, we have the short exact sequence

$$0 \rightarrow M \simeq \mathbb{Z}^2 \rightarrow \text{Div}_{T_N}(X_\Sigma) \simeq \mathbb{Z}^r \rightarrow \text{Pic}(X_\Sigma) \rightarrow 0,$$

from which we obtain that $\text{Pic}_{\mathbb{R}}(X_\Sigma) \simeq \mathbb{R}^{r-2}$. Fix a basis $\{g_0, \dots, g_{r-3}\}$ of $\text{Pic}(X_\Sigma)$, and define the intersection matrix Q by $Q_{ij} := g_i \cdot g_j$. Furthermore, we can describe each torus invariant divisor $D_{\tau_i} = D_i$ associated to $\tau_i \in \Sigma(1)$, in this basis,

$$D_i = \sum_{n=0}^{r-3} d_n^{(i)} g_n, \quad \text{for some } d_n^{(i)} \in \mathbb{R}.$$

Let $d^{(i)}$ denote the coefficient vector in $\mathbb{R}^{r-2}$, we then define $v_i := Q d^{(i)}$. Similarly for an arbitrary divisor $\alpha \in \text{Pic}(X_\Sigma)$, we let $a \in \mathbb{R}^{r-2}$ denote the coefficients of α , and write $v_\alpha := Qa$. For a fixed divisor $\alpha \in \text{Pic}(X_\Sigma)$ let $l_\alpha(\cdot) := \langle \cdot, v_\alpha \rangle$. This defines linear functionals $l \in (\mathbb{R}^{r-2})^*$. Then for $\beta = \sum_{n=0}^{r-3} b_n g_n$,

$$l_\alpha(\beta) = \langle \beta, v_\alpha \rangle = b^T Qa = \beta \cdot \alpha = \alpha \cdot \beta = l_\beta(\alpha),$$

thus intersection numbers are recovered via these linear functionals. Note for a divisor α to be Kähler we must have,

$$l_{D_i}(\alpha) > 0, \quad \forall i.$$

We can then prove Theorem 5.5 by the following.

Proof. Using the above notation the linear functionals l_{D_i} ,

$$\mathcal{K}^\vee := \overline{\text{NE}}(X_\Sigma) = \{l \in (\mathbb{R}^{r-2}) \mid l(\alpha) \geq 0, \forall \alpha \in \mathcal{K}\},$$

which is generated by the torus-invariant curves, i.e. the functionals l_{D_i} .

Using the fact for $\alpha \in \mathcal{K}(X_\Sigma)$ that $\alpha^2 > 0$ and $-K$ is an effective divisor, hence $\alpha \cdot (-K) > 0$, we can then rewrite,

$$\begin{aligned} I(\alpha) &= -\max_{D_i} \left(2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} \right) \\ &= -\left(2 \frac{l_\alpha(-K)}{l_\alpha(\alpha)} - \min_{D_i} \left(\frac{l_{D_i}(-K)}{l_{D_i}(\alpha)} \right) \right) \\ &= \min_{D_i} \left(\frac{l_{D_i}(-K)}{l_{D_i}(\alpha)} \right) - 2 \frac{l_\alpha(-K)}{l_\alpha(\alpha)} \end{aligned}$$

Any $l \in \mathcal{K}^\vee$ can be described as

$$l = \sum_i c_i l_{D_i}, \quad c_i \geq 0.$$

Then

$$\frac{l(-K)}{l(\alpha)} = \sum_i \frac{c_i l_{D_i}(\alpha)}{\sum_j c_j l_{D_j}(\alpha)} \cdot \frac{l_{D_i}(-K)}{l_{D_i}(\alpha)},$$

It is hence a weighted average of the values on the D_i , therefore the infimum is the minimum over the generators. Consider the quantity to be minimized.

$$\min_{D_i} \left(\frac{l_{D_i}(-K)}{l_{D_i}(\alpha)} \right) = \inf_{l \in \mathcal{K}^\vee} \left(\frac{l(-K)}{l(\alpha)} \right),$$

for $l(\alpha) > 0$, which is automatically satisfied as α is a Kähler class. Define the functional $\tilde{l}_\alpha(\cdot) := \frac{l_\alpha(\cdot)}{l_\alpha(\alpha)}$. Since $-K$ is effective and α is Kähler,

$$\tilde{l}_\alpha(-K) = \frac{l_\alpha(-K)}{l_\alpha(\alpha)} = \frac{\alpha \cdot (-K)}{\alpha^2} > 0.$$

We must then have

$$I(\alpha) = \inf_{l \in \mathcal{K}^\vee} \left(\frac{l(-K)}{l(\alpha)} \right) - 2\tilde{l}_\alpha(-K) \leq \tilde{l}_\alpha(-K) - 2\tilde{l}_\alpha(-K) = -\tilde{l}_\alpha(-K) < 0$$

Hence we conclude for any smooth complete toric surface that $I(\alpha) \leq 0$. $\square$

In this setting we can also obtain information about the decomposition of the Kähler cone with respect to the torus invariant divisors D_i . Namely a pointed, strictly convex cone $C \subset V$ induces a partial ordering $\leq_C$ on V . In our case consider the Kähler cone $\mathcal{K}(X_\Sigma)$, the partial ordering $\leq_{\mathcal{K}}$ on $\text{Pic}(X_\Sigma)_{\mathbb{R}} \simeq \mathbb{R}^{r-2}$ is then given by,

$$\gamma \leq_{\mathcal{K}} \omega \iff \omega - \gamma \in \mathcal{K}.$$

We propose that

$$\inf_{l \in \mathcal{K}^\vee} \left(\frac{l(\gamma)}{l(\omega)} \right) = \sup(\rho > 0 \mid \rho\omega \leq_{\mathcal{K}} \gamma).$$

Suppose that $\rho > 0$ and $\rho\omega \leq_{\mathcal{K}} \gamma$, then

$$\gamma - \rho\omega \in \mathcal{K} \iff l_{D_i}(\gamma - \rho\omega) > 0, \forall i \iff l_{D_i}(\gamma) - \rho l_{D_i}(\omega) > 0, \forall i \iff \frac{l_{D_i}(\gamma)}{l_{D_i}(\omega)} > \rho, \forall i,$$

hence

$$\inf_{l \in \mathcal{K}^\vee} \left(\frac{l(\gamma)}{l(\omega)} \right) > \rho \implies \inf_{l \in \mathcal{K}^\vee} \left(\frac{l(\gamma)}{l(\omega)} \right) > \sup(\rho > 0 \mid \rho\omega \leq_{\mathcal{K}} \gamma).$$

Conversely suppose that $\inf_{l \in \mathcal{K}^\vee} \left(\frac{l(\gamma)}{l(\omega)} \right) > \rho$ for some $\rho \in \mathbb{R}_{>0}$, then

$$\frac{l(\gamma)}{l(\omega)} > \rho, \forall l \in \mathcal{K}^\vee \iff l(\gamma) > \rho l(\omega), \forall l \in \mathcal{K}^\vee \iff l(\gamma - \rho\omega) > 0, \forall l \in \mathcal{K}^\vee.$$

Since $\mathcal{K}^\vee$ is generated by the l_{D_i} , we conclude that $l_{D_i}(\gamma - \rho\omega) > 0, \forall i$, hence $\gamma - \rho\omega \in \mathcal{K}$ and then we conclude $\rho\omega \leq_{\mathcal{K}} \gamma$. Therefore,

$$\inf_{l \in \mathcal{K}^\vee} \left(\frac{l(\gamma)}{l(\omega)} \right) < \sup(\rho > 0 \mid \rho\omega \leq_{\mathcal{K}} \gamma),$$

and in total we have

$$\inf_{l \in \mathcal{K}^\vee} \left(\frac{l(\gamma)}{l(\omega)} \right) = \sup(\rho > 0 \mid \rho\omega \leq_{\mathcal{K}} \gamma).$$

We can then rewrite $I(\alpha)$,

$$\begin{aligned} I(\alpha) &= \inf_{l \in \mathcal{K}^\vee} \left(\frac{l(-K)}{l(\alpha)} \right) - 2\tilde{l}_\alpha(-K) \\ &= \sup(\rho > 0 \mid \rho\alpha \leq_{\mathcal{K}} (-K)) - 2\tilde{l}_\alpha(-K). \end{aligned}$$

Where one can see that the decomposition is then governed by the coefficients of the anticanonical divisor $-K$.

Example 5.6. Consider the same example as done in Subsection Figure 5.2, the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 0 : 1]$. In this example we have $\text{Pic}(X_\Sigma)_\mathbb{R} \simeq \mathbb{R}^3$, and the anticanonical divisor is given by

$$-K = 2g_0 + 2g_1 + 1g_2,$$

and after rescaling, to follow the same procedure as we have done so far, we get

$$-K = 1g_0 + 1g_1 + \frac{1}{2}g_2.$$

Let $\alpha \in \mathcal{K}(X_\Sigma)$ be given by $\alpha = ag_0 + bg_1 + cg_2$. Then

$$(-K) - \rho\alpha = (1 - \rho a)g_0 + (1 - \rho b)g_1 + \left(\frac{1}{2} - \rho c\right)g_2.$$

The criteria on the coefficients for a divisor to lie within the Kähler cone are $a, b, c > 0$, $b > c$ and $a + c > b$. Hence for $((-K) - \rho\alpha)$ to be Kähler we must have,

$$\begin{array}{llll} a > 0 : & 1 - \rho a > 0 & \implies & \frac{1}{a} > \rho \\ b > 0 : & 1 - \rho b > c & \implies & \frac{1}{b} > \rho \\ c > 0 : & \frac{1}{2} - \rho > c & \implies & \frac{1}{2c} > \rho \\ b > c : & 1 - \rho b > \frac{1}{2} - \rho c & \implies & \frac{1}{2(b-c)} > \rho \\ a + c > b : & (1 - \rho a) + \left(\frac{1}{2} - \rho c\right) > 1 - \rho b & \implies & \frac{1}{2(a-b+c)} > \rho \end{array}$$

One can then find the supremum by determining the following,

$$\min \left(\frac{1}{a}, \frac{1}{b}, \frac{1}{2c}, \frac{1}{2(b-c)}, \frac{1}{2(a-b+c)} \right)$$

◇

So far we have discussed the special case of $I(\alpha) := J(\alpha, K)$ associated to the cscK problem. If we instead considered $J(\alpha, \beta)$, namely

$$J(\alpha, \beta) = \min_{D_i} \left(2 \frac{\alpha \cdot \beta}{\alpha^2} - \frac{\beta \cdot D_i}{\alpha \cdot D_i} \right).$$

Following similar arguments as we have done in this section, we can then rewrite $J(\alpha, \beta)$ as the following,

$$J(\alpha, \beta) = 2\tilde{l}_\alpha(\beta) - \sup_{l \in \mathcal{K}^\vee} \left(\frac{l(\beta)}{l(\alpha)} \right).$$

Further, one can show that

$$\sup_{l \in \mathcal{K}^\vee} \left(\frac{l(\beta)}{l(\alpha)} \right) = \inf(\rho > 0 \mid \beta \leq_{\mathcal{K}} \rho\alpha).$$

Hence the decomposition of $\mathcal{K}(X_\Sigma)$, with respect to $J(\alpha, \beta)$, is governed by the coefficients of β .

5.3 A Combinatorial Observation

Throughout the development of this thesis, we made an observation, that, to the best of our knowledge, have not yet been documented in the literature.

Unique Fans and Catalan k -fold Convolution

By Definition 2.38 performing a blow-up of a 2-dimensional cone $\sigma = \text{Cone}(v_1, v_2)$, corresponds to adding a ray $v = v_1 + v_2$. This can also be seen as subdividing the region of σ into two regions. The fact the added ray v is exactly the sum of v_1 and v_2 , is explicitly such that we obtain the blow-up by Proposition 2.40. If we are considering the possible number of blow-ups obtained from a starting fan Σ , this addition of the ray v can be thought of as splitting a region perfectly into two. We then noted a relation between subdividing a region into two, and full rooted binary trees, defined in (Meszaros and St. Dizier n.d.).

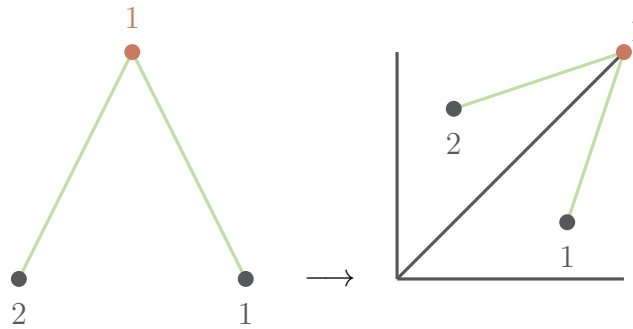
Definition 5.7. A full rooted binary tree consists of vertices and edges, where one of the vertices is the root of the tree. We can define binary trees recursively:

- (i) The empty graph, no vertices or edges, is a binary tree.
- (ii) A nonempty binary tree consists of a root vertex v_0 , a left-subtree T_l , and a right-subtree T_r such that both T_l and T_r are binary trees, and v_0 has an edge to the roots of both.

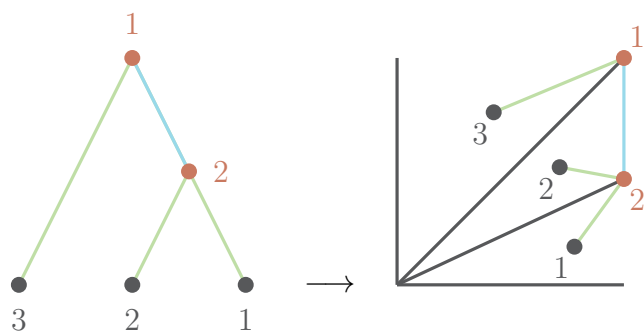
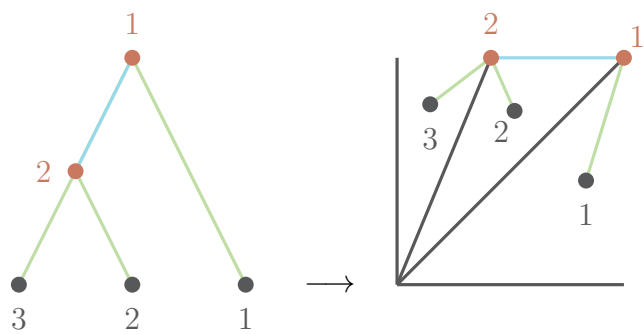
Being full means that every vertex connects via edges to two vertices or connect to no vertices.

It is a known fact that the number of full rooted binary trees on $2n + 1$ vertices are counted by the Catalan numbers C_n . We then observe a correspondance between the full rooted binary trees on $2n + 1$ vertices, and blowing up an area n times. Namely that the roots of the tree corresponds to the regions obtained by blowing up. Furthermore, the structure of the full rooted tree with $n > 1$ has a correspondence with whether the blow-up is performed to the *right* or *left* of the initial blow-up. Consider the following figures for $n = 1, 2, 3$, for a clearer picture of the idea.

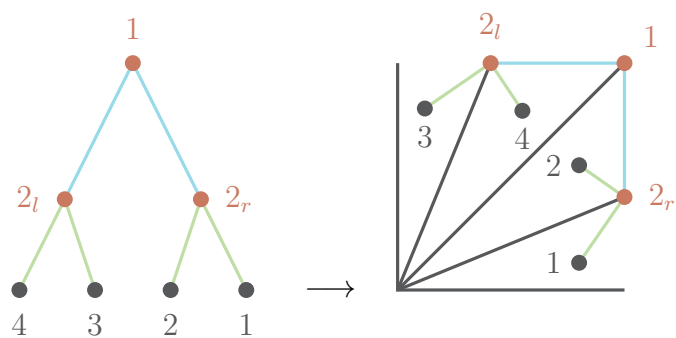
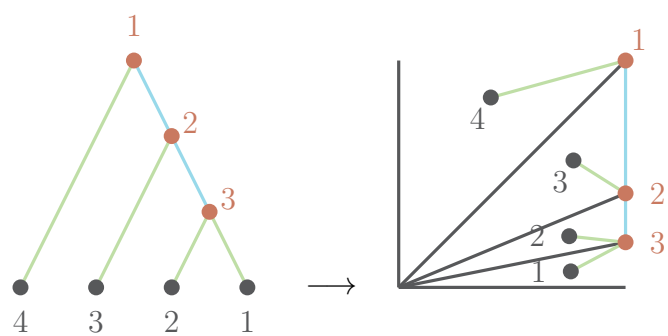
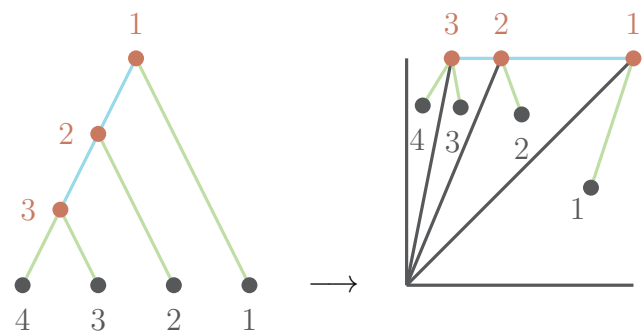
$n = 1$:

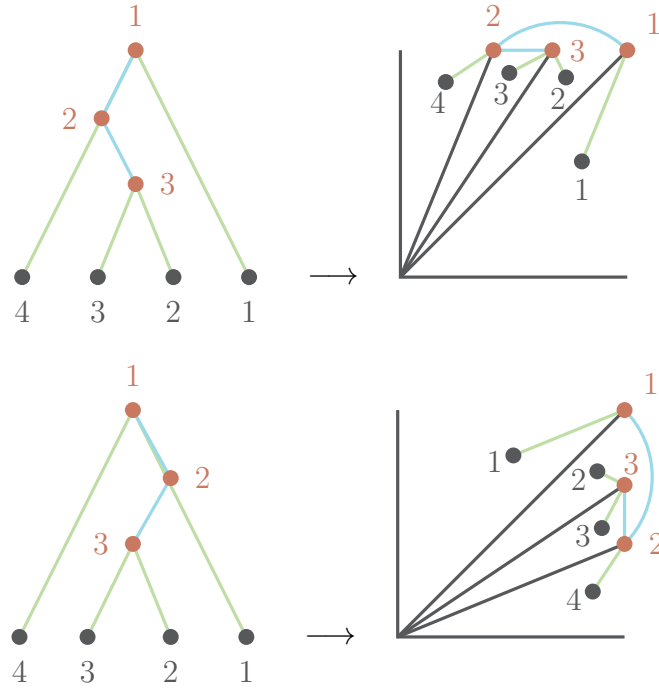


$n = 2$:



$n = 3$:





Note that the possible number of fans obtained by a finite sequence of blow-ups are not determined by the relation of the rays of the fan, but merely the number of rays of the fan. Using the script in Appendix C, we obtained the following number of unique fans for fans consisting of 3 or 4 rays, as by Theorem 2.59 every smooth complete toric surface is obtained by such fans.

# of blow-ups	$\mathbb{P}^2(3 \text{ rays})$	$\mathbb{P}^1 \times \mathbb{P}^1$ or $\mathcal{H}_r(4 \text{ rays})$
0	1	1
1	3	4
2	9	14
3	28	48
4	90	165
5	297	572
6	1.001	2.002
7	3.432	7.072
8	11.934	25.194
9	41.990	90.440
10	149.226	326.876

These sequences of integers seemed familiar, as they are indeed the first terms of the Catalan k -fold convolution,

$$C_n^{(k)} = \frac{k}{2n+k} \cdot \binom{2n+k}{n} = \frac{k}{2n+k} \frac{(2n+k)!}{n!(n+k)!}.$$

This coincide with the scripts results where n is the number of blow-ups performed, and k is the number of 2-dimensional cones of the initial fan Σ .

A Appendix: Detailed Example of Smooth Complete Toric Surface

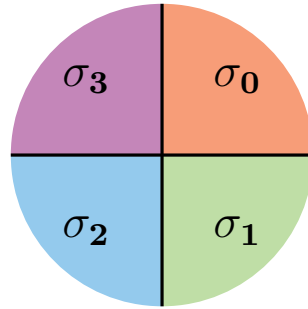
In this appendix we will construct another example of an affine toric variety arising from the theory of cones and fans from Chapter 2. We will construct the affine toric variety $\mathbb{P}^1 \times \mathbb{P}^1$, and leave out the example of the Hirzebruch surfaces $\mathcal{H}_r$ for $r \geq 2$, as Lemma 5.4 effeciently rules out $\mathcal{H}_r$ in our main case of interest.

A.1 $\mathbb{P}^1 \times \mathbb{P}^1$

Let $N \simeq \mathbb{Z}^2$ be a fixed lattice, with dual lattice $M \simeq \mathbb{Z}^2$, such that we have the pair of dual vector spaces respectively $N_{\mathbb{R}} \simeq \mathbb{R}^2$ and $M_{\mathbb{R}} \simeq \mathbb{R}^2$. Let e_1, e_2 and e_1^*, e_2^* be the canonical basis and dual basis. Consider the vectors $v_1 = e_1, v_2 = e_2, v_3 = -e_1$ and $v_4 = -e_2$, we can then create the following cones:

$$\sigma_0 = \text{Cone}(v_1, v_2), \sigma_1 = \text{Cone}(v_1, v_4), \sigma_2 = \text{Cone}(v_4, v_3) \text{ and } \sigma_4 = \text{Cone}(v_3, v_2).$$

The cones can then be visualised as the following:



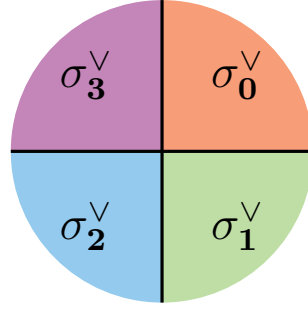
Let $u = (x, y) \in \mathbb{R}^2$, then

$$\langle u, v_1 \rangle = x, \langle u, v_2 \rangle = y, \langle u, v_3 \rangle = -x \text{ and } \langle u, v_4 \rangle = -y.$$

The dual cones of $\sigma_0, \sigma_1, \sigma_2$ and σ_3 are then respectively

$$\begin{aligned} \sigma_0^\vee &= \{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \text{ and } y \geq 0\} = \text{Cone}(e_1^*, e_2^*), \\ \sigma_1^\vee &= \{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \text{ and } y \leq 0\} = \text{Cone}(e_1^*, -e_2^*), \\ \sigma_2^\vee &= \{(x, y) \in \mathbb{R}^2 \mid x \leq 0 \text{ and } y \leq 0\} = \text{Cone}(-e_1^*, -e_2^*), \\ \sigma_3^\vee &= \{(x, y) \in \mathbb{R}^2 \mid x \leq 0 \text{ and } y \geq 0\} = \text{Cone}(-e_1^*, e_2^*). \end{aligned}$$

The dual cones can be visualised as follows:



In this example the cones and their dual cones coincide, but it is worth noting that they live in different spaces, namely $N_{\mathbb{R}}$ and $M_{\mathbb{R}}$.

We will now find the faces of the above cones. Consider $e_1^* \in \sigma_0^\vee$ and let $u = (x, y) \in N_{\mathbb{R}} \simeq \mathbb{R}^2$. Then $\langle e_1, u \rangle = x$, hence

$$H_{e_1} = \{(x, y) \in \mathbb{R}^2 \mid x = 0\}.$$

We therefore obtain

$$H_{e_1} \cap \sigma_0 = \{(x, y) \in \mathbb{R}^2 \mid x = 0 \text{ and } y \geq 0\} = \text{Cone}(v_2)$$

i.e. the positive y -axis. If we instead considered $e_2^* \in \sigma_0^\vee$, we get $\langle e_2, u \rangle = y$, and obtain

$$H_{e_2} \cap \sigma_0 = \{(x, y) \in \mathbb{R}^2 \mid y = 0 \text{ and } x \geq 0\} = \text{Cone}(v_1)$$

i.e. the positive x -axis. Likewise, doing the same computations for the generators of the other cones σ_1, σ_2 and σ_3 , we obtain

$$\begin{aligned} H_{e_1} \cap \sigma_1 &= \{(x, y) \in \mathbb{R}^2 \mid x = 0 \text{ and } -y \geq 0\} = \text{Cone}(-e_2), \\ H_{-e_2} \cap \sigma_1 &= \{(x, y) \in \mathbb{R}^2 \mid -y = 0 \text{ and } x \geq 0\} = \text{Cone}(e_1), \\ H_{-e_2} \cap \sigma_2 &= \{(x, y) \in \mathbb{R}^2 \mid -y = 0 \text{ and } -x \geq 0\} = \text{Cone}(-e_1), \\ H_{-e_1} \cap \sigma_2 &= \{(x, y) \in \mathbb{R}^2 \mid -x = 0 \text{ and } -y \geq 0\} = \text{Cone}(-e_2), \\ H_{-e_1} \cap \sigma_3 &= \{(x, y) \in \mathbb{R}^2 \mid -x = 0 \text{ and } y \geq 0\} = \text{Cone}(e_2), \\ H_{e_2} \cap \sigma_3 &= \{(x, y) \in \mathbb{R}^2 \mid y = 0 \text{ and } -x \geq 0\} = \text{Cone}(-e_1). \end{aligned}$$

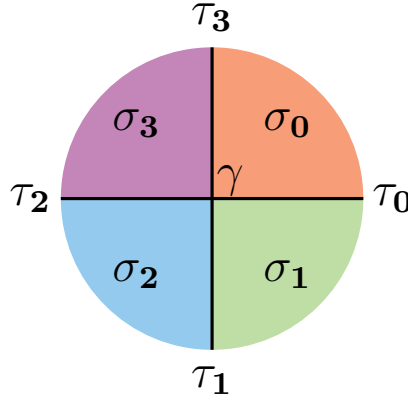
We note again that the faces of the cones can be seen as the lines separating the cones. We note $\sigma_0, \sigma_1, \sigma_2, \sigma_3$ are strongly convex, rational, smooth and simplicial according to the definitions and discussion in Section 2.1.

Let $\gamma = \{0\}$, $\tau_0 = \text{Cone}(v_1)$, $\tau_1 = \text{Cone}(v_2)$, $\tau_2 = \text{Cone}(v_3)$, $\tau_3 = \text{Cone}(v_4)$, the origin and the faces of σ_i . Consider the collection of the mentioned cones, and denote it by Σ . We wish to show it is a fan according to Definition 2.17.

(i) By the note of Definition 2.14 every cone in Σ is rational.

(ii) Every face of σ_i is described by a τ_j , and γ is a face of every cone in Σ .

(iii) Consider the figure



Here we can see that (iii) is satisfied. For example consider $\sigma_0 \cap \sigma_1$, from the figure we can see that $\sigma_0 \cap \sigma_1 = \tau_0$.

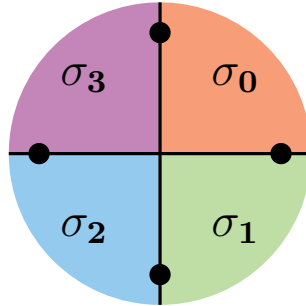
Recall that for a rational polyhedral cone $\sigma \subseteq N_{\mathbb{R}}$, the lattice points given by

$$S_{\sigma} = \sigma^{\vee} \cap M \subseteq M$$

forms a semigroup. Hence for our maximal cones $\sigma_i, i = 0, 1, 2, 3$ in Σ , we obtain

$$\begin{aligned} S_{\sigma_0} &= \sigma_0^{\vee} \cap M = \{(x, y) \in \mathbb{Z}^2 \mid x \geq 0 \text{ and } y \geq 0\} \\ S_{\sigma_1} &= \sigma_1^{\vee} \cap M = \{(x, y) \in \mathbb{Z}^2 \mid x \geq 0 \text{ and } -y \geq 0\} \\ S_{\sigma_2} &= \sigma_2^{\vee} \cap M = \{(x, y) \in \mathbb{Z}^2 \mid -y \geq 0 \text{ and } -x \geq 0\} \\ S_{\sigma_3} &= \sigma_3^{\vee} \cap M = \{(x, y) \in \mathbb{Z}^2 \mid -x \geq 0 \text{ and } y \geq 0\}. \end{aligned}$$

Considering the figure below, we can easily describe the generators of the semigroups.



The generators of S_{σ_0} are e_1^* and e_2^* , S_{σ_1} has generators e_1^* and $-e_2^*$, S_{σ_2} has generators $-e_1^*$ and $-e_2^*$, lastly S_{σ_3} has generators $-e_1^*$ and e_2^* . By Theorem 2.20 the following are affine varieties

$$\begin{aligned} U_{\sigma_0} &= \text{Spec}(\mathbb{C}[S_{\sigma_0}]) \\ U_{\sigma_1} &= \text{Spec}(\mathbb{C}[S_{\sigma_1}]) \\ U_{\sigma_2} &= \text{Spec}(\mathbb{C}[S_{\sigma_2}]) \\ U_{\sigma_3} &= \text{Spec}(\mathbb{C}[S_{\sigma_3}]). \end{aligned}$$

The torus of U_{σ_i} is $T_N = (\mathbb{C}^*)^2$. As in the Ongoing Example in Chapter 2, we can elaborate on the semigroups. We have the character lattice $M \simeq \mathbb{Z}^2$ of $(\mathbb{C}^*)^2$, then a character

$m \in M$ gives a character χ^m . We have the description $S_{\sigma_0} = \mathbb{N}\mathcal{A}$ for $\mathcal{A} = \{e_1^*, e_2^*\} \subseteq S_{\sigma_0}$. We then get that $\mathbb{C}[S_{\sigma_0}] = \mathbb{C}[\chi^{e_1^*}, \chi^{e_2^*}]$, setting $x_i = \chi^{e_i^*}$ we obtain the polynomial ring $\mathbb{C}[x_1, x_2]$. For the rest of the maximal cones, through similar descriptions, we obtain the polynomial rings $\mathbb{C}[x_1, x_2^{-1}]$, $\mathbb{C}[x_1^{-1}, x_2^{-1}]$ and $\mathbb{C}[x_1^{-1}, x_2]$ for $S_{\sigma_1}, S_{\sigma_2}$ and S_{σ_3} respectively.

Consider $x_1 \in \mathbb{C}[x_1, x_2]$ and let

$$(U_{\sigma_0})_{x_1} = \{p \in U_{\sigma_0} \mid x_1(p) \neq 0\} \subseteq U_{\sigma_0}.$$

For each of the variables of the above defined polynomial rings, let the following be defined as above;

$$(U_{\sigma_0})_{x_2}, (U_{\sigma_1})_{x_1}, (U_{\sigma_1})_{x_2^{-1}}, (U_{\sigma_2})_{x_1^{-1}}, (U_{\sigma_2})_{x_2^{-1}}, (U_{\sigma_3})_{x_1^{-1}}, (U_{\sigma_3})_{x_2}.$$

From the discussion Section 2.1, the above are Zariski open. If U_0 and U_1 are the standard affine open sets in $\mathbb{P}^1$, then $U_{\sigma_0} \simeq U_0 \times U_0$, $U_{\sigma_1} \simeq U_0 \times U_1$, $U_{\sigma_2} \simeq U_1 \times U_1$ and $U_{\sigma_3} \simeq U_1 \times U_0$. Denote by X_Σ the toric variety associated to the fan Σ , then the gluing makes $X_\Sigma \simeq \mathbb{P}^1 \times \mathbb{P}^1$.

Consider the before mentioned action of $(t_0, t_1) \in (\mathbb{C}^*)^2$ on $\mathbb{P}^1 \times \mathbb{P}^1$ with homogeneous coordinates $([x_0 : x_1], [y_0 : y_1])$ given by;

$$(t_0, t_1) \cdot ([x_0 : x_1], [y_0 : y_1]) = ([x_0 : t_0 x_1], [y_0 : t_1 y_1]).$$

We can clearly see that $([1 : 0], [1 : 0])$ is a fixed point. Since we are working with homogeneous coordinates, we can rescale with either t_0 or t_1 , and observe that the remaining fixed points must be

$$\begin{aligned} ([1 : 0], [0 : t_1]) &\sim ([1 : 0], [0 : 1]) \\ ([0 : t_0], [1 : 0]) &\sim ([0 : 1], [1 : 0]) \\ ([0 : t_0], [0 : t_1]) &\sim ([0 : 1], [0 : 1]). \end{aligned}$$

As in the Ongoing Example in Chapter 2, we can describe the orbits of the action by the combinations of the coordinates $x_0, x_1, y_0, y_1 = 0$ or $x_0, x_1, y_0, y_1 \neq 0$, taking in respect that $[x_0 : x_1]$ is in the first $\mathbb{P}^1$ and $[y_0 : y_1]$ is in the second $\mathbb{P}^1$. In other words we have the orbits;

$$\begin{aligned} O_1 &= \{([x_0 : x_1], [y_0 : y_1]) \in \mathbb{P}^1 \times \mathbb{P}^1 \mid x_0, x_1, y_0, y_1 \neq 0\} \ni ([1 : 1], [1 : 1]) \\ O_2 &= \{([x_0 : x_1], [y_0 : y_1]) \in \mathbb{P}^1 \times \mathbb{P}^1 \mid x_0 = 0, x_1, y_0, y_1 \neq 0\} \ni ([0 : 1], [1 : 1]) \\ O_3 &= \{([x_0 : x_1], [y_0 : y_1]) \in \mathbb{P}^1 \times \mathbb{P}^1 \mid x_1 = 0, x_0, y_0, y_1 \neq 0\} \ni ([1 : 0], [1 : 1]) \\ O_4 &= \{([x_0 : x_1], [y_0 : y_1]) \in \mathbb{P}^1 \times \mathbb{P}^1 \mid y_0 = 0, y_1, x_0, x_1 \neq 0\} \ni ([1 : 1], [0 : 1]) \\ O_5 &= \{([x_0 : x_1], [y_0 : y_1]) \in \mathbb{P}^1 \times \mathbb{P}^1 \mid y_1 = 0, y_0, x_0, x_1 \neq 0\} \ni ([1 : 1], [1 : 0]) \\ O_6 &= \{([x_0 : x_1], [y_0 : y_1]) \in \mathbb{P}^1 \times \mathbb{P}^1 \mid x_0, y_0 = 0, x_1, y_1 \neq 0\} \ni ([0 : 1], [0 : 1]) \\ O_7 &= \{([x_0 : x_1], [y_0 : y_1]) \in \mathbb{P}^1 \times \mathbb{P}^1 \mid x_0, y_1 = 0, x_1, y_0 \neq 0\} \ni ([0 : 1], [1 : 0]) \\ O_8 &= \{([x_0 : x_1], [y_0 : y_1]) \in \mathbb{P}^1 \times \mathbb{P}^1 \mid x_1, y_0 = 0, x_0, y_1 \neq 0\} \ni ([1 : 0], [0 : 1]) \\ O_9 &= \{([x_0 : x_1], [y_0 : y_1]) \in \mathbb{P}^1 \times \mathbb{P}^1 \mid x_1, y_1 = 0, x_0, y_0 \neq 0\} \ni ([1 : 0], [1 : 0]). \end{aligned}$$

Where we note that the last 4 orbits actually contains the fixed points of the action.

We now view the torus slightly different from before. The torus $(\mathbb{C}^*)^2 \subseteq \mathbb{P}^1 \times \mathbb{P}^1$ consists of points $([1 : s], [1 : t]), s, t \neq 0$. If we consider a point $(a, b) \in \mathbb{Z}^2$, this yields a curve in $\mathbb{P}^1 \times \mathbb{P}^1$, as in the Ongoing Example in Chapter 2, truly a curve in $(\mathbb{C}^*)^2$, but we view it as a curve in $\mathbb{P}^1 \times \mathbb{P}^1$:

$$\lambda^u(t) = ([1 : t^a], [1 : t^b]).$$

Consider the limit of $\lambda^u(t)$ as $t \rightarrow 0$. Clearly the limit points depend on (a, b) . If $a, b > 0$ we obtain

$$\lim_{t \rightarrow 0} ([1 : t^a], [1 : t^b]) = ([1 : 0], [1 : 0]),$$

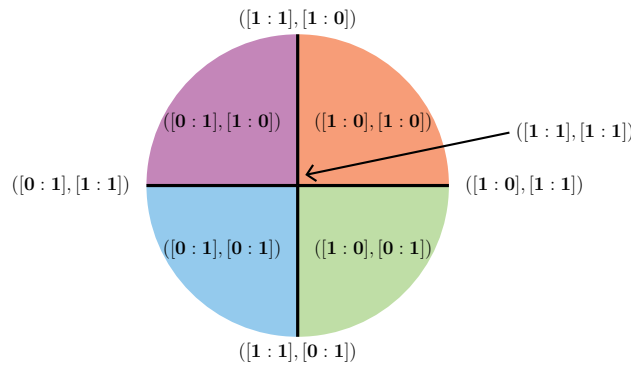
i.e. one of the fixed points that correspond to an orbit. Considering the fan Σ , one would suspect the other possible combinations of a and b to yield the other limit points, as we are working with homogeneous coordinates. If we instead considered (a, b) with $a > 0$ and $b < 0$, we would obtain;

$$\lim_{t \rightarrow 0} ([1 : t^a], [1 : t^b]) \sim ([1 : 0], [t^{-b} : 1]) = ([1 : 0], [0 : 1]).$$

The other combinations would yield:

$$\begin{aligned} a < 0, b > 0 : \lim_{t \rightarrow 0} ([1 : t^a], [1 : t^b]) &\sim ([t^{-a} : 1], [1 : 0]) = ([0 : 1], [1 : 0]), \\ a, b < 0 : \lim_{t \rightarrow 0} ([1 : t^a], [1 : t^b]) &\sim ([t^{-a} : 1], [t^{-b} : 1]) = ([0 : 1], [0 : 1]), \\ a = 0, b > 0 : \lim_{t \rightarrow 0} ([1 : t^a], [1 : t^b]) &= ([1 : 1], [1 : 0]), \\ a = 0, b < 0 : \lim_{t \rightarrow 0} ([1 : t^a], [1 : t^b]) &\sim ([1 : 1], [t^{-b} : 1]) = ([1 : 1], [0 : 1]), \\ b = 0, a > 0 : \lim_{t \rightarrow 0} ([1 : t^a], [1 : t^b]) &= ([1 : 0], [1 : 1]), \\ b = 0, a < 0 : \lim_{t \rightarrow 0} ([1 : t^a], [1 : t^b]) &\sim ([t^{-a} : 1], [1 : 1]) = ([0 : 1], [1 : 1]), \\ a, b = 0 : \lim_{t \rightarrow 0} ([1 : t^a], [1 : t^b]) &= ([1 : 1], [1 : 1]). \end{aligned}$$

We then observe that the above conditions on (a, b) actually correspond to exactly the cones of the fan Σ .



We now wish to construct the blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ at the distinguished point $([1 : 0], [1 : 0])$, following Definition 2.40. The distinguished point corresponds to the cone $\sigma_0 = \text{Cone}(v_1, v_2)$,

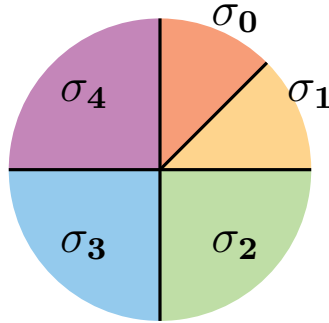
hence we let $v_5 = e_1 + e_2$. Furthermore, let $\Sigma'(\sigma_0)$ denote the set of all cones generated by subsets of $\{v_5, v_1, v_2\}$ not containing $\{v_1, v_2\}$, i.e. the set

$$\Sigma'(\sigma_0) = \{\text{Cone}(v_5), \text{Cone}(v_1), \text{Cone}(v_2), \text{Cone}(v_5, v_1), \text{Cone}(v_5, v_2), \text{Cone}(\emptyset)\}.$$

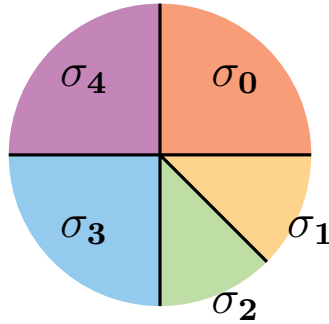
Consider

$$\begin{aligned} \Sigma^*(\sigma_0) &= (\Sigma \setminus \{\sigma_0\}) \cup \Sigma'(\sigma_0) \\ &= \{\text{Cone}(v_1), \text{Cone}(v_2), \text{Cone}(v_3), \text{Cone}(v_4), \text{Cone}(v_5), \text{Cone}(v_1, v_5) \\ &\quad \text{Cone}(v_2, v_5), \text{Cone}(v_1, v_4), \text{Cone}(v_3, v_4), \text{Cone}(v_3, v_2), \text{Cone}(\emptyset)\}. \end{aligned}$$

This is the star subdivision of Σ along the distinguished point $([1 : 0], [1 : 0])$. The new fan $\Sigma^*(\sigma_0)$ is then



As in the Ongoing Example in Chapter 2, we could have done the procedure along one of the other distinguished points. For example along the point $([1 : 0], [0 : 1])$ corresponding to the cone $\sigma_1 \in \Sigma$, would yield the fan;

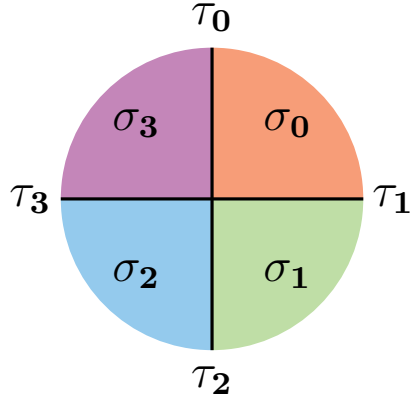


The distinction between the fans, becomes clear when we consider the divisors of the fan. Recall that each 1-dimensional cone of the fan corresponds to an orbit through the Orbit-Cone correspondence Theorem 2.30, and hence a 1-dimensional toric variety, which is a toric variety of codimension 1. Therefore we have four divisors, one for each ray of the fan Σ . We now wish to describe the class group $\text{Cl}(X_\Sigma)$ through the divisors of the characters. The first map of Theorem 2.48 is given by $m \mapsto \text{div}(\chi^m)$, and we have the short exact sequence

$$0 \rightarrow M \simeq \mathbb{Z}^2 \xrightarrow{f} \text{div}_{T_N}(X_\Sigma) \xrightarrow{g} \text{Cl}(X_\Sigma) \rightarrow 0.$$

This gives us $\text{Cl}(X_\Sigma) \simeq \text{Div}_{T_N}(X_\Sigma)/\text{im}(f) \simeq \text{Div}_{T_N}(X_\Sigma)/\text{ker}(g)$.

For the sake of clarity, we will relabel the cones to



and the vectors defining them to $v_0 = e_2, v_1 = e_1, v_2 = -e_2$ and $v_3 = -e_1$.

Then $\text{Cl}(X_\Sigma)$ is generated by classes of the divisors D_0, D_1, D_2, D_3 corresponding to $\tau_0, \tau_1, \tau_2, \tau_3 \in \Sigma(1)$, subject to the relations

$$\begin{aligned} 0 \sim \text{div}(\chi^{e_1}) &= \langle e_1, v_0 \rangle D_0 + \langle e_1, v_1 \rangle D_1 + \langle e_1, v_2 \rangle D_2 + \langle e_1, v_3 \rangle D_3 \\ &= D_1 - D_3, \\ 0 \sim \text{div}(\chi^{e_2}) &= \langle e_2, v_0 \rangle D_0 + \langle e_2, v_1 \rangle D_1 + \langle e_2, v_2 \rangle D_2 + \langle e_2, v_3 \rangle D_3 \\ &= D_0 - D_2. \end{aligned}$$

Thus the class group has generators $[D_1] = [D_3]$ and $[D_0] = [D_2]$, and hence $\text{Cl}(X_\Sigma) \simeq \mathbb{Z}^2$. As every cone of Σ are smooth, Σ is smooth, hence X_Σ is smooth and $\text{Cl}(X_\Sigma) = \text{Pic}(X_\Sigma)$. Following the procedure of Equation 2.56 and Theorem 2.57, we have the following equations;

$$\begin{aligned} v_0 + v_2 &= e_2 + (-e_2) = 0 \cdot e_1 = 0 \cdot v_1, \\ v_1 + v_3 &= e_1 + (-e_1) = 0 \cdot (-e_2) = 0 \cdot v_2, \\ v_2 + v_0 &= -e_2 + e_2 = 0 \cdot (-e_1) = 0 \cdot v_3, \\ v_3 + v_1 &= -e_1 + e_1 = 0 \cdot e_2 = 0 \cdot v_0. \end{aligned}$$

Furthermore;

$$\begin{aligned} \tau_0 &= \text{Cone}(v_0) = \text{Cone}(v_0, v_1) \cap \text{Cone}(v_0, v_3) = \sigma_0 \cap \sigma_3, \\ \tau_1 &= \text{Cone}(v_1) = \text{Cone}(v_1, v_0) \cap \text{Cone}(v_1, v_2) = \sigma_0 \cap \sigma_1, \\ \tau_2 &= \text{Cone}(v_2) = \text{Cone}(v_2, v_1) \cap \text{Cone}(v_2, v_3) = \sigma_1 \cap \sigma_2, \\ \tau_3 &= \text{Cone}(v_3) = \text{Cone}(v_3, v_2) \cap \text{Cone}(v_3, v_0) = \sigma_2 \cap \sigma_3. \end{aligned}$$

These relations then gives us the intersection matrix;

	D_0	D_1	D_2	D_3
D_0	0	1	0	1
D_1	1	0	1	0
D_2	0	1	0	1
D_3	1	0	1	0

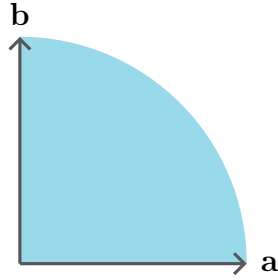
To describe the Nef and Kähler cone of $\mathbb{P}^1 \times \mathbb{P}^1$, we consider the description

$$\text{Pic}(X_\Sigma) = \{a[D_0] + b[D_1] \mid a, b \in \mathbb{Z}\},$$

coming from the relations A.1. We further need to check for nefness against the generators, i.e. let $D = a[D_0] + b[D_1] \in \text{Pic}(X_\Sigma)$, then

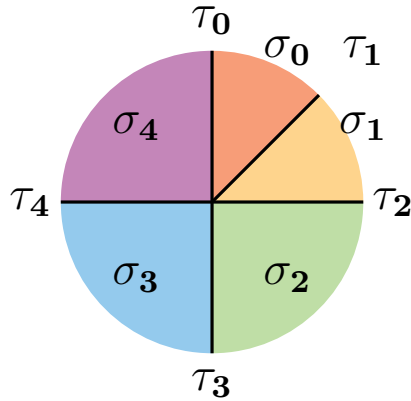
$$\begin{aligned} D \cdot [D_0] &= (a[D_0] + b[D_1]) \cdot [D_0] = a[D_0]^2 + b[D_0] \cdot [D_1] = b \geq 0, \\ D \cdot [D_1] &= (a[D_0] + b[D_1]) \cdot [D_1] = a[D_0] \cdot [D_1] + b[D_1]^2 = a \geq 0. \end{aligned}$$

Which is visualised as the first quadrant,



hence the Kähler cone is the open first quadrant.

What if we considered the blow-up along one of the distinguished points? Consider the blow-up along the point $([1 : 0], [1 : 0])$, for clarity we denote the cones of the fan as the following;



and the vectors defining them as $v_0 = e_2, v_1 = e_1 + e_2, v_2 = e_1, v_3 = -e_2$ and $v_4 = -e_1$. Following the same procedure as before, yields us the relations;

$$\begin{aligned} 0 \sim \text{div}(\chi^{e_1}) &= \sum_{i=0}^4 \langle e_1, v_i \rangle D_i \\ &= D_1 + D_2 - D_4, \\ 0 \sim \text{div}(\chi^{e_2}) &= \sum_{i=0}^4 \langle e_2, v_i \rangle D_i \\ &= D_0 + D_1 - D_3. \end{aligned}$$

Thus the Picard group has generators $[D_1], [D_0] = [D_2]$ and $[D_3] = [D_4]$.

$$\begin{aligned}
v_0 + v_2 &= e_2 + e_1 = 1 \cdot (e_1 + e_2) = 1 \cdot v_1, \\
v_1 + v_3 &= (e_1 + e_2) + (-e_2) = 1 \cdot e_1 = 1 \cdot v_2, \\
v_2 + v_4 &= e_1 + (-e_1) = 0 \cdot (-e_2) = 0 \cdot v_3, \\
v_3 + v_0 &= (-e_2) + e_2 = 0 \cdot (-e_1) = 0 \cdot v_4, \\
v_4 + v_1 &= (-e_1) + (e_1 + e_2) = 1 \cdot e_2 = 1 \cdot v_0.
\end{aligned}$$

Then since

$$\begin{aligned}
\tau_0 &= \text{Cone}(v_0) = \text{Cone}(v_0, v_1) \cap \text{Cone}(v_0, v_4) = \sigma_0 \cap \sigma_4, \\
\tau_1 &= \text{Cone}(v_1) = \text{Cone}(v_1, v_0) \cap \text{Cone}(v_1, v_2) = \sigma_0 \cap \sigma_1, \\
\tau_2 &= \text{Cone}(v_2) = \text{Cone}(v_2, v_1) \cap \text{Cone}(v_2, v_3) = \sigma_1 \cap \sigma_2, \\
\tau_3 &= \text{Cone}(v_3) = \text{Cone}(v_3, v_2) \cap \text{Cone}(v_3, v_4) = \sigma_2 \cap \sigma_3, \\
\tau_4 &= \text{Cone}(v_4) = \text{Cone}(v_4, v_3) \cap \text{Cone}(v_4, v_0) = \sigma_3 \cap \sigma_4,
\end{aligned}$$

we obtain the intersection matrix

	D_0	D_1	D_2	D_3	D_4
D_0	-1	1	0	0	1
D_1	1	-1	1	0	0
D_2	0	1	-1	1	0
D_3	0	0	1	0	1
D_4	1	0	0	1	0

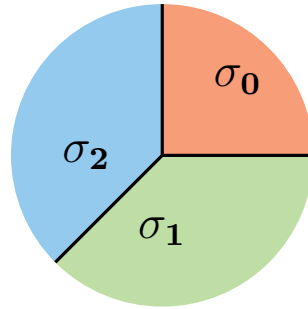
One could consider blowing up the other distinguished points, which would yield a similar, but noteworthy quite different intersection matrix.

B Appendix: Complete List of Exceptional Cases for Theorem 5.1

B.1 Exceptional Cases of $\mathbb{P}^2$ and their Wall-Chamber Decomposition

$\mathbb{P}^2$

Fan



Intersection Matrix.

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|ccc} & D_0 & D_1 & D_2 \\ \hline D_0 & 1 & 1 & 1 \\ D_1 & 1 & 1 & 1 \\ D_2 & 1 & 1 & 1 \end{array}.$$

Picard Group Generators.

The generator of the Picard group is

$$g_0 = [D_2].$$

Kähler Cone Generators.

The generator of the Kähler cone is

$$\tilde{g}_0 = g_0.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 \\ &= x_0 g_0 \end{aligned}$$

gives $x_0 = 1$. Hence any divisor

$$\alpha = ag_0 \in \text{Pic}(X_\Sigma), \quad a \in \mathbb{R},$$

can be written as

$$\alpha = a\tilde{g}_0.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a > 0.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [aD_2], \quad \alpha' = [0, 0, a].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^2 D_i = [3D_2] = 3g_0.$$

After rescaling, we set

$$-K' = [0, 0, 1].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [a, a, a] \cdot (\alpha')^T \\ &= a^2, \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= a. \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{2a}{a^2} = \frac{2}{a} > 0.$$

Intersections with Torus Divisors.

For each $i = 0, 1, 2$,

$$\begin{aligned} (-K) \cdot D_i &= (-K') \cdot IM \cdot D_i^T \\ &= 1, \end{aligned}$$

$$\begin{aligned} \alpha \cdot D_i &= \alpha' \cdot IM \cdot D_i^T \\ &= a. \end{aligned}$$

Hence,

$$\frac{(-K).D_i}{\alpha.D_i} = \frac{1}{a} > 0.$$

Numerical Invariant

We consider the intersection numbers

$$2\frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\}.$$

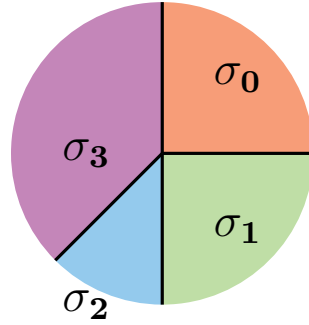
In this case, all divisors D_0, D_1, D_2 give identical values, so for each i we obtain

$$2\frac{\alpha.(-K)}{\alpha^2} - \frac{(-K).D_i}{\alpha.D_i} = \frac{2}{a} - \frac{1}{a} = \frac{1}{a} > 0.$$

Conclusion.

Thus, in every region where D_i is optimizing, we obtain

$$- \left(2\frac{\alpha.(-K)}{\alpha^2} - \frac{(-K).D_i}{\alpha.D_i} \right) < 0, \quad i = 0, 1, 2.$$

Blow-up of $\mathbb{P}^2$ along the point $[0 : 0 : 1]$ **Fan****Intersection Matrix.**

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|cccc} & D_0 & D_1 & D_2 & D_3 \\ \hline D_0 & 1 & 1 & 0 & 1 \\ D_1 & 1 & 0 & 1 & 0 \\ D_2 & 0 & 1 & -1 & 1 \\ D_3 & 1 & 0 & 1 & 0 \end{array}$$

Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2 + D_3] \text{ and } g_1 = [D_3].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_1 \text{ and } \tilde{g}_1 = g_0.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 \\ &= x_1 g_0 + x_0 g_1, \end{aligned}$$

$$\begin{aligned} g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 \\ &= y_1 g_0 + y_0 g_1, \end{aligned}$$

gives

$$\begin{aligned} x_0 &= 0, & x_1 &= 1, \\ y_0 &= 1, & y_1 &= 0. \end{aligned}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 \in \text{Pic}(X_\Sigma), \quad a, b \in \mathbb{R},$$

can be written as

$$\alpha = b\tilde{g}_0 + a\tilde{g}_1.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b > 0.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [aD_2 + (a+b)D_3], \quad \alpha' = [0, 0, a, a+b].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^3 D_i = [2D_2 + 3D_3] = 2g_0 + g_1.$$

After rescaling, we set

$$-K' = \left[0, 0, 1, \frac{3}{2}\right].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [a+b, a, b, a] \cdot (\alpha')^T \\ &= a^2 + 2ab, \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= \frac{3a+2b}{2}, \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{3a+2b}{a^2+2ab} > 0.$$

Intersections with Torus Divisors.

We further compute,

$$(-K).D_i = (-K') \cdot IM \cdot D_i^T = \begin{cases} \frac{3}{2}, & \text{if } i = 0, \\ 1, & \text{if } i = 1, 3, \\ \frac{1}{2}, & \text{if } i = 2. \end{cases}$$

$$\alpha.D_i = \alpha' \cdot IM \cdot D_i^T = \begin{cases} a+b, & \text{if } i = 0, \\ a, & \text{if } i = 1, 3, \\ b, & \text{if } i = 2. \end{cases}$$

hence

$$\frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{3}{2(a+b)}, & \text{if } i = 0, \\ \frac{1}{a}, & \text{if } i = 1, 3, \\ \frac{1}{2b}, & \text{if } i = 2. \end{cases}$$

We note that for each i the above is positive.

Numerical Invariant

We consider the intersection numbers

$$2\frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\}.$$

Assume $i = 0$ minimizes, hence

$$\begin{aligned} \frac{3}{2(a+b)} < \frac{1}{a} &\implies a < 2b, \\ \frac{3}{2(a+b)} < \frac{1}{2b} &\implies 2b < a. \end{aligned}$$

where we get a contradiction, hence D_0 is never minimizing.

Assume $i = 1$ minimizes, hence

$$\begin{aligned} \frac{1}{a} < \frac{3}{2(a+b)} &\implies 2b < a, \\ \frac{1}{a} < \frac{1}{2b} &\implies 2b < a. \end{aligned}$$

hence D_1 , and therefore also D_3 , is optimizing under the above restrictions.

Assume $i = 2$ minimizes, hence

$$\begin{aligned} \frac{1}{2b} < \frac{3}{2(a+b)} &\implies 2b > a, \\ \frac{1}{2b} < \frac{1}{a} &\implies 2b > a. \end{aligned}$$

hence D_2 is optimizing under the above restrictions.

We now have 2 areas within $\mathcal{K}(X_\Sigma)$ with optimizing divisors D_2 or D_1, D_3 . Consider now the entire expression. Recall

$$2\frac{\alpha.(-K)}{\alpha^2} - \frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{3a+2b}{a^2+2ab} - \frac{1}{a}, & \text{if } i = 1, 3, \\ \frac{3a+2b}{a^2+2ab} - \frac{1}{2b}, & \text{if } i = 2. \end{cases}$$

Using the fraction rule $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$, we can conclude that the denominator is positive in each case.

Then for $i = 1, 3$, we get

$$\frac{3a+2b}{a^2+2ab} - \frac{1}{a} = \frac{2}{a+2b} > 0.$$

For $i = 2$, we get

$$\frac{3a + 2b}{a^2 + 2ab} - \frac{1}{2b} = \frac{4ab + 4b^2 - a^2}{2b(a^2 + 2ab)}.$$

Consider the numerator, under the restriction $2b > a \implies 4b^2 > a^2$, we conclude the numerator is positive.

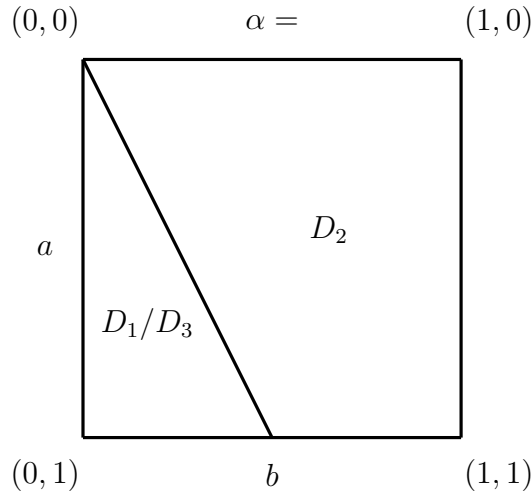
Conclusion.

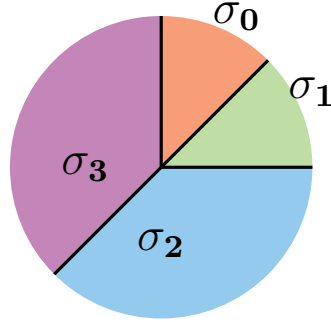
Thus, in every region where D_i is optimizing, we obtain

$$- \left(2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} \right) < 0, \quad i = 1, 2, 3.$$

D_0 will never be optimizing.

Decomposition



Blow-up of $\mathbb{P}^2$ along the point $[1 : 0 : 0]$ **Fan****Intersection Matrix.**

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|cccc} & D_0 & D_1 & D_2 & D_3 \\ \hline D_0 & 0 & 1 & 0 & 1 \\ D_1 & 1 & -1 & 1 & 0 \\ D_2 & 0 & 1 & 0 & 1 \\ D_3 & 1 & 0 & 1 & 1 \end{array}$$

Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2] \text{ and } g_1 = [-D_2 + D_3].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_0 \text{ and } \tilde{g}_1 = g_0 + g_1.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 \\ &= (x_0 + x_1)g_0 + x_1 g_1, \end{aligned}$$

$$\begin{aligned} g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 \\ &= (y_0 + y_1)g_0 + y_1 g_1 \end{aligned}$$

gives

$$\begin{aligned} x_0 &= 1, & x_1 &= 0, \\ y_0 &= -1, & y_1 &= 1. \end{aligned}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 \in \text{Pic}(X_\Sigma), \quad a, b \in \mathbb{R},$$

can be written as

$$\alpha = (a - b)\tilde{g}_0 + b\tilde{g}_1.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b > 0, \quad a > b.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [(a - b)D_2 + bD_3], \quad \alpha' = [0, 0, a - b, b].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^3 D_i = [D_2 + 2D_3] = 3g_0 + 2g_1.$$

After rescaling, we set

$$-K' = \left[0, 0, \frac{1}{3}, \frac{2}{3}\right].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [b, a - b, b, a] \cdot (\alpha')^T \\ &= -b^2 + 2ab, \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= \frac{2a + b}{3}. \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{4a + 2b}{-3b^2 + 6ab} = \frac{4a + 2b}{3b(-b + 2a)} > 0.$$

Intersections with Torus Divisors.

We further compute,

$$(-K) \cdot D_i = (-K') \cdot IM \cdot D_i^T = \begin{cases} \frac{2}{3}, & \text{if } i = 0, 2, \\ \frac{1}{3}, & \text{if } i = 1, \\ 1, & \text{if } i = 3. \end{cases}$$

$$\alpha \cdot D_i = \alpha' \cdot IM \cdot D_i^T = \begin{cases} b, & \text{if } i = 0, 2, \\ a - b, & \text{if } i = 1, \\ a, & \text{if } i = 3. \end{cases}$$

hence

$$\frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{2}{3b}, & \text{if } i = 0, 2, \\ \frac{1}{3(a-b)}, & \text{if } i = 1, \\ \frac{1}{a}, & \text{if } i = 3. \end{cases}$$

We note that for each i the above is positive.

Numerical Invariant

We consider the intersection numbers

$$2\frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\}.$$

Assume $i = 0$ minimizes, hence

$$\begin{aligned} \frac{2}{3b} &< \frac{1}{3(a-b)} \implies 2a < 3b, \\ \frac{2}{3b} &< \frac{1}{a} \implies 2a < 3b, \end{aligned}$$

hence D_0 , and therefore also D_2 , is optimizing under the above restrictions.

Assume $i = 1$ minimizes, hence

$$\begin{aligned} \frac{1}{3(a-b)} &< \frac{2}{3b} \implies 3b < 2a, \\ \frac{1}{3(a-b)} &< \frac{1}{a} \implies 3b < 2a, \end{aligned}$$

hence D_1 is optimizing under the above restrictions.

Assume $i = 3$ minimizes, hence

$$\begin{aligned} \frac{1}{a} &< \frac{1}{3(a-b)} \implies 2a < 3b, \\ \frac{1}{a} &< \frac{2}{3b} \implies 3b < 2a, \end{aligned}$$

which is a contradiction, hence D_3 is never minimizing.

We now have 2 areas within $\mathcal{K}(X_\Sigma)$ with optimizing divisors D_1 or D_0, D_2 . Consider now the entire expression. Recall

$$2\frac{\alpha.(-K)}{\alpha^2} - \frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{4a+2b}{-3b^2+6ab} - \frac{2}{3b}, & \text{if } i = 0, 2, \\ \frac{4a+2b}{-3b^2+6ab} - \frac{1}{3(a-b)}, & \text{if } i = 1. \end{cases}$$

Using the fraction rule $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$, we can conclude that the denominator is positive in each case.

Then for $i = 0, 2$, we get

$$\frac{4a+2b}{-3b^2+6ab} - \frac{2}{3b} = \frac{4}{3(-b+2a)} > 0.$$

For $i = 1$, we get

$$\frac{4a + 2b}{-3b^2 + 6ab} - \frac{1}{3(a - b)} = \frac{4a^2 - 4ab - b^2}{(a - b)(-3b^2 + 6ab)}$$

Consider the numerator, this is a quadratic in a . Then

$$d = 16b^2 - 4 \cdot 4 \cdot (-b^2) = 32b^2,$$

hence the roots are

$$z = \frac{4b \pm \sqrt{32b^2}}{8} = \frac{1 \pm \sqrt{2}}{2}b \approx \begin{cases} 1, 207b, & \text{for } +, \\ -0, 207b, & \text{for } -. \end{cases}$$

Since the leading coefficient $4 > 0$, we know for $a \in (-0, 207b, 1, 207b)$, that $4a^2 - 4ab - b^2 < 0$, and for $a > 1, 207b$, that $4a^2 - 4ab - b^2 > 0$. Since we have the restriction $2a > 3b \implies a > \frac{3}{2}b$, we conclude that the quadratic, hence the entire expression, is always positive.

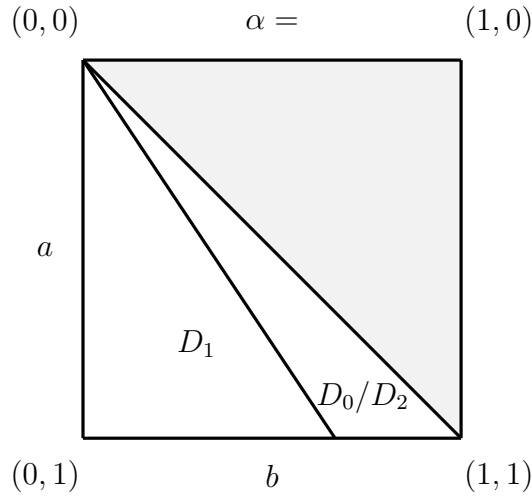
Conclusion.

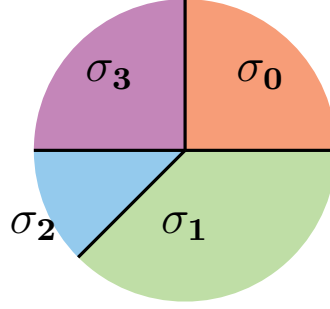
Thus, in every region where D_i is optimizing, we obtain

$$-\left(2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i}\right) < 0, \quad i = 0, 1, 2.$$

D_3 will never be optimizing.

Decomposition



Blow-up of $\mathbb{P}^2$ along the point $[0 : 1 : 0]$ **Fan****Intersection Matrix.**

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|cccc} & D_0 & D_1 & D_2 & D_3 \\ \hline D_0 & 0 & 1 & 0 & 1 \\ D_1 & 1 & 1 & 1 & 0 \\ D_2 & 0 & 1 & 0 & 1 \\ D_3 & 1 & 0 & 1 & -1 \end{array}$$

Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2] \text{ and } g_1 = [D_2 + D_3].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_0 \text{ and } \tilde{g}_1 = g_1.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 \\ &= x_0 g_0 + x_1 g_1, \end{aligned}$$

$$\begin{aligned} g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 \\ &= y_0 g_0 + y_1 g_1 \end{aligned}$$

gives

$$\begin{aligned} x_0 &= 1, & x_1 &= 0, \\ y_0 &= 0, & y_1 &= 1. \end{aligned}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 \in \text{Pic}(X_\Sigma), \quad a, b \in \mathbb{R},$$

can be written as

$$\alpha = a\tilde{g}_0 + b\tilde{g}_1.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b > 0.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [(a+b)D_2 + bD_3], \quad \alpha' = [0, 0, a+b, b].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^3 D_i = [3D_2 + 2D_3] = 1g_0 + 2g_1.$$

After rescaling, we set

$$-K' = \left[0, 0, \frac{3}{2}, 1\right].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [b, a+b, b, a] \cdot (\alpha')^T \\ &= b^2 + 2ab, \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= \frac{2a+3b}{2}, \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{2a+3b}{b^2+2ab} > 0.$$

Intersections with Torus Divisors.

We further compute,

$$(-K).D_i = (-K') \cdot IM \cdot D_i^T = \begin{cases} 1, & \text{if } i = 0, 2, \\ \frac{3}{2}, & \text{if } i = 1, \\ \frac{1}{2}, & \text{if } i = 3. \end{cases}$$

$$\alpha.D_i = \alpha' \cdot IM \cdot D_i^T = \begin{cases} b, & \text{if } i = 0, 2, \\ a+b, & \text{if } i = 1, \\ a, & \text{if } i = 3. \end{cases}$$

hence

$$\frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{1}{b}, & \text{if } i = 0, 2, \\ \frac{3}{2(a+b)}, & \text{if } i = 1, \\ \frac{1}{2a}, & \text{if } i = 3. \end{cases}$$

We note that for each i the above is positive.

Numerical Invariant

We consider the intersection numbers

$$2\frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\}.$$

Assume $i = 0, 2$ minimizes, hence

$$\begin{aligned} \frac{1}{b} < \frac{3}{2(a+b)} &\implies 2a < b, \\ \frac{1}{b} < \frac{1}{2a} &\implies 2a < b, \end{aligned}$$

hence D_0, D_2 are optimizing under the above restrictions.

Assume $i = 1$ minimizes, hence

$$\begin{aligned} \frac{3}{2(a+b)} < \frac{1}{b} &\implies b < 2a, \\ \frac{3}{2(a+b)} < \frac{1}{2a} &\implies 2a < b, \end{aligned}$$

which is a contradiction, hence D_1 never optimizing.

Assume $i = 3$ minimizes, hence

$$\begin{aligned} \frac{1}{2a} < \frac{3}{2(a+b)} &\implies b < 2a, \\ \frac{1}{2a} < \frac{1}{b} &\implies b < 2a, \end{aligned}$$

hence D_3 is optimizing under the above restrictions.

We now have 2 areas within $\mathcal{K}(X_\Sigma)$ with optimizing divisors D_3 or D_0, D_2 . Consider now the entire expression. Recall

$$2\frac{\alpha.(-K)}{\alpha^2} - \frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{2a+3b}{b^2+2ab} - \frac{1}{b}, & \text{if } i = 0, 2, \\ \frac{2a+3b}{b^2+2ab} - \frac{1}{2a}, & \text{if } i = 3. \end{cases}$$

Using the fraction rule $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$, we can conclude that the denominator is positive in each case.

Then for $i = 0, 2$, we get

$$\frac{2a+3b}{b^2+2ab} - \frac{1}{b} = \frac{2}{b+2a} > 0.$$

For $i = 3$, we get

$$\frac{2a + 3b}{b^2 + 2ab} - \frac{1}{2a} = \frac{4a^2 + 4ab - b^2}{2a(b^2 + 2ab)} > 0,$$

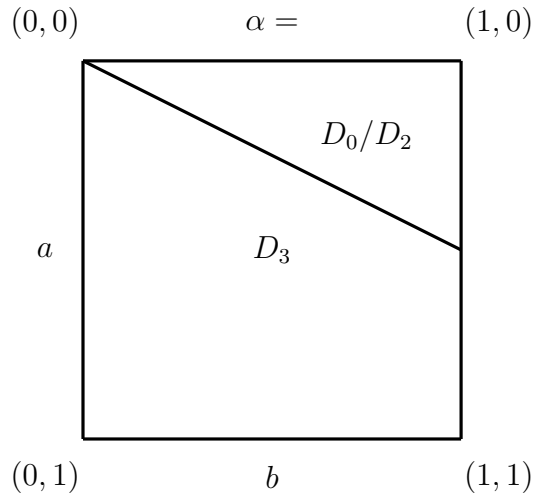
using the restriction $b < 2a \implies b^2 < 4a^2$.

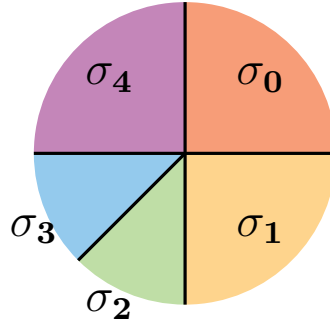
Conclusion.

Thus, in every region where D_i is optimizing, we obtain

$$- \left(2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} \right) < 0, \quad i = 0, 2, 3.$$

Decomposition



Blow-up of $\mathbb{P}^2$ along the points $[0 : 0 : 1]$ and $[0 : 1 : 0]$ **Fan****Intersection Matrix.**

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|ccccc} & D_0 & D_1 & D_2 & D_3 & D_4 \\ \hline D_0 & 0 & 1 & 0 & 0 & 1 \\ D_1 & 1 & 0 & 1 & 0 & 0 \\ D_2 & 0 & 1 & -1 & 1 & 0 \\ D_3 & 0 & 0 & 1 & -1 & 1 \\ D_4 & 1 & 0 & 0 & 1 & -1 \end{array}$$
Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2 + D_3], \quad g_1 = [D_3 + D_4], \quad g_2 = [D_2].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_1, \quad \tilde{g}_1 = g_0, \quad \tilde{g}_2 = g_1 + g_2$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 + x_2 \tilde{g}_2 \\ &= x_1 g_0 + (x_0 + x_2) g_1 + x_2 g_2, \end{aligned}$$

$$\begin{aligned} g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 + y_2 \tilde{g}_2 \\ &= y_1 g_0 + (y_0 + y_2) g_1 + y_2 g_2, \end{aligned}$$

$$\begin{aligned} g_2 &= z_0 \tilde{g}_0 + z_1 \tilde{g}_1 + z_2 \tilde{g}_2 \\ &= z_1 g_0 + (z_0 + z_2) g_1 + z_2 g_2, \end{aligned}$$

gives

$$\begin{array}{lll} x_0 = 0, & x_1 = 1, & x_2 = 0, \\ y_0 = 1, & y_1 = 0, & y_2 = 0, \\ z_0 = -1, & z_1 = 0, & z_2 = 1. \end{array}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 + cg_2 \in \text{Pic}(X_\Sigma), \quad a, b, c \in \mathbb{R},$$

can be written as

$$\alpha = (b - c)\tilde{g}_0 + a\tilde{g}_1 + c\tilde{g}_2.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b, c > 0, \quad b > c.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [(a + c)D_2 + (a + b)D_3 + bD_4], \quad \alpha' = [0, 0, a + c, a + b, b].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^4 D_i = [2D_2 + 3D_3 + 2D_4] = 1g_0 + 2g_1 + 1g_2.$$

After rescaling, we set

$$-K' = \left[0, 0, 1, \frac{3}{2}, 1\right].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [b, a + c, b - c, c, a] \cdot (\alpha')^T \\ &= -c^2 + 2(ab + cb), \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= \frac{c + 2(a + b)}{2}. \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{c + 2(a + b)}{-c^2 + 2(ab + cb)} > 0.$$

Intersections with Torus Divisors.

We further compute,

$$(-K) \cdot D_i = (-K') \cdot IM \cdot D_i = \begin{cases} 1, & \text{if } i = 0, 1, \\ \frac{1}{2}, & \text{if } i = 2, 3, 4. \end{cases}$$

$$\alpha.D_i = \alpha' \cdot IM \cdot D_i = \begin{cases} b, & \text{if } i = 0, \\ a + c, & \text{if } i = 1, \\ b - c, & \text{if } i = 2, \\ c, & \text{if } i = 3, \\ a, & \text{if } i = 4. \end{cases}$$

Hence

$$\frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{1}{b}, & \text{if } i = 0, \\ \frac{1}{a+c}, & \text{if } i = 1, \\ \frac{1}{2(b-c)}, & \text{if } i = 2, \\ \frac{1}{2c}, & \text{if } i = 3, \\ \frac{1}{2a}, & \text{if } i = 4. \end{cases}$$

We note that for each i the above is positive.

Numerical Invariant

We consider the intersection numbers

$$2 \frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\}.$$

Assume $i = 0$ minimizes, hence

$$\begin{aligned} b &> a + c, \\ b &> 2(b - c) \implies 2c > b, \\ b &> 2c, \\ b &> 2a, \end{aligned}$$

which is a contradiction, hence D_0 is never optimizing.

Assume $i = 1$ minimizes, hence

$$\begin{aligned} a + c &> b, \\ a + c &> 2(b - c), \\ a + c &> 2c \implies a > c, \\ a + c &> 2a \implies c > a, \end{aligned}$$

which is a contradiction, hence D_1 is never optimizing.

Assume $i = 2$ minimizes, hence

$$\begin{aligned} 2b - 2c &> b \implies b > 2c, \\ 2b - 2c &> a + c \implies 2b > a + 3c, \\ 2b - 2c &> 2c \implies b > 2c, \\ 2b - 2c &> 2a \implies b > a + c, \end{aligned}$$

hence D_2 is optimizing under the above restrictions.

Assume $i = 3$ minimizes, hence

$$\begin{aligned} 2c &> b, \\ 2c &> a + c \implies c > a, \\ 2c &> 2b - 2c \implies 2c > b, \\ 2c &> 2a, \end{aligned}$$

hence D_3 is optimizing under the above restrictions.

Assume $i = 4$ minimizes, hence

$$\begin{aligned} 2a &> b, \\ 2a &> a + c \implies a > c, \\ 2a &> 2b - 2c \implies a > b - c, \\ 2a &> 2c \implies a > c, \end{aligned}$$

hence D_4 is optimizing under the above restrictions.

We now have 2 areas within $\mathcal{K}(X_\Sigma)$ with optimizing divisors D_2, D_3 or D_4 . Consider now the entire expression. Recall

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} = \begin{cases} \frac{c+2(a+b)}{-c^2+2(ab+cb)} - \frac{1}{2(b-c)}, & \text{if } i = 2, \\ \frac{c+2(a+b)}{-c^2+2(ab+cb)} - \frac{1}{2c}, & \text{if } i = 3, \\ \frac{c+2(a+b)}{-c^2+2(ab+cb)} - \frac{1}{2a}, & \text{if } i = 4. \end{cases}$$

Using the fraction rule $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$, we can conclude that the denominator is positive in each case.

Then for $i = 2$ we get

$$\begin{aligned} 2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_2}{\alpha \cdot D_2} &= \frac{c + 2(a + b)}{-c^2 + 2(ab + cb)} - \frac{1}{2(b - c)} \\ &= \frac{-c^2 - 4cb - 4ca + 4b^2 + 2ba}{-6c^2b - 4cba + 2c^3 + 4cb^2 + 4b^2a}. \end{aligned}$$

Consider the numerator and refactor,

$$2ba - 4ca + 4b^2 - 4cb - c^2 = 2a(b - 2c) + 4b^2 - 4cb - c^2$$

using the restrictions we derive $2a(b - 2c) > 0$, and the remaining is a quadratic in b .

Then

$$d = (-4c)^2 - 4 \cdot 4 \cdot (-c^2) = 32c^2 > 0,$$

hence it has 2 roots at

$$x = \frac{4c \pm \sqrt{32c^2}}{8} = \frac{c(1 \pm \sqrt{2})}{2} \approx \begin{cases} 1, 207c, & \text{for } +, \\ -0.207c, & \text{for } -. \end{cases}$$

Since the leading coefficient is $4 > 0$, we know for $b \in (-0, 207c, 1, 207c)$ that $4b^2 - 4cb - c^2 < 0$, hence for $b > 1, 207c$ we have $4b^2 - 4cb - c^2 > 0$. Since we have the restriction that $b > 2c$, we conclude that the numerator is positive. In total

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_2}{\alpha \cdot D_2} > 0$$

.

Consider $i = 3$, we then have

$$\begin{aligned} 2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_3}{\alpha \cdot D_3} &= \frac{c + 2(a + b)}{-c^2 + 2(ab + cb)} - \frac{1}{2c} \\ &= \frac{3c^2 + 2cb + 4ca - 2ab}{4c^2b - 2c^3 + 4cab}. \end{aligned}$$

Consider the numerator and refactor,

$$2b(c - a) + 3c^2 + 4ca$$

since we have the restrictions $a, b, c > 0$, $2c > b$ and $c > a$, we conclude that the numerator is positive. In total

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_3}{\alpha \cdot D_3} > 0.$$

Consider $i = 4$, we then have

$$\begin{aligned} 2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_4}{\alpha \cdot D_4} &= \frac{c + 2(a + b)}{-c^2 + 2(ab + cb)} - \frac{1}{2a} \\ &= \frac{2ca + 2ab + 4a^2 + c^2 - 2cb}{2a(-c^2 + 2cb + 2ab)}. \end{aligned}$$

Consider the numerator and refactor,

$$2ca + 4a^2 + c^2 + 2b(a - c)$$

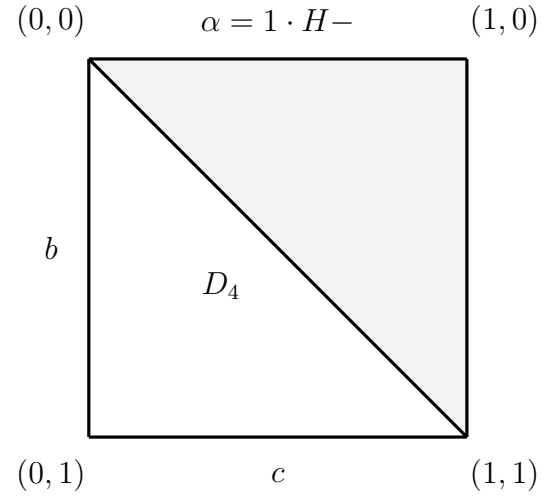
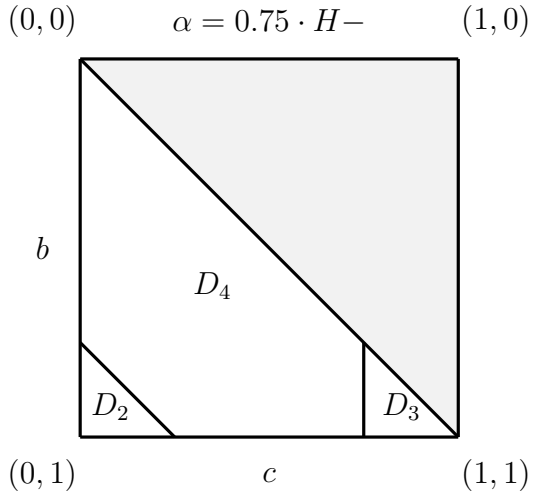
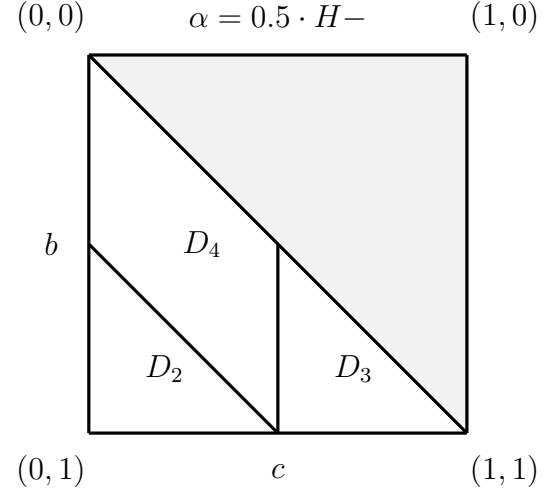
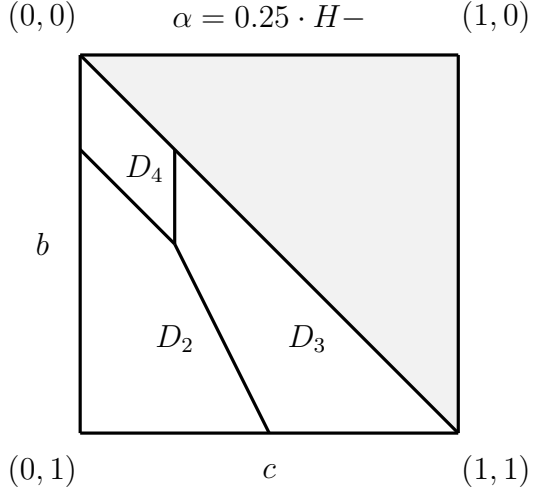
since we have the restrictions $a, b, c > 0$, $2a > b$, $a > c$, $a > b - c$, we conclude that the numerator is positive. In total

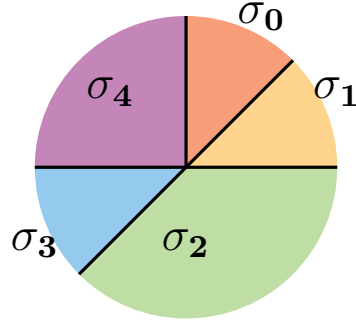
$$2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_4}{\alpha \cdot D_4} > 0.$$

Conclusion.

Thus, in every region where D_i is optimizing, we obtain

$$- \left(2 \frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} \right) < 0, \quad i = 2, 3, 4.$$

Decomposition

Blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$ and $[0 : 1 : 0]$ **Fan****Intersection Matrix.**

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|ccccc} & D_0 & D_1 & D_2 & D_3 & D_4 \\ \hline D_0 & -1 & 1 & 0 & 0 & 1 \\ D_1 & 1 & -1 & 1 & 0 & 0 \\ D_2 & 0 & 1 & 0 & 1 & 0 \\ D_3 & 0 & 0 & 1 & 0 & 1 \\ D_4 & 1 & 0 & 0 & 1 & -1 \end{array}$$

Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2 - D_4], g_1 = [-D_2 + D_3 + D_4] \text{ and } g_2 = [D_2].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_0 + g_1, \tilde{g}_1 = g_2 \text{ and } \tilde{g}_2 = g_1 + g_2.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 + x_2 \tilde{g}_2 \\ &= x_0 g_0 + (x_0 + x_2) g_1 + (x_1 + x_2) g_2, \end{aligned}$$

$$\begin{aligned} g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 + y_2 \tilde{g}_2 \\ &= y_0 g_0 + (y_0 + y_2) g_1 + (y_1 + y_2) g_2, \end{aligned}$$

$$\begin{aligned} g_2 &= z_0 \tilde{g}_0 + z_1 \tilde{g}_1 + z_2 \tilde{g}_2 \\ &= z_0 g_0 + (z_0 + z_2) g_1 + (z_1 + z_2) g_2, \end{aligned}$$

gives

$$\begin{array}{lll} x_0 = 1, & x_1 = 1, & x_2 = -1, \\ y_0 = 0, & y_1 = -1, & y_2 = 1, \\ z_0 = 0, & z_1 = 1, & z_2 = 0. \end{array}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 + cg_2 \in \text{Pic}(X_\Sigma), \quad a, b, c \in \mathbb{R},$$

can be written as

$$\alpha = a\tilde{g}_0 + (a - b + c)\tilde{g}_1 + (b - a)\tilde{g}_2.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b, c > 0, \quad b > a, \quad a + c > b.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [(a - b + c)D_2 + bD_3 + (b - a)D_4], \quad \alpha' = [0, 0, a - b + c, b, b - a].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^4 D_i = [D_2 + 2D_3 + D_4] = 1g_0 + 2g_1 + 2g_2.$$

After rescaling, we set

$$-K' = \left[0, 0, \frac{1}{2}, 1, \frac{1}{2}\right].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [b - a, a - b + c, b, c, a] \cdot (\alpha')^T \\ &= 2ab + 2bc - a^2 - b^2, \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= \frac{2c + a + b}{2}. \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{2c + a + b}{2ab + 2bc - a^2 - b^2}.$$

Clearly the numerator is positive, consider the denominator.

$$\begin{aligned} 2ab + 2bc - a^2 - b^2 &= 2ab - a^2 + b(2c - b) \\ &> 2ab - a^2 + b(2(b - a) - b) \\ &= b^2 - a^2 > 0, \end{aligned}$$

using the restrictions $a, b, c > 0$, $c > b - a$ and $b > a$.

Intersections with Torus Divisors.

We further compute,

$$(-K).D_i = (-K') \cdot IM \cdot D_i^T = \begin{cases} \frac{1}{2}, & \text{if } i = 0, 1, 4, \\ 1, & \text{if } i = 2, 3. \end{cases}$$

$$\alpha.D_i = \alpha' \cdot IM \cdot D_i^T = \begin{cases} b - a, & \text{if } i = 0, \\ a - b + c, & \text{if } i = 1, \\ b, & \text{if } i = 2, \\ c, & \text{if } i = 3, \\ a, & \text{if } i = 4. \end{cases}$$

Hence

$$\frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{1}{2(b-a)}, & \text{if } i = 0, \\ \frac{1}{2(a-b+c)}, & \text{if } i = 1, \\ \frac{1}{b}, & \text{if } i = 2, \\ \frac{1}{c}, & \text{if } i = 3, \\ \frac{1}{2a}, & \text{if } i = 4. \end{cases}$$

We note that for each i the above is positive.

Numerical Invariant

We consider the intersection numbers

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\}.$$

Assume $i = 0$ minimizes, hence

$$\begin{aligned} \frac{1}{2(b-a)} &< \frac{1}{2(a-b+c)} \implies 2a + c < 2b, \\ \frac{1}{2(b-a)} &< \frac{1}{b} \implies 2a < b, \\ \frac{1}{2(b-a)} &< \frac{1}{c} \implies 2a + c < 2b, \\ \frac{1}{2(b-a)} &< \frac{1}{2a} \implies 2a < b, \end{aligned}$$

hence D_0 is optimizing under the above restrictions.

Assume $i = 1$ minimizes, hence

$$\begin{aligned} \frac{1}{2(a-b+c)} &< \frac{1}{2(b-a)} \implies 2b < 2a + c, \\ \frac{1}{2(a-b+c)} &< \frac{1}{b} \implies 3b < 2(a + c), \\ \frac{1}{2(a-b+c)} &< \frac{1}{c} \implies 2b < 2a + c, \\ \frac{1}{2(a-b+c)} &< \frac{1}{2a} \implies b < c, \end{aligned}$$

hence D_1 is optimizing under the above restrictions.

Assume $i = 2$ minimizes, hence

$$\begin{aligned}\frac{1}{b} &< \frac{1}{2(b-a)} \implies b < 2a, \\ \frac{1}{b} &< \frac{1}{2(a-b+c)} \implies 2(a+c) < 3b, \\ \frac{1}{b} &< \frac{1}{c} \implies c < b, \\ \frac{1}{b} &< \frac{1}{2a} \implies 2a < b,\end{aligned}$$

which is a contradiction, hence D_2 is never optimizing.

Assume $i = 3$ minimizes, hence

$$\begin{aligned}\frac{1}{c} &< \frac{1}{2(b-a)} \implies 2b < 2a + c, \\ \frac{1}{c} &< \frac{1}{2(a-b+c)} \implies 2a + c < 2b, \\ \frac{1}{c} &< \frac{1}{b} \implies b < c, \\ \frac{1}{c} &< \frac{1}{2a} \implies 2a < c,\end{aligned}$$

which is a contradiction, hence D_3 is never optimizing.

Assume $i = 4$ minimizes, hence

$$\begin{aligned}\frac{1}{2a} &< \frac{1}{2(b-a)} \implies b < 2a, \\ \frac{1}{2a} &< \frac{1}{2(a-b+c)} \implies c < b, \\ \frac{1}{2a} &< \frac{1}{b} \implies b < 2a, \\ \frac{1}{2a} &< \frac{1}{c} \implies c < 2a,\end{aligned}$$

hence D_4 is optimizing under the above restrictions.

We now have 3 areas within $\mathcal{K}(X_\Sigma)$ with optimizing divisors D_0, D_1 or D_4 . Consider now the entire expression. Recall

$$2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} = \begin{cases} \frac{2c+a+b}{2ab+2bc-a^2-b^2} - \frac{1}{2(b-a)}, & \text{if } i = 0, \\ \frac{2c+a+b}{2ab+2bc-a^2-b^2} - \frac{1}{2(a-b+c)}, & \text{if } i = 1, \\ \frac{2c+a+b}{2ab+2bc-a^2-b^2} - \frac{1}{2a}, & \text{if } i = 4. \end{cases}$$

Using the fraction rule $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$, we can conclude that the denominator is positive in each case.

Then for $i = 0$, we get

$$\frac{2c+a+b}{2ab+2bc-a^2-b^2} - \frac{1}{2(b-a)} = \frac{3b^2 - a^2 + 2bc - 2ab - 4ac}{2(b-a)(2ab+2bc-a^2-b^2)}.$$

Consider the numerator

$$\begin{aligned}
3b^2 - a^2 + 2bc - 2ab - 4ac &> 3b^2 - b^2 + 2bc - 2ab - 4ac \\
&= 2b^2 + 2bc - 2ab - 4ac \\
&> 2b^2 - 2b^2 + 2bc - 4ac \\
&= 2bc - 4ac \\
&= 2(c(b - 2a)) > 0,
\end{aligned}$$

using the restrictions $a, b, c > 0$, $b > a$ and $b > 2a$.

For $i = 1$, we get

$$\frac{2c + a + b}{2ab + 2bc - a^2 - b^2} - \frac{1}{2(a - b + c)} = \frac{4c^2 + 3a^2 - b^2 + 6ac - 2ab - 4bc}{2(a - b + c)(2ab + 2bc - a^2 - b^2)}.$$

Consider the numerator.

$$\begin{aligned}
4c^2 + 3a^2 - b^2 + 6ac - 2ab - 4bc &= 4c^2 - b^2 - 4bc + a(3a + 6c - 2b) \\
&> 4c^2 - b^2 - 4bc + a(6c - 2b + a + 2b - c) \\
&= 4c^2 - b^2 - 4bc + a(5c + a) \\
&= 4c^2 - b^2 - 4bc + 5ac + a^2 \\
&= c(4c - 4b + 5a) + a^2 - b^2 \\
&> c(4c - 4b + a + 4b - 2c) + a^2 - b^2 \\
&= c(2c + a) + a^2 - b^2 \\
&= 2c^2 + ac + a^2 - b^2 \\
&> c^2 + ac + a^2 > 0.
\end{aligned}$$

Using the restrictions $a, b, c > 0$, $2a + c > 2b$ and $c > b$.

For $i = 4$, we get

$$\frac{2c + a + b}{2ab + 2bc - a^2 - b^2} - \frac{1}{2a} = \frac{3a^2 + b^2 + 4ac - 2bc}{2a(2ab + 2bc - a^2 - b^2)}.$$

Consider the numerator.

$$3a^2 + b^2 + 4ac - 2bc = 3a^2 + b^2 + 2c(2a - b) > 0.$$

using the restrictions $a, b, c > 0$ and $2a > b$.

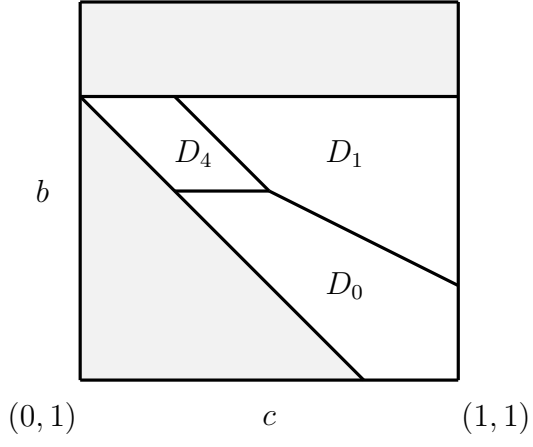
Conclusion.

Thus, in every region where D_i is optimizing, we obtain

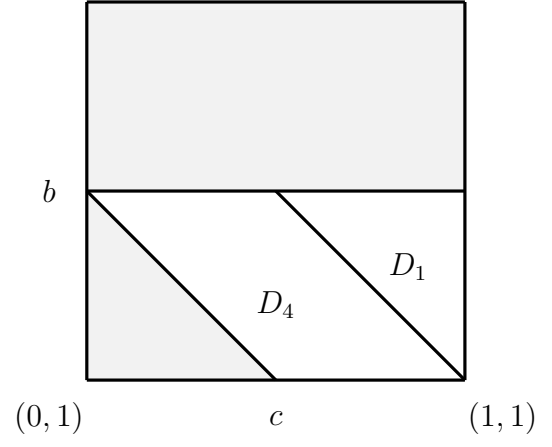
$$-\left(2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i}\right) < 0, \quad i = 0, 1, 4.$$

Decomposition

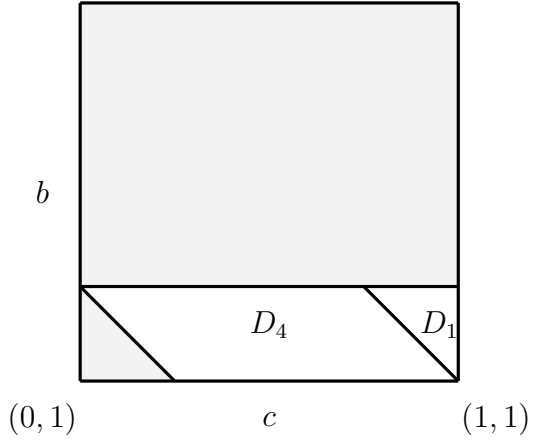
$(0,0)$ $\alpha = 0.25 \cdot H -$ $(1,0)$



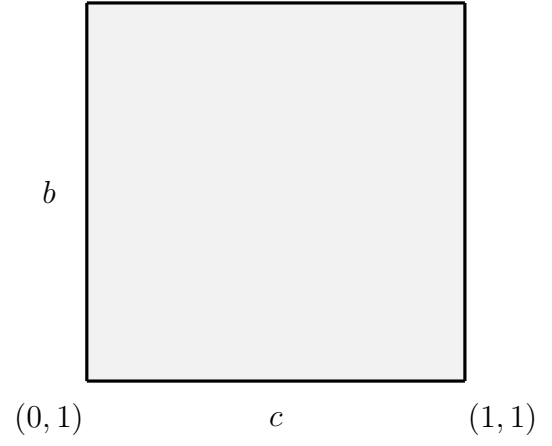
$(0,0)$ $\alpha = 0.5 \cdot H -$ $(1,0)$

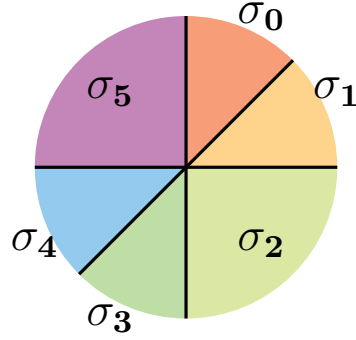


$(0,0)$ $\alpha = 0.75 \cdot H -$ $(1,0)$



$(0,0)$ $\alpha = 1 \cdot H -$ $(1,0)$



Blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0]$, $[0 : 1 : 0]$ and $[0 : 0 : 1]$ **Fan****Intersection Matrix.**

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|cccccc} & D_0 & D_1 & D_2 & D_3 & D_4 & D_5 \\ \hline D_0 & -1 & 1 & 0 & 0 & 0 & 1 \\ D_1 & 1 & -1 & 1 & 0 & 0 & 0 \\ D_2 & 0 & 1 & -1 & 1 & 0 & 0 \\ D_3 & 0 & 0 & 1 & -1 & 1 & 0 \\ D_4 & 0 & 0 & 0 & 1 & -1 & 1 \\ D_5 & 1 & 0 & 0 & 0 & 1 & -1 \end{array}$$
Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2 + D_3 - D_5], g_1 = [-D_2 + D_4 + D_5], g_2 = [D_2] \text{ and } g_3 = [D_3].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_1 + g_2, \tilde{g}_1 = g_2 + g_3, \tilde{g}_2 = g_0 + g_1, \tilde{g}_3 = g_0 + g_1 + g_2 \text{ and } \tilde{g}_4 = g_1 + g_2 + g_3.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 + x_2 \tilde{g}_2 + x_3 \tilde{g}_3 + x_4 \tilde{g}_4 \\ &= (x_2 + x_3)g_0 + (x_0 + x_2 + x_3 + x_4)g_1 + (x_0 + x_1 + x_3 + x_4)g_2 + (x_1 + x_4)g_3, \end{aligned}$$

$$\begin{aligned} g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 + y_2 \tilde{g}_2 + y_3 \tilde{g}_3 + y_4 \tilde{g}_4 \\ &= (y_2 + y_3)g_0 + (y_0 + y_2 + y_3 + y_4)g_1 + (y_0 + y_1 + y_3 + y_4)g_2 + (y_1 + y_4)g_3, \end{aligned}$$

$$\begin{aligned} g_2 &= z_0 \tilde{g}_0 + z_1 \tilde{g}_1 + z_2 \tilde{g}_2 + z_3 \tilde{g}_3 + z_4 \tilde{g}_4 \\ &= (z_2 + z_3)g_0 + (z_0 + z_2 + z_3 + z_4)g_1 + (z_0 + z_1 + z_3 + z_4)g_2 + (z_1 + z_4)g_3, \end{aligned}$$

$$\begin{aligned} g_3 &= w_0 \tilde{g}_0 + w_1 \tilde{g}_1 + w_2 \tilde{g}_2 + w_3 \tilde{g}_3 + w_4 \tilde{g}_4 \\ &= (w_2 + w_3)g_0 + (w_0 + w_2 + w_3 + w_4)g_1 + (w_0 + w_1 + w_3 + w_4)g_2 + (w_1 + w_4)g_3, \end{aligned}$$

gives

$$\begin{array}{ccccc} x_0 = -1, & x_1 = 0, & x_2 = 0, & x_3 = 1, & x_4 = 0, \\ y_0 = 1, & y_1 = 0, & y_2 = 1, & y_3 = -1, & y_4 = 0, \\ z_0 = 0, & z_1 = 0, & z_2 = -1, & z_3 = 1, & z_4 = 0, \\ w_0 = 0, & w_1 = 1, & w_2 = 1, & w_3 = -1, & w_4 = 0 \end{array}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 + cg_2 + dg_3 \in \text{Pic}(X_\Sigma), \quad a, b, c, d \in \mathbb{R},$$

can be written as

$$\alpha = (b - a)\tilde{g}_0 + d\tilde{g}_1 + (b - c + d)\tilde{g}_2 + (a - b + c - d)\tilde{g}_3.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b, c, d > 0, \quad b > a, \quad b + d > c, \quad c > d, \quad a + c > b + d.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [(a - b + c)D_2 + (a + d)D_3 + bD_4 + (b - a)D_5], \quad \alpha' = [0, 0, a - b + c, a + d, b, b - a].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^5 D_i = [D_2 + 2D_3 + 2D_4 + D_5] = 1g_0 + 2g_1 + 2g_2 + 1g_3.$$

After rescaling, we set

$$-K' = \left[0, 0, \frac{1}{2}, 1, 1, \frac{1}{2}\right].$$

Intersections with Torus Divisors.

We further compute,

$$(-K).D_i = (-K') \cdot IM \cdot D_i^T = \left\{ \frac{1}{2}, \quad \text{if } i = 0, 1, 2, 3, 4, 5. \right.$$

$$\alpha.D_i = \alpha' \cdot IM \cdot D_i^T = \begin{cases} b - a, & \text{if } i = 0, \\ a - b + c, & \text{if } i = 1, \\ b - c + d, & \text{if } i = 2, \\ c - d, & \text{if } i = 3, \\ d, & \text{if } i = 4, \\ a, & \text{if } i = 5. \end{cases}$$

Hence

$$\frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{1}{2(b-a)}, & \text{if } i = 0, \\ \frac{1}{2(a-b+c)}, & \text{if } i = 1, \\ \frac{1}{2(b-c+d)}, & \text{if } i = 2, \\ \frac{1}{2(c-d)}, & \text{if } i = 3, \\ \frac{1}{2d}, & \text{if } i = 4, \\ \frac{1}{2a}, & \text{if } i = 5. \end{cases}$$

We note that for each i the above is positive.

Numerical Invariant

We consider the intersection numbers

$$2\frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\} = 2\frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{1}{2(\alpha.D_i)} \right\}.$$

Let us describe α in terms of torus-invariant divisors, where we denote the coefficients by x_i , then

$$\alpha = \sum_{i=0}^5 x_i D_i.$$

By bilinearity we have

$$\alpha^2 = \sum_{i=0}^5 x_i (\alpha.D_i).$$

note that since $\alpha \in \mathcal{K}c(X_\Sigma)$, we have $\alpha.D_i > 0$. Therefore $\alpha^2 > 0$. Hence for D_k to minimize, we must have that

$$\alpha.D_k \geq \alpha.D_j, \quad \text{for } j \neq k.$$

Therefore

$$\alpha^2 = \sum_{i=0}^5 x_i (\alpha.D_i) \leq \sum_{i=0}^5 x_i (\alpha.D_k),$$

we further have

$$\begin{aligned} \sum_{i=0}^5 x_i &= a - b + c + a + d + b + b - a \\ &= a + b + c + d, \end{aligned}$$

hence for D_k to minimize we must have

$$\alpha^2 \leq (a + b + c + d)(\alpha.D_k).$$

Consider the entire intersection numbers

$$2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_k}{\alpha \cdot D_k} = \frac{a+b+c+d}{\alpha^2} - \frac{1}{2(\alpha \cdot D_k)}.$$

For the intersection numbers to be positive we must have

$$\frac{a+b+c+d}{\alpha^2} > \frac{1}{2(\alpha \cdot D_k)} \implies 2(\alpha \cdot D_k)(a+b+c+d) > \alpha^2$$

which follows from the assumption of D_k minimizing. We conclude that when D_k is minimizing, then

$$-\left(2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_k}{\alpha \cdot D_k}\right) < 0.$$

Assume $i = 0$ minimizes, hence

$$\begin{aligned} \frac{1}{2(b-a)} &< \frac{1}{2(a-b+c)} \implies 2a+c < 2b, \\ \frac{1}{2(b-a)} &< \frac{1}{2(b-c+d)} \implies a+d < c, \\ \frac{1}{2(b-a)} &< \frac{1}{2(c-d)} \implies a+c < b+d, \\ \frac{1}{2(b-a)} &< \frac{1}{2d} \implies a+d < b, \\ \frac{1}{2(b-a)} &< \frac{1}{2a} \implies 2a < b, \end{aligned}$$

hence D_0 is optimizing under the above restrictions.

Assume $i = 1$ minimizes, hence

$$\begin{aligned} \frac{1}{2(a-b+c)} &< \frac{1}{2(b-a)} \implies 2b < 2a+c, \\ \frac{1}{2(a-b+c)} &< \frac{1}{2(b-c+d)} \implies 2b+d < a+2c, \\ \frac{1}{2(a-b+c)} &< \frac{1}{2(c-d)} \implies b < a+d, \\ \frac{1}{2(a-b+c)} &< \frac{1}{2d} \implies b+d < a+c, \\ \frac{1}{2(a-b+c)} &< \frac{1}{2a} \implies b < c, \end{aligned}$$

hence D_1 is optimizing under the above restrictions.

Assume $i = 2$ minimizes, hence

$$\begin{aligned}
\frac{1}{2(b-c+d)} &< \frac{1}{2(b-a)} \implies c < a+d, \\
\frac{1}{2(b-c+d)} &< \frac{1}{2(a-b+c)} \implies a+2c < 2b+d, \\
\frac{1}{2(b-c+d)} &< \frac{1}{2(c-d)} \implies 2c < b+2d, \\
\frac{1}{2(b-c+d)} &< \frac{1}{2d} \implies c < b, \\
\frac{1}{2(b-c+d)} &< \frac{1}{2a} \implies a+c < b+d,
\end{aligned}$$

hence D_2 is optimizing under the above restrictions.

Assume $i = 3$ minimizes, hence

$$\begin{aligned}
\frac{1}{2(c-d)} &< \frac{1}{2(b-a)} \implies b+d < a+c, \\
\frac{1}{2(c-d)} &< \frac{1}{2(a-b+c)} \implies a+d < b, \\
\frac{1}{2(c-d)} &< \frac{1}{2(b-c+d)} \implies b+2d < 2c, \\
\frac{1}{2(c-d)} &< \frac{1}{2d} \implies 2d < c, \\
\frac{1}{2(c-d)} &< \frac{1}{2a} \implies a+d < c,
\end{aligned}$$

hence D_3 is optimizing under the above restrictions.

Assume $i = 4$ minimizes, hence

$$\begin{aligned}
\frac{1}{2d} &< \frac{1}{2(b-a)} \implies b < a+d, \\
\frac{1}{2d} &< \frac{1}{2(a-b+c)} \implies a+c < b+d, \\
\frac{1}{2d} &< \frac{1}{2(b-c+d)} \implies b < c, \\
\frac{1}{2d} &< \frac{1}{2(c-d)} \implies c < 2d, \\
\frac{1}{2d} &< \frac{1}{2a} \implies a < d,
\end{aligned}$$

hence D_4 is optimizing under the above restrictions.

Assume $i = 5$ minimizes, hence

$$\begin{aligned}\frac{1}{2a} &< \frac{1}{2(b-a)} \implies b < 2a, \\ \frac{1}{2a} &< \frac{1}{2(a-b+c)} \implies c < b, \\ \frac{1}{2a} &< \frac{1}{2(b-c+d)} \implies b+d < a+c, \\ \frac{1}{2a} &< \frac{1}{2(c-d)} \implies c < a+d, \\ \frac{1}{2a} &< \frac{1}{2d} \implies d < a,\end{aligned}$$

hence D_5 is optimizing under the above restrictions.

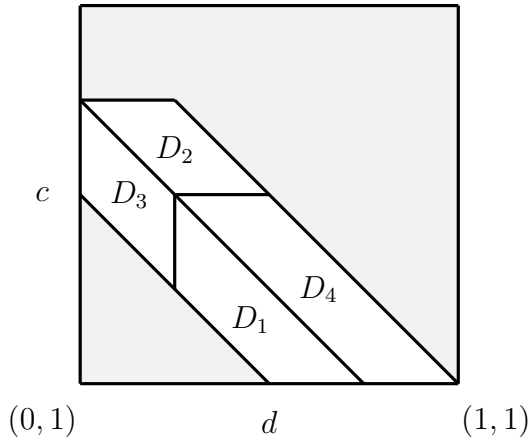
Conclusion.

Thus, in every region where D_i is optimizing, we obtain

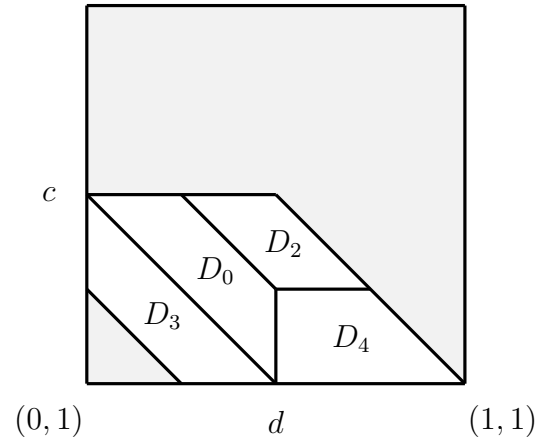
$$-\left(2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i}\right) < 0, \quad i = 0, 1, 2, 3, 4, 5.$$

Decomposition

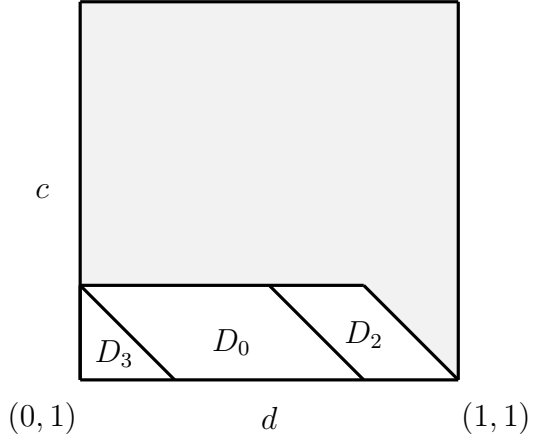
$$(0,0) \quad \alpha = 0.25 \cdot H - 0.5 \cdot E_1 - \quad (1,0)$$



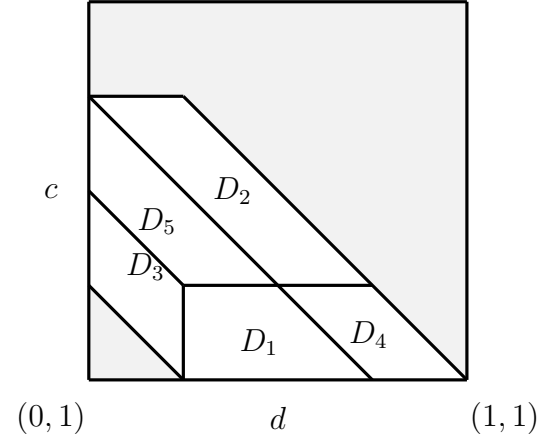
$$(0,0) \quad \alpha = 0.25 \cdot H - 0.75 \cdot E_1 - \quad (1,0)$$



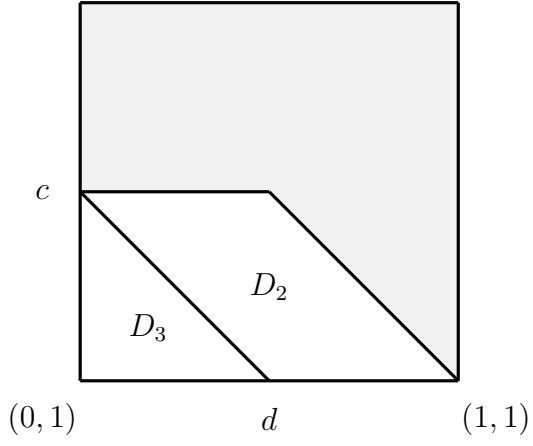
$$(0,0) \quad \alpha = 0.25 \cdot H - 1 \cdot E_1 - \quad (1,0)$$



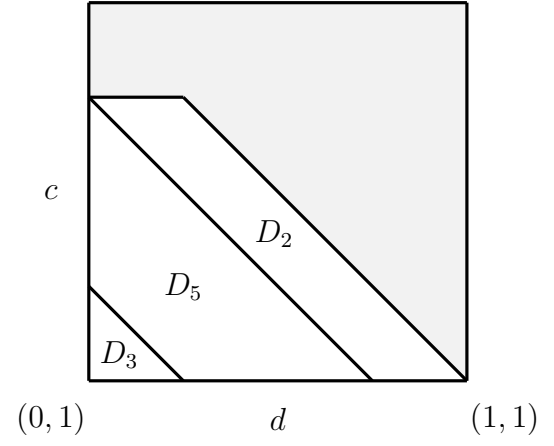
$$(0,0) \quad \alpha = 0.5 \cdot H - 0.75 \cdot E_1 - \quad (1,0)$$



$$(0,0) \quad \alpha = 0.5 \cdot H - 1 \cdot E_1 - \quad (1,0)$$



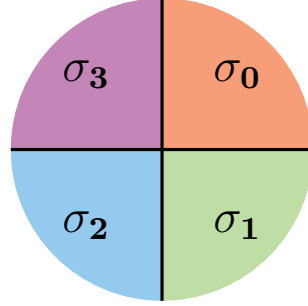
$$(0,0) \quad \alpha = 0.75 \cdot H - 1 \cdot E_1 - \quad (1,0)$$



B.2 Exceptional Cases of $\mathbb{P}^1 \times \mathbb{P}^1$ and their Wall-Chamber Decomposition

$\mathbb{P}^1 \times \mathbb{P}^1$

Fan



Intersection Matrix.

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|cccc} & D_0 & D_1 & D_2 & D_3 \\ \hline D_0 & 0 & 1 & 0 & 1 \\ D_1 & 1 & 0 & 1 & 0 \\ D_2 & 0 & 1 & 0 & 1 \\ D_3 & 1 & 0 & 1 & 0 \end{array}$$

Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2] \text{ and } g_1 = [D_3].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_1 \text{ and } \tilde{g}_1 = g_0.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 \\ &= x_0 g_1 + x_1 g_0, \end{aligned}$$

$$\begin{aligned} g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 \\ &= y_0 g_1 + y_1 g_0, \end{aligned}$$

gives

$$\begin{aligned} x_0 &= 0, & x_1 &= 1, \\ y_0 &= 1, & y_1 &= 0. \end{aligned}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 \in \text{Pic}(X_\Sigma), \quad a, b \in \mathbb{R},$$

can be written as

$$\alpha = b\tilde{g}_0 + a\tilde{g}_1.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b > 0.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [aD_2 + bD_3], \quad \alpha' = [0, 0, a, b].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^3 D_i = [2D_2 + 2D_3] = 2g_0 + 2g_1.$$

After rescaling, we set

$$-K' = [0, 0, 1, 1].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [b, a, b, a] \cdot (\alpha')^T \\ &= 2ab, \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= a + b. \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{2(a+b)}{2ab} = \frac{a+b}{ab} > 0.$$

Intersections with Torus Divisors.

We further compute,

$$(-K) \cdot D_i = (-K') \cdot IM \cdot D_i^T = \begin{cases} 1, & \text{if } i = 0, 1, 2, 3. \end{cases}$$

$$\alpha \cdot D_i = \alpha' \cdot IM \cdot D_i^T = \begin{cases} b, & \text{if } i = 0, 2, \\ a, & \text{if } i = 1, 3. \end{cases}$$

hence

$$\frac{(-K) \cdot D_i}{\alpha \cdot D_i} = \begin{cases} \frac{1}{b}, & \text{if } i = 0, 2, \\ \frac{1}{a}, & \text{if } i = 1, 3. \end{cases}$$

We note that for each i the above is positive.

Numerical Invariant

We consider the intersection numbers

$$2\frac{\alpha \cdot (-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K) \cdot D_i}{\alpha \cdot D_i} \right\}.$$

Assume $i = 0, 2$ minimizes, hence

$$\frac{1}{b} < \frac{1}{a} \implies b > a,$$

hence D_0, D_2 is optimizing under the above restrictions.

Assume $i = 1, 3$ minimizes, hence

$$\frac{1}{a} < \frac{1}{b} \implies a > b,$$

hence D_1, D_3 is optimizing under the above restrictions.

We now have 2 areas within $\mathcal{K}(X_\Sigma)$ with optimizing divisors D_0, D_2 or D_1, D_3 . Consider now the entire expression. Recall

$$2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} = \begin{cases} \frac{a+b}{ab} - \frac{1}{b}, & \text{if } i = 0, 2, \\ \frac{a+b}{ab} - \frac{1}{a}, & \text{if } i = 1, 3. \end{cases}$$

Using the fraction rule $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$, we can conclude that the denominator is positive in each case.

Then for $i = 0, 2$, we get

$$\frac{a+b}{ab} - \frac{1}{b} = \frac{1}{a} > 0.$$

For $i = 1, 3$, we get

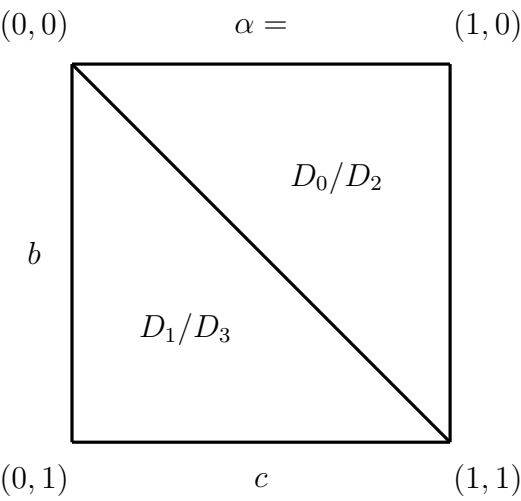
$$\frac{a+b}{ab} - \frac{1}{a} = \frac{1}{b} > 0.$$

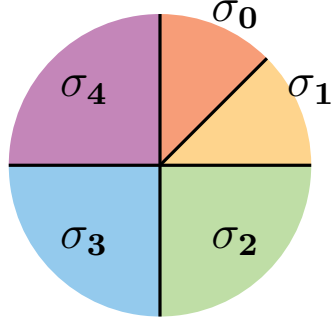
Conclusion.

Thus, in every region where D_i is optimizing, we obtain

$$- \left(2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} \right) < 0, \quad i = 0, 1, 2, 3.$$

Decomposition



Blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along the point $([1 : 0], [1 : 0])$ **Fan****Intersection Matrix.**

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|ccccc} & D_0 & D_1 & D_2 & D_3 & D_4 \\ \hline D_0 & -1 & 1 & 0 & 0 & 1 \\ D_1 & 1 & -1 & 1 & 0 & 0 \\ D_2 & 0 & 1 & -1 & 1 & 0 \\ D_3 & 0 & 0 & 1 & 0 & 1 \\ D_4 & 1 & 0 & 0 & 1 & 0 \end{array}$$
Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2 + D_3 - D_4], g_1 = [-D_2 + D_4] \text{ and } g_2 = [D_2].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_1 + g_2, \tilde{g}_1 = g_0 + g_1 \text{ and } \tilde{g}_2 = g_0 + g_1 + g_2.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 + x_2 \tilde{g}_2 \\ &= (x_1 + x_2)g_0 + (x_0 + x_1 + x_2)g_1 + (x_0 + x_2)g_2 \\ &, \\ g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 + y_2 \tilde{g}_2 \\ &= (y_1 + y_2)g_0 + (y_0 + y_1 + y_2)g_1 + (y_0 + y_2)g_2 \\ &, \\ g_2 &= z_0 \tilde{g}_0 + z_1 \tilde{g}_1 + z_2 \tilde{g}_2 \\ &= (z_1 + z_2)g_0 + (z_0 + z_1 + z_2)g_1 + (z_0 + z_2)g_2, \end{aligned}$$

gives

$$\begin{array}{lll} x_0 = -1, & x_1 = 0, & x_2 = 1, \\ y_0 = 1, & y_1 = 1, & y_2 = -1, \\ z_0 = 0, & z_1 = -1, & z_2 = 1. \end{array}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 + cg_2 \in \text{Pic}(X_\Sigma), \quad a, b, c \in \mathbb{R},$$

can be written as

$$\alpha = (b - a)\tilde{g}_0 + (b - c)\tilde{g}_1 + (a - b + c)\tilde{g}_2.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b, c > 0, \quad b > a, \quad b > c, \quad a + c > b.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [(a - b + c)D_2 + aD_3 + (b - a)D_4], \quad \alpha' = [0, 0, a - b + c, a, b - a].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^4 D_i = [D_2 + 2D_3 + D_4] = 2g_0 + 3g_1 + 2g_2.$$

After rescaling, we set

$$-K' = \left[0, 0, \frac{1}{3}, \frac{2}{3}, \frac{1}{3}\right].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [b - a, a - b + c, b - c, c, a] \cdot (\alpha')^T \\ &= 2ab + 2bc - a^2 - b^2 - c^2, \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= \frac{a + b + c}{3}. \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{2(a + b + c)}{3(2ab + 2bc - a^2 - b^2 - c^2)}.$$

Clearly the numerator is positive, consider the denominator.

$$\begin{aligned} 2ab + 2bc - a^2 - b^2 - c^2 &= 2ab - a^2 + b(2c - b) - c^2 \\ &> 2ab - a^2 + b(2c - a - c) - c^2 \\ &= ab - a^2 - c^2 + bc \\ &> a^2 - a^2 - c^2 + c^2 \\ &= 0 \end{aligned}$$

using the restrictions $a, b, c > 0$, $b > a$, $b > c$ and $a + c > b$.

Intersections with Torus Divisors.

We further compute,

$$(-K).D_i = (-K') \cdot IM \cdot D_i^T = \begin{cases} \frac{1}{3}, & \text{if } i = 0, 1, 2, \\ \frac{2}{3}, & \text{if } i = 3, 4. \end{cases}$$

$$\alpha.D_i = \alpha' \cdot IM \cdot D_i^T = \begin{cases} b - a, & \text{if } i = 0, \\ a - b + c, & \text{if } i = 1, \\ b - c, & \text{if } i = 2, \\ c, & \text{if } i = 3, \\ a, & \text{if } i = 4. \end{cases}$$

hence

$$\frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{1}{3(b-a)}, & \text{if } i = 0, \\ \frac{1}{3(a-b+c)}, & \text{if } i = 1, \\ \frac{1}{3(b-c)}, & \text{if } i = 2, \\ \frac{2}{3c}, & \text{if } i = 3, \\ \frac{2}{3a}, & \text{if } i = 4. \end{cases}$$

We note that for each i the above is positive.

Numerical Invariant

We consider the intersection numbers

$$2 \frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\}.$$

Assume $i = 0$ minimizes, hence

$$\begin{aligned} \frac{1}{3(b-a)} &< \frac{1}{3(a-b+c)} \implies 2a + c < 2b, \\ \frac{1}{3(b-a)} &< \frac{1}{3(b-c)} \implies a < c, \\ \frac{1}{3(b-a)} &< \frac{2}{3c} \implies 2a + c < 2b, \\ \frac{1}{3(b-a)} &< \frac{2}{3a} \implies 3a < 2b, \end{aligned}$$

hence D_0 is optimizing under the above restrictions.

Assume $i = 1$ minimizes, hence

$$\begin{aligned}\frac{1}{3(a-b+c)} &< \frac{1}{3(b-a)} \implies 2b < 2a+c, \\ \frac{1}{3(a-b+c)} &< \frac{1}{3(b-c)} \implies 2b < a+2c, \\ \frac{1}{3(a-b+c)} &< \frac{2}{3c} \implies 2b < 2a+c, \\ \frac{1}{3(a-b+c)} &< \frac{2}{3a} \implies 2b < a+2c,\end{aligned}$$

hence D_1 is optimizing under the above restrictions.

Assume $i = 2$ minimizes, hence

$$\begin{aligned}\frac{1}{3(b-c)} &< \frac{1}{3(b-a)} \implies c < a, \\ \frac{1}{3(b-c)} &< \frac{1}{3(1-b+c)} \implies a+2c < 2b, \\ \frac{1}{3(b-c)} &< \frac{2}{3c} \implies 3c < 2b, \\ \frac{1}{3(b-c)} &< \frac{2}{3a} \implies a+2c < 2b,\end{aligned}$$

hence D_2 is optimizing under the above restrictions.

Assume $i = 3$ minimizes, hence

$$\begin{aligned}\frac{2}{3c} &< \frac{1}{3(b-a)} \implies 2b < 2a+c, \\ \frac{2}{3c} &< \frac{1}{3(a-b+c)} \implies 2a+c < 2b, \\ \frac{2}{3c} &< \frac{1}{3(b-c)} \implies 2b < 3c, \\ \frac{2}{3c} &< \frac{2}{3a} \implies a < c,\end{aligned}$$

which is a contradiction, hence D_3 is never optimizing.

Assume $i = 4$ minimizes, hence

$$\begin{aligned}\frac{2}{3a} &< \frac{1}{3(b-a)} \implies 2b < 3a, \\ \frac{2}{3a} &< \frac{1}{3(a-b+c)} \implies a+2c < 2b, \\ \frac{2}{3a} &< \frac{1}{3(b-c)} \implies 2b < a+2c, \\ \frac{2}{3a} &< \frac{2}{3c} \implies c < a,\end{aligned}$$

which is a contradiction, hence D_4 is never optimizing.

We now have 3 areas within $\mathcal{K}(X_\Sigma)$ with optimizing divisors D_0, D_1 or D_2 . Consider now the entire expression. Recall

$$2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} = \begin{cases} \frac{2(a+b+c)}{3(2ab+2bc-a^2-b^2-c^2)} - \frac{1}{3(b-a)}, & \text{if } i = 0, \\ \frac{2(a+b+c)}{3(2ab+2bc-a^2-b^2-c^2)} - \frac{1}{3(a-b+c)}, & \text{if } i = 1, \\ \frac{2(a+b+c)}{3(2ab+2bc-a^2-b^2-c^2)} - \frac{1}{3(b-c)}, & \text{if } i = 2. \end{cases}$$

Using the fraction rule $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$, we can conclude that the denominator is positive in each case.

Then for $i = 0$, we get

$$\frac{2(a+b+c)}{3(2ab+2bc-a^2-b^2-c^2)} - \frac{1}{3(b-a)} = \frac{3b^2 - a^2 + c^2 - 2ac - 2ab}{3(b-a)(2ab+2bc-a^2-b^2-c^2)},$$

consider the numerator.

$$\begin{aligned} 3b^2 - a^2 + c^2 - 2ac - 2ab &= 3\left(b^2 - \frac{2ab}{3}\right) + (c^2 + a^2 - 2ac) - 2a^2 \\ &= 3\left(\left(b - \frac{a}{3}\right)^2 - \frac{a^2}{9}\right) + (c-a)^2 - 2a^2. \end{aligned}$$

From the restriction $2b > 3a \implies b > \frac{3a}{2} \implies b - \frac{a}{3} > \frac{3a}{2} - \frac{a}{3} = \frac{7a}{6}$, hence continuing the above

$$\begin{aligned} &> 3\left(\left(\frac{7a}{6}\right)^2 - \frac{a^2}{9}\right) + (c-a)^2 - 2a^2 \\ &= \frac{49a^2}{12} - \frac{a^2}{3} - 2a^2 + (c-a)^2 \\ &= \frac{7a^2}{4} + (c-a)^2 > 0, \end{aligned}$$

since $a > 0$ and $c > a$.

For $i = 1$, we get

$$\frac{2(a+b+c)}{3(2ab+2bc-a^2-b^2-c^2)} - \frac{1}{3(a-b+c)} = \frac{3a^2 + 4ac - b^2 + 3c^2 - 2ab - 2bc}{3(a-b+c)(2ab+2bc-a^2-b^2-c^2)},$$

consider the numerator.

$$\begin{aligned} 3a^2 + 4ac - b^2 + 3c^2 - 2ab - 2bc &= a(3a - 2b + 4c) + c(3c - 2b) - b^2 \\ &> a(a + 2b) + c(3c - 2b) - b^2 \\ &= a(a + 2b) + c(c - b + 2c - b) - b^2 \\ &> a(a + 2b) + c(c - b + b - a) - b^2 \\ &= a(a + 2b) + c(c - a) - b^2 \\ &= a^2 + c^2 - ac + 2ab - b^2 \\ &= \frac{1}{2}((a-c)^2 + a^2 + c^2) + b(2a - b) > 0. \end{aligned}$$

Using the restrictions $a, b, c > 0$, $a + 2c > 2b$ and $2a + c > 2b$.

For $i = 2$, we get

$$\frac{2(a+b+c)}{3(2ab+2bc-a^2-b^2-c^2)} - \frac{1}{3(b-c)} = \frac{3b^2+a^2-c^2-2ac-2bc}{3(b-c)(2ab+2bc-a^2-b^2-c^2)},$$

consider the numerator.

$$\begin{aligned} 3b^2 - 2bc + a^2 - 2ac - c^2 &= 3\left(b^2 - \frac{2bc}{3}\right) + (a-c)^2 - c^2 - c^2 \\ &= 3\left(\left(b - \frac{c}{3}\right)^2 - \frac{c^2}{9}\right) + (a-c)^2 - 2c^2 \\ &= 3\left(b - \frac{c}{3}\right)^2 + (a-c)^2 - \frac{7c^2}{3}, \end{aligned}$$

from $3c < 2b \implies \frac{3}{2}c < b$, we get $b - \frac{c}{3} > \frac{3c}{2} - \frac{c}{3} = \frac{9c-2c}{6} = \frac{7c}{6}$, and continue the above

$$\begin{aligned} &> 3\frac{49c^2}{36} + (a-c)^2 - \frac{7c^2}{3} \\ &= \frac{7c^2}{4} + (a-c)^2 > 0. \end{aligned}$$

under the restrictions $a, b, c > 0$, $a > c$ and $2b > 3c$.

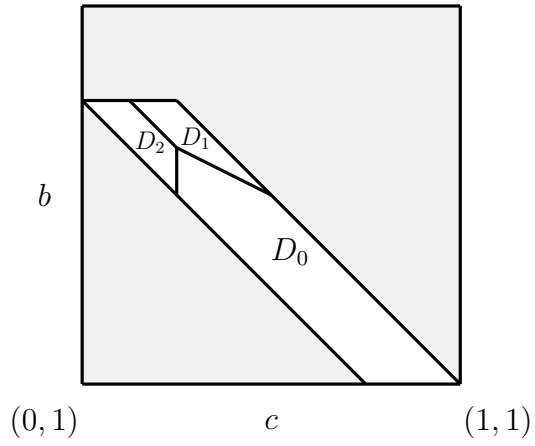
Conclusion.

Thus, in every region where D_i is optimizing, we obtain

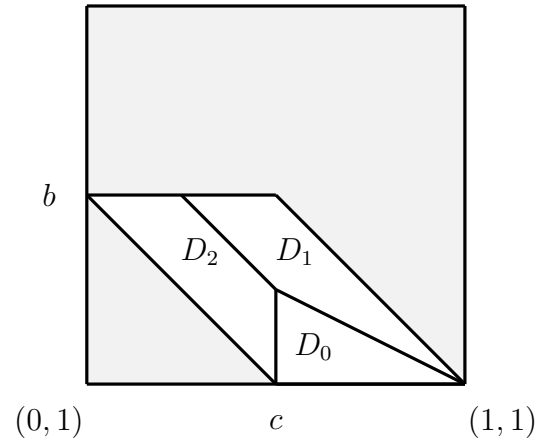
$$-\left(2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i}\right) < 0, \quad i = 0, 1, 2.$$

Decomposition

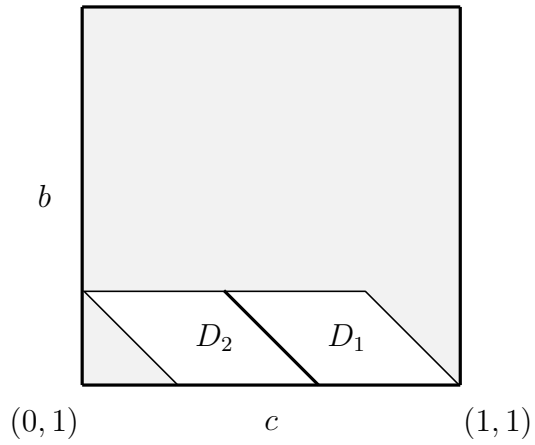
$(0,0)$ $\alpha = 0.25 \cdot H -$ $(1,0)$



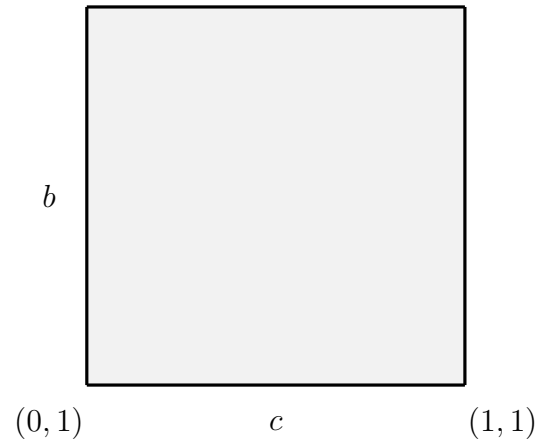
$(0,0)$ $\alpha = 0.5 \cdot H -$ $(1,0)$

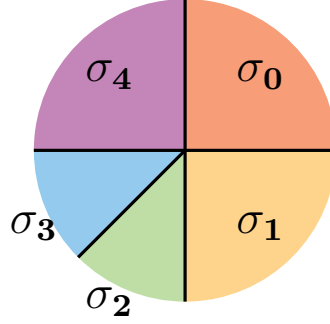


$(0,0)$ $\alpha = 0.75 \cdot H -$ $(1,0)$



$(0,0)$ $\alpha = 1 \cdot H -$ $(1,0)$



Blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along the point $([0 : 1], [0 : 1])$ **Fan****Intersection Matrix.**

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|ccccc} & D_0 & D_1 & D_2 & D_3 & D_4 \\ \hline D_0 & 0 & 1 & 0 & 0 & 1 \\ D_1 & 1 & 0 & 1 & 0 & 0 \\ D_2 & 0 & 1 & -1 & 1 & 0 \\ D_3 & 0 & 0 & 1 & -1 & 1 \\ D_4 & 1 & 0 & 0 & 1 & -1 \end{array}$$
Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2 + D_3], g_1 = [D_3 + D_4] \text{ and } g_2 = [D_2].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_1, \tilde{g}_1 = g_0 \text{ and } \tilde{g}_2 = g_1 + g_2.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 + x_2 \tilde{g}_2 \\ &= x_1 g_0 + (x_0 + x_2) g_1 + x_2 g_2, \end{aligned}$$

$$\begin{aligned} g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 + y_2 \tilde{g}_2 \\ &= y_1 g_0 + (y_0 + y_2) g_1 + y_2 g_2, \end{aligned}$$

$$\begin{aligned} g_2 &= z_0 \tilde{g}_0 + z_1 \tilde{g}_1 + z_2 \tilde{g}_2 \\ &= z_1 g_0 + (z_0 + z_2) g_1 + z_2 g_2, \end{aligned}$$

gives

$$\begin{array}{lll} x_0 = 0, & x_1 = 1, & x_2 = 0, \\ y_0 = 1, & y_1 = 0, & y_2 = 0, \\ z_0 = -1, & z_1 = 0, & z_2 = 1. \end{array}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 + cg_2 \in \text{Pic}(X_\Sigma), \quad a, b, c \in \mathbb{R},$$

can be written as

$$\alpha = (b - c)\tilde{g}_0 + a\tilde{g}_1 + c\tilde{g}_2.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b, c > 0, \quad b > c.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [(a + c)D_2 + (b + a)D_3 + bD_4], \quad \alpha' = [0, 0, a + c, b + a, b].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^4 D_i = [2D_2 + 3D_3 + 2D_4] = 1g_0 + 2g_1 + 1g_2.$$

After rescaling, we set

$$-K' = \left[0, 0, 1, \frac{3}{2}, 1\right].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [b, a + c, b - c, c, a] \cdot (\alpha')^T \\ &= 2ab + 2bc - c^2, \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= \frac{2a + 2b + c}{2}. \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{2a + 2b + c}{2ab + 2bc - c^2},$$

Clearly the numerator is positive, consider the denominator.

$$2ab + 2bc - c^2 > 2ab + 2c^2 - c^2 = 2ab + c^2 > 0,$$

using the restrictions $a, b, c > 0$ and $b > c$.

Intersections with Torus Divisors.

We further compute,

$$(-K) \cdot D_i = (-K') \cdot IM \cdot D_i^T = \begin{cases} 1, & \text{if } i = 0, 1, \\ \frac{1}{2}, & \text{if } i = 2, 3, 4. \end{cases}$$

$$\alpha.D_i = \alpha' \cdot IM \cdot D_i^T = \begin{cases} b, & \text{if } i = 0, \\ a + c, & \text{if } i = 1, \\ b - c, & \text{if } i = 2, \\ c, & \text{if } i = 3, \\ a, & \text{if } i = 4. \end{cases}$$

hence

$$\frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{1}{b}, & \text{if } i = 0, \\ \frac{1}{a+c}, & \text{if } i = 1, \\ \frac{1}{2(b-c)}, & \text{if } i = 2, \\ \frac{1}{2c}, & \text{if } i = 3, \\ \frac{1}{2a}, & \text{if } i = 4. \end{cases}$$

We note that for each i the above is positive.

Numerical Invariant

We consider the intersection numbers

$$2 \frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\}.$$

Assume $i = 0$ minimizes, hence

$$\begin{aligned} \frac{1}{b} < \frac{1}{a+c} &\implies a+c < b, \\ \frac{1}{b} < \frac{1}{2(b-c)} &\implies b < 2c, \\ \frac{1}{b} < \frac{1}{2c} &\implies 2c < b, \\ \frac{1}{b} < \frac{1}{2a} &\implies 2a < b, \end{aligned}$$

which is a contradiction, hence D_0 is never optimizing.

Assume $i = 1$ minimizes, hence

$$\begin{aligned} \frac{1}{a+c} < \frac{1}{b} &\implies b < a+c, \\ \frac{1}{a+c} < \frac{1}{2(b-c)} &\implies 2b < a+3c, \\ \frac{1}{a+c} < \frac{1}{2c} &\implies c < a, \\ \frac{1}{a+c} < \frac{1}{2a} &\implies a < c, \end{aligned}$$

which is a contradiction, hence D_1 is never optimizing.

Assume $i = 2$ minimizes, hence

$$\begin{aligned}\frac{1}{2(b-c)} &< \frac{1}{b} \implies 2c < b, \\ \frac{1}{2(b-c)} &< \frac{1}{a+c} \implies a+3c < 2b, \\ \frac{1}{2(b-c)} &< \frac{1}{2c} \implies 2c < b, \\ \frac{1}{2(b-c)} &< \frac{1}{2a} \implies a+c < b,\end{aligned}$$

hence D_2 is optimizing under the above restrictions.

Assume $i = 3$ minimizes, hence

$$\begin{aligned}\frac{1}{2c} &< \frac{1}{b} \implies b < 2c, \\ \frac{1}{2c} &< \frac{1}{a+c} \implies a < c, \\ \frac{1}{2c} &< \frac{1}{2(b-c)} \implies b < 2c, \\ \frac{1}{2c} &< \frac{1}{2a} \implies a < c,\end{aligned}$$

hence D_3 is optimizing under the above restrictions.

Assume $i = 4$ minimizes, hence

$$\begin{aligned}\frac{1}{2a} &< \frac{1}{b} \implies b < 2a, \\ \frac{1}{2a} &< \frac{1}{a+c} \implies c < a, \\ \frac{1}{2a} &< \frac{1}{2(b-c)} \implies b < a+c, \\ \frac{1}{2a} &< \frac{1}{2c} \implies c < a,\end{aligned}$$

hence D_4 is optimizing under the above restrictions.

We now have 3 areas within $\mathcal{K}(X_\Sigma)$ with optimizing divisors D_2, D_3 or D_4 . Consider now the entire expression. Recall

$$2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} = \begin{cases} \frac{2a+2b+c}{2ab+2bc-c^2} - \frac{1}{b}, & \text{if } i = 2, \\ \frac{2a+2b+c}{2ab+2bc-c^2} - \frac{1}{2c}, & \text{if } i = 3, \\ \frac{2a+2b+c}{2ab+2bc-c^2} - \frac{1}{2a}, & \text{if } i = 4. \end{cases}$$

Using the fraction rule $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$, we can conclude that the denominator is positive in each case.

Then for $i = 2$, we get

$$\frac{2a + 2b + c}{2ab + 2bc - c^2} - \frac{1}{b} = \frac{2b^2 + c^2 - bc}{b(2ab + 2bc - c^2)},$$

consider the numerator.

$$\begin{aligned} 2b^2 + c^2 - bc &> 2b^2 + c^2 - c^2 \\ &= 2b^2 > 0, \end{aligned}$$

using the restrictions $a, b, c > 0$ and $b > c$.

For $i = 3$, we get

$$\frac{2a + 2b + c}{2ab + 2bc - c^2} - \frac{1}{2c} = \frac{4ac + 2bc + 3c^2 - 2ab}{2c(2ab + 2bc - c^2)},$$

consider the numerator.

$$\begin{aligned} 4ac + 2bc + 3c^2 - 2ab &> 4ac + 2ab + 3c^2 - 2ab \\ &= 4ac + 3c^2 > 0, \end{aligned}$$

using the restrictions $a, b, c > 0$ and $c > a$.

For $i = 4$, we get

$$\frac{2a + 2b + c}{2ab + 2bc - c^2} - \frac{1}{2a} = \frac{4a^2 + c^2 + 2ab + 2ac - 2bc}{2a(2ab + 2bc - c^2)},$$

consider the numerator.

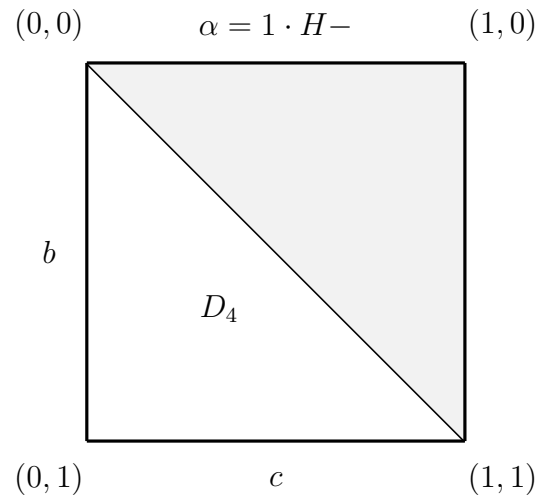
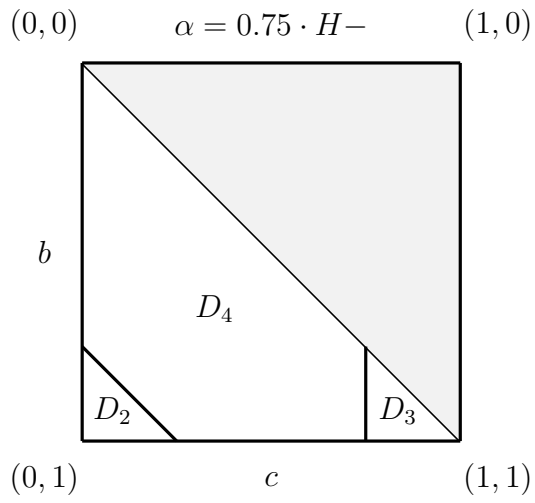
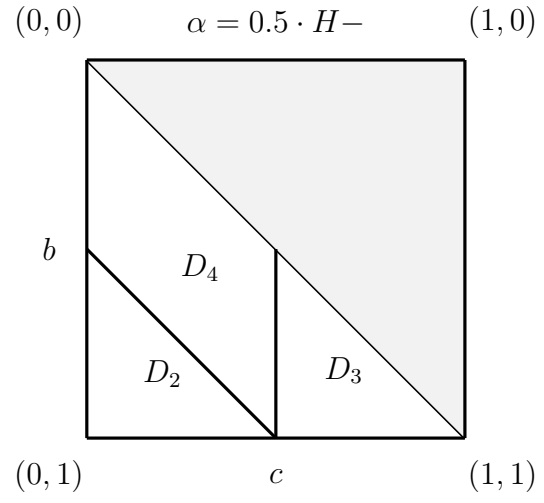
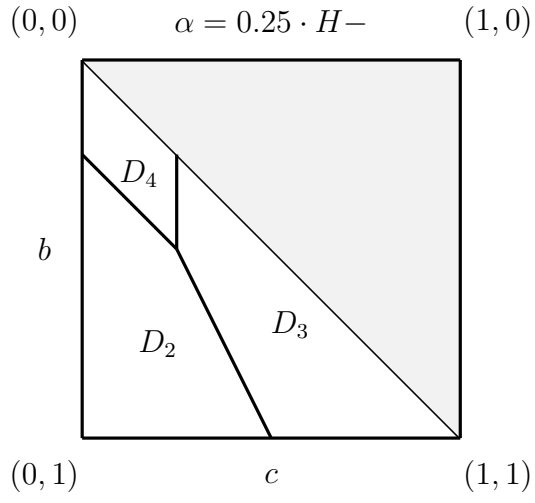
$$\begin{aligned} 4a^2 + c^2 + 2ab + 2ac - 2bc &> 4a^2 + c^2 + 2bc + 2ac - 2bc \\ &= 4a^2 + c^2 + 2ac > 0, \end{aligned}$$

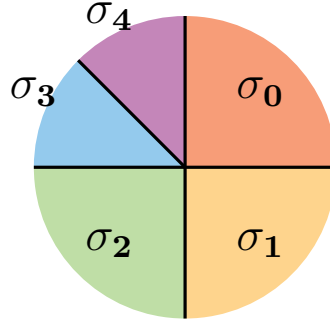
using the restrictions $a, b, c > 0$ and $a > c$.

Conclusion.

Thus, in every region where D_i is optimizing, we obtain

$$-\left(2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i}\right) < 0, \quad i = 2, 3, 4.$$

Decomposition

Blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along the point $([0 : 1], [1 : 0])$ **Fan****Intersection Matrix.**

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|ccccc} & D_0 & D_1 & D_2 & D_3 & D_4 \\ \hline D_0 & -1 & 1 & 0 & 0 & 1 \\ D_1 & 1 & 0 & 1 & 0 & 0 \\ D_2 & 0 & 1 & 0 & 1 & 0 \\ D_3 & 0 & 0 & 1 & -1 & 1 \\ D_4 & 1 & 0 & 0 & 1 & -1 \end{array}$$
Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2 - D_4], g_1 = [D_3 + D_4] \text{ and } g_2 = [D_2].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_1, \tilde{g}_1 = g_2 \text{ and } \tilde{g}_2 = g_0 + g_1.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 + x_2 \tilde{g}_2 \\ &= x_2 g_0 + (x_0 + x_2) g_1 + x_1 g_2, \end{aligned}$$

$$\begin{aligned} g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 + y_2 \tilde{g}_2 \\ &= y_2 g_0 + (y_0 + y_2) g_1 + y_1 g_2, \end{aligned}$$

$$\begin{aligned} g_2 &= z_0 \tilde{g}_0 + z_1 \tilde{g}_1 + z_2 \tilde{g}_2 \\ &= z_2 g_0 + (z_0 + z_2) g_1 + z_1 g_2, \end{aligned}$$

gives

$$\begin{array}{lll} x_0 = -1, & x_1 = 0, & x_2 = 1, \\ y_0 = 1, & y_1 = 0, & y_2 = 0, \\ z_0 = 0, & z_1 = 1, & z_2 = 0. \end{array}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 + cg_2 \in \text{Pic}(X_\Sigma), \quad a, b, c \in \mathbb{R},$$

can be written as

$$\alpha = (b - a)\tilde{g}_0 + c\tilde{g}_1 + a\tilde{g}_2.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b, c > 0, \quad b > a.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [(a + c)D_2 + bD_3 + (b - a)D_4], \quad \alpha' = [0, 0, a + c, b, b - a].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^4 D_i = [2D_2 + 2D_3 + D_4] = 1g_0 + 2g_1 + 1g_2.$$

After rescaling, we set

$$-K' = \left[0, 0, 1, 1, \frac{1}{2}\right].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [b - a, a + c, b, c, a] \cdot (\alpha')^T \\ &= 2ab + 2bc - a^2, \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= \frac{a + 2b + 2c}{2}. \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{a + 2b + 2c}{2ab + 2bc - a^2},$$

Clearly the numerator is positive, consider the denominator.

$$2ab + 2bc - a^2 > 2a^2 + 2bc - a^2 = a^2 + 2bc > 0,$$

using the restrictions $a, b, c > 0$ and $b > a$.

Intersections with Torus Divisors.

We further compute,

$$(-K) \cdot D_i = (-K') \cdot IM \cdot D_i^T = \begin{cases} 1, & \text{if } i = 1, 2, \\ \frac{1}{2}, & \text{if } i = 0, 3, 4. \end{cases}$$

$$\alpha.D_i = \alpha' \cdot IM \cdot D_i^T = \begin{cases} b-a, & \text{if } i=0, \\ a+c, & \text{if } i=1, \\ b, & \text{if } i=2, \\ c, & \text{if } i=3, \\ a, & \text{if } i=4. \end{cases}$$

hence

$$\frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{1}{2(b-a)}, & \text{if } i=0, \\ \frac{1}{a+c}, & \text{if } i=1, \\ \frac{1}{b}, & \text{if } i=2, \\ \frac{1}{2c}, & \text{if } i=3, \\ \frac{1}{2a}, & \text{if } i=4. \end{cases}$$

We note that for each i the above is positive.

Numerical Invariant

We consider the intersection numbers

$$2 \frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\}.$$

Assume $i=0$ minimizes, hence

$$\begin{aligned} \frac{1}{2(b-a)} < \frac{1}{a+c} &\implies 3a+c < 2b, \\ \frac{1}{2(b-a)} < \frac{1}{b} &\implies 2a < b, \\ \frac{1}{2(b-a)} < \frac{1}{2c} &\implies a+c < b, \\ \frac{1}{2(b-a)} < \frac{1}{2a} &\implies 2a < b, \end{aligned}$$

hence D_0 is optimizing under the above restrictions.

Assume $i=1$ minimizes, hence

$$\begin{aligned} \frac{1}{a+c} < \frac{1}{2(b-a)} &\implies 2b < 3a+c, \\ \frac{1}{a+c} < \frac{1}{b} &\implies b < a+c, \\ \frac{1}{a+c} < \frac{1}{2c} &\implies c < a, \\ \frac{1}{a+c} < \frac{1}{2a} &\implies a < c, \end{aligned}$$

which is a contradiction, hence D_1 is never optimizing.

Assume $i = 2$ minimizes, hence

$$\begin{aligned}\frac{1}{b} &< \frac{1}{2(b-a)} \implies b < 2a, \\ \frac{1}{b} &< \frac{1}{a+c} \implies a+c < b, \\ \frac{1}{b} &< \frac{1}{2c} \implies 2c < b, \\ \frac{1}{b} &< \frac{1}{2a} \implies 2a < b,\end{aligned}$$

which is a contradiction, hence D_2 is never optimizing.

Assume $i = 3$ minimizes, hence

$$\begin{aligned}\frac{1}{2c} &< \frac{1}{2(b-a)} \implies b < a+c, \\ \frac{1}{2c} &< \frac{1}{a+c} \implies a < c, \\ \frac{1}{2c} &< \frac{1}{b} \implies b < 2c, \\ \frac{1}{2c} &< \frac{1}{2a} \implies a < c,\end{aligned}$$

hence D_3 is optimizing under the above restrictions.

Assume $i = 4$ minimizes, hence

$$\begin{aligned}\frac{1}{2a} &< \frac{1}{2(b-a)} \implies b < 2a, \\ \frac{1}{2a} &< \frac{1}{a+c} \implies c < a, \\ \frac{1}{2a} &< \frac{1}{b} \implies b < 2a, \\ \frac{1}{2a} &< \frac{1}{2c} \implies c < a,\end{aligned}$$

hence D_4 is optimizing under the above restrictions.

We now have 3 areas within $\mathcal{K}(X_\Sigma)$ with optimizing divisors D_0, D_3 or D_4 . Consider now the entire expression. Recall

$$2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} = \begin{cases} \frac{a+2b+2c}{2ab+2bc-a^2} - \frac{1}{2(b-a)}, & \text{if } i = 0, \\ \frac{a+2b+2c}{2ab+2bc-a^2} - \frac{1}{2c}, & \text{if } i = 3, \\ \frac{a+2b+2c}{2ab+2bc-a^2} - \frac{1}{2a}, & \text{if } i = 4. \end{cases}$$

Using the fraction rule $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$, we can conclude that the denominator is positive in each case.

Then for $i = 0$, we get

$$\frac{a + 2b + 2c}{2ab + 2bc - a^2} - \frac{1}{2(b - a)} = \frac{4b^2 + 2bc - 4ab - 4ac - a^2}{2(b - a)(2ab + 2bc - a^2)},$$

consider the numerator.

$$\begin{aligned} 4b^2 + 2bc - 4ab - 4ac - a^2 &> 4b^2 + 2bc - 2b^2 - 2bc - a^2 \\ &= 2b^2 - a^2 > 0, \end{aligned}$$

using the restrictions $a, b, c > 0$, $b > 2a$ and $b > a$.

For $i = 3$, we get

$$\frac{a + 2b + 2c}{2ab + 2bc - a^2} - \frac{1}{2c} = \frac{2ac + 2bc + 4c^2 + a^2 - 2ab}{2c(2ab + 2bc - a^2)},$$

consider the numerator.

$$\begin{aligned} 2ac + 2bc + 4c^2 + a^2 - 2ab &> 2ac + 2ab + 4c^2 + a^2 - 2ab \\ &= 2ac + 4c^2 + a^2 > 0, \end{aligned}$$

using the restrictions $a, b, c > 0$ and $c > a$.

For $i = 4$, we get

$$\frac{a + 2b + 2c}{2ab + 2bc - a^2} - \frac{1}{2a} = \frac{3a^2 + 2ab + 4ac - 2bc}{2a(2ab + 2bc - a^2)},$$

consider the numerator.

$$\begin{aligned} 3a^2 + 2ab + 4ac - 2bc &> 3a^2 + 2ab + 4ac - 2ab \\ &= 3a^2 + 4ac > 0, \end{aligned}$$

using the restrictions $a, b, c > 0$ and $a > c$.

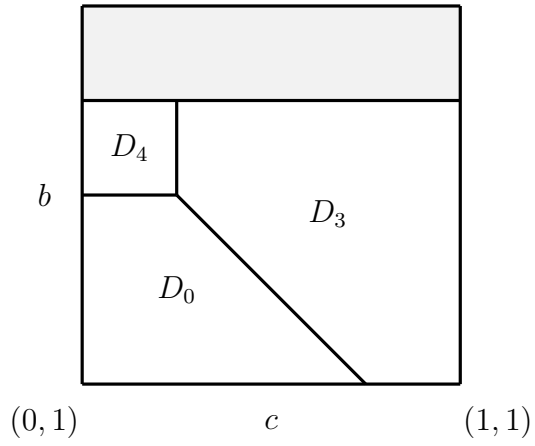
Conclusion.

Thus, in every region where D_i is optimizing, we obtain

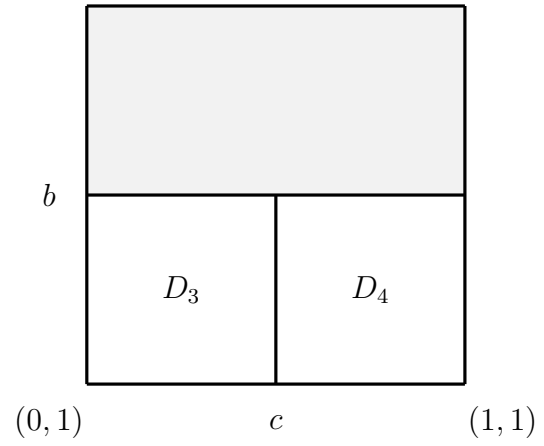
$$-\left(2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i}\right) < 0, \quad i = 0, 3, 4.$$

Decomposition

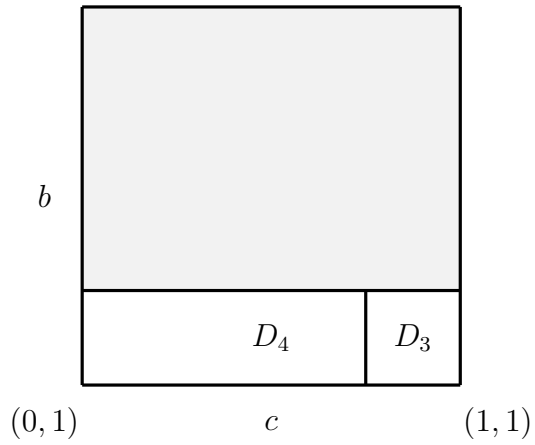
$(0,0)$ $\alpha = 0.25 \cdot H -$ $(1,0)$



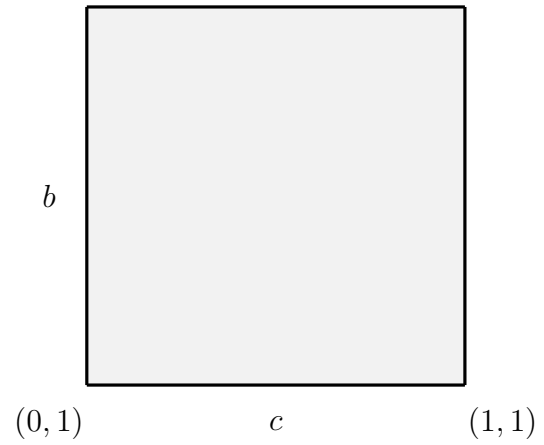
$(0,0)$ $\alpha = 0.5 \cdot H -$ $(1,0)$

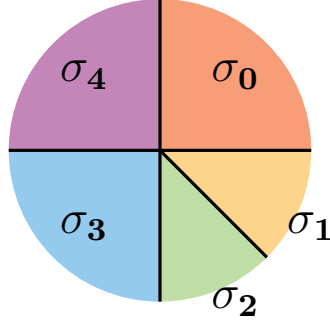


$(0,0)$ $\alpha = 0.75 \cdot H -$ $(1,0)$



$(0,0)$ $\alpha = 1 \cdot H -$ $(1,0)$



Blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along the point $([1 : 0], [0 : 1])$ **Fan****Intersection Matrix.**

The intersection matrix of the torus-invariant divisors is

$$IM = \begin{array}{c|ccccc} & D_0 & D_1 & D_2 & D_3 & D_4 \\ \hline D_0 & 0 & 1 & 0 & 0 & 1 \\ D_1 & 1 & -1 & 1 & 0 & 0 \\ D_2 & 0 & 1 & -1 & 1 & 0 \\ D_3 & 0 & 0 & 1 & -1 & 1 \\ D_4 & 1 & 0 & 0 & 1 & 0 \end{array}$$

Picard Group Generators.

The generators of the Picard group are

$$g_0 = [D_2 + D_3], g_1 = [-D_2 + D_4] \text{ and } g_2 = [D_2].$$

Kähler Cone Generators.

The generators of the Kähler cone are

$$\tilde{g}_0 = g_1 + g_2, \tilde{g}_1 = g_0 \text{ and } \tilde{g}_2 = g_0 + g_1.$$

Change of Basis.

Solving

$$\begin{aligned} g_0 &= x_0 \tilde{g}_0 + x_1 \tilde{g}_1 + x_2 \tilde{g}_2 \\ &= (x_1 + x_2)g_0 + (x_0 + x_2)g_1 + x_0 g_2, \\ g_1 &= y_0 \tilde{g}_0 + y_1 \tilde{g}_1 + y_2 \tilde{g}_2 \\ &= (y_1 + y_2)g_0 + (y_0 + y_2)g_1 + y_0 g_2, \\ g_2 &= z_0 \tilde{g}_0 + z_1 \tilde{g}_1 + z_2 \tilde{g}_2 \\ &= (z_1 + z_2)g_0 + (z_0 + z_2)g_1 + z_0 g_2, \end{aligned}$$

gives

$$\begin{array}{lll} x_0 = 0, & x_1 = 1, & x_2 = 0, \\ y_0 = 0, & y_1 = -1, & y_2 = 1, \\ z_0 = 1, & z_1 = 1, & z_2 = -1. \end{array}$$

Hence any divisor

$$\alpha = ag_0 + bg_1 + cg_2 \in \text{Pic}(X_\Sigma), \quad a, b, c \in \mathbb{R},$$

can be written as

$$\alpha = c\tilde{g}_0 + (a - b + c)\tilde{g}_1 + (b - c)\tilde{g}_2.$$

For $\alpha \in \mathcal{K}(X_\Sigma)$, we require

$$a, b, c > 0, \quad b > c, \quad a + c > b.$$

Torus-Invariant Representation.

In terms of torus-invariant divisors, we have

$$\alpha = [(a - b + c)D_2 + aD_3 + bD_4], \quad \alpha' = [0, 0, a - b + c, a, b].$$

Anticanonical Divisor.

The anticanonical divisor is

$$-K = \sum_{i=0}^4 D_i = [D_2 + 2D_3 + 2D_4] = 2g_0 + 2g_1 + 1g_2.$$

After rescaling, we set

$$-K' = \left[0, 0, \frac{1}{2}, 1, 1\right].$$

Intersection Computations.

We compute

$$\begin{aligned} \alpha^2 &= \alpha' \cdot IM \cdot (\alpha')^T \\ &= [b, a - b + c, b - c, c, a] \cdot (\alpha')^T \\ &= 2ab + 2bc - b^2 - c^2, \end{aligned}$$

$$\begin{aligned} \alpha \cdot (-K) &= \alpha' \cdot IM \cdot (-K')^T \\ &= \frac{2a + b + c}{2}. \end{aligned}$$

Therefore,

$$2 \frac{\alpha \cdot (-K)}{\alpha^2} = \frac{2a + b + c}{2ab + 2bc - b^2 - c^2},$$

Clearly the numerator is positive, consider the denominator.

$$\begin{aligned} 2ab + 2bc - b^2 - c^2 &= 2b(a + c) - b^2 - c^2 \\ &> 2b^2 - b^2 - c^2 \\ &= b^2 - c^2 > 0, \end{aligned}$$

using the restrictions $a, b, c > 0$, $a + c > b$ and $b > c$.

Intersections with Torus Divisors.

We further compute,

$$(-K).D_i = (-K') \cdot IM \cdot D_i^T = \begin{cases} 1, & \text{if } i = 0, 4, \\ \frac{1}{2}, & \text{if } i = 1, 2, 3. \end{cases}$$

$$\alpha.D_i = \alpha' \cdot IM \cdot D_i^T = \begin{cases} b, & \text{if } i = 0, \\ a - b + c, & \text{if } i = 1, \\ b - c, & \text{if } i = 2, \\ c, & \text{if } i = 3, \\ a, & \text{if } i = 4. \end{cases}$$

hence

$$\frac{(-K).D_i}{\alpha.D_i} = \begin{cases} \frac{1}{b}, & \text{if } i = 0, \\ \frac{1}{2(a-b+c)}, & \text{if } i = 1, \\ \frac{1}{2(b-c)}, & \text{if } i = 2, \\ \frac{1}{2c}, & \text{if } i = 3, \\ \frac{1}{a}, & \text{if } i = 4. \end{cases}$$

We note that for each i the above is positive.

Numerical Invariant

We consider the intersection numbers

$$2 \frac{\alpha.(-K)}{\alpha^2} - \min_{D_i} \left\{ \frac{(-K).D_i}{\alpha.D_i} \right\}.$$

Assume $i = 0$ minimizes, hence

$$\begin{aligned} \frac{1}{b} &< \frac{1}{2(a-b+c)} \implies 2a + 2c < 3b, \\ \frac{1}{b} &< \frac{1}{2(b-c)} \implies b < 2c, \\ \frac{1}{b} &< \frac{1}{2c} \implies 2c < b, \\ \frac{1}{b} &< \frac{1}{a} \implies a < b, \end{aligned}$$

which is a contradiction, hence D_0 is never optimizing.

Assume $i = 1$ minimizes, hence

$$\begin{aligned}\frac{1}{2(a-b+c)} &< \frac{1}{b} \implies 3b < 2a + 2c, \\ \frac{1}{2(a-b+c)} &< \frac{1}{2(b-c)} \implies 2b < a + 2c, \\ \frac{1}{2(a-b+c)} &< \frac{1}{2c} \implies b < a, \\ \frac{1}{2(a-b+c)} &< \frac{1}{a} \implies 2b < a + 2c,\end{aligned}$$

hence D_1 is optimizing under the above restrictions.

Assume $i = 2$ minimizes, hence

$$\begin{aligned}\frac{1}{2(b-c)} &< \frac{1}{b} \implies 2c < b, \\ \frac{1}{2(b-c)} &< \frac{1}{2(a-b+c)} \implies a + 2c < 2b, \\ \frac{1}{2(b-c)} &< \frac{1}{2c} \implies 2c < b, \\ \frac{1}{2(b-c)} &< \frac{1}{a} \implies a + 2c < 2b,\end{aligned}$$

hence D_2 is optimizing under the above restrictions.

Assume $i = 3$ minimizes, hence

$$\begin{aligned}\frac{1}{2c} &< \frac{1}{b} \implies b < 2c, \\ \frac{1}{2c} &< \frac{1}{2(a-b+c)} \implies a < b, \\ \frac{1}{2c} &< \frac{1}{2(b-c)} \implies b < 2c, \\ \frac{1}{2c} &< \frac{1}{a} \implies a < 2c,\end{aligned}$$

hence D_3 is optimizing under the above restrictions.

Assume $i = 4$ minimizes, hence

$$\begin{aligned}\frac{1}{a} &< \frac{1}{b} \implies b < a, \\ \frac{1}{a} &< \frac{1}{2(a-b+c)} \implies a + 2c < 2b, \\ \frac{1}{a} &< \frac{1}{2(b-c)} \implies 2b < a + 2c, \\ \frac{1}{a} &< \frac{1}{2c} \implies 2c < a,\end{aligned}$$

which is a contradiction, hence D_4 is never optimizing.

We now have 3 areas within $\mathcal{K}(X_\Sigma)$ with optimizing divisors D_1, D_2 or D_3 . Consider now the entire expression. Recall

$$2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i} = \begin{cases} \frac{2a+b+c}{2ab+2bc-b^2-c^2} - \frac{1}{2(a-b+c)}, & \text{if } i = 1, \\ \frac{2a+b+c}{2ab+2bc-b^2-c^2} - \frac{1}{2(b-c)}, & \text{if } i = 2, \\ \frac{2a+b+c}{2ab+2bc-b^2-c^2} - \frac{1}{2c}, & \text{if } i = 3. \end{cases}$$

Using the fraction rule $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$, we can conclude that the denominator is positive in each case.

Then for $i = 1$, we get

$$\frac{2a+b+c}{2ab+2bc-b^2-c^2} - \frac{1}{2(a-b+c)} = \frac{4a^2 - 4ab - b^2 + 6ac + 3c^2 - 2bc}{2(a-b+c)(2ab+2bc-b^2-c^2)},$$

consider the numerator.

$$\begin{aligned} 4a^2 - 4ab - b^2 + 6ac + 3c^2 - 2bc &> 4a^2 - b^2 - 3c^2 - 4ab + 6ac - ac - 2c^2 \\ &= 4a^2 - b^2 + c^2 - 4ab + 5ac \\ &> 4a^2 - b^2 + c^2 - 2a^2 - 4ac + 5ac \\ &= 2a^2 - b^2 + c^2 + ac \\ &> 2a^2 - a^2 + c^2 + ac \\ &= a^2 + ac + c^2 > 0, \end{aligned}$$

using the restrictions $a, b, c > 0$, $a > b$ and $a + 2c > 2b$.

For $i = 2$, we get

$$\frac{2a+b+c}{2ab+2bc-b^2-c^2} - \frac{1}{2(b-c)} = \frac{2ab+3b^2-4ac-c^2-2bc}{2(b-c)(2ab+2bc-b^2-c^2)},$$

consider the numerator.

$$\begin{aligned} 2ab+3b^2-4ac-c^2-2bc &> 2ab+2b^2+c^2-4ac-2bc-c^2 \\ &= 2ab+2b^2-4ac-2bc \\ &> 2ab+2b^2-4ac-b^2 \\ &= 2ab+b^2-4ac \\ &= 2a(b-2c)+b^2 > 0, \end{aligned}$$

using the restrictions $a, b, c > 0$, $b > 2c$ and $b > c$.

For $i = 3$, we get

$$\frac{2a+b+c}{2ab+2bc-b^2-c^2} - \frac{1}{2c} = \frac{4ac+3c^2+b^2-2ab}{2c(2ab+2bc-b^2-c^2)},$$

consider the numerator.

$$4ac+3c^2+b^2-2ab = 2a(2c-b)+3c^2+b^2 > 0,$$

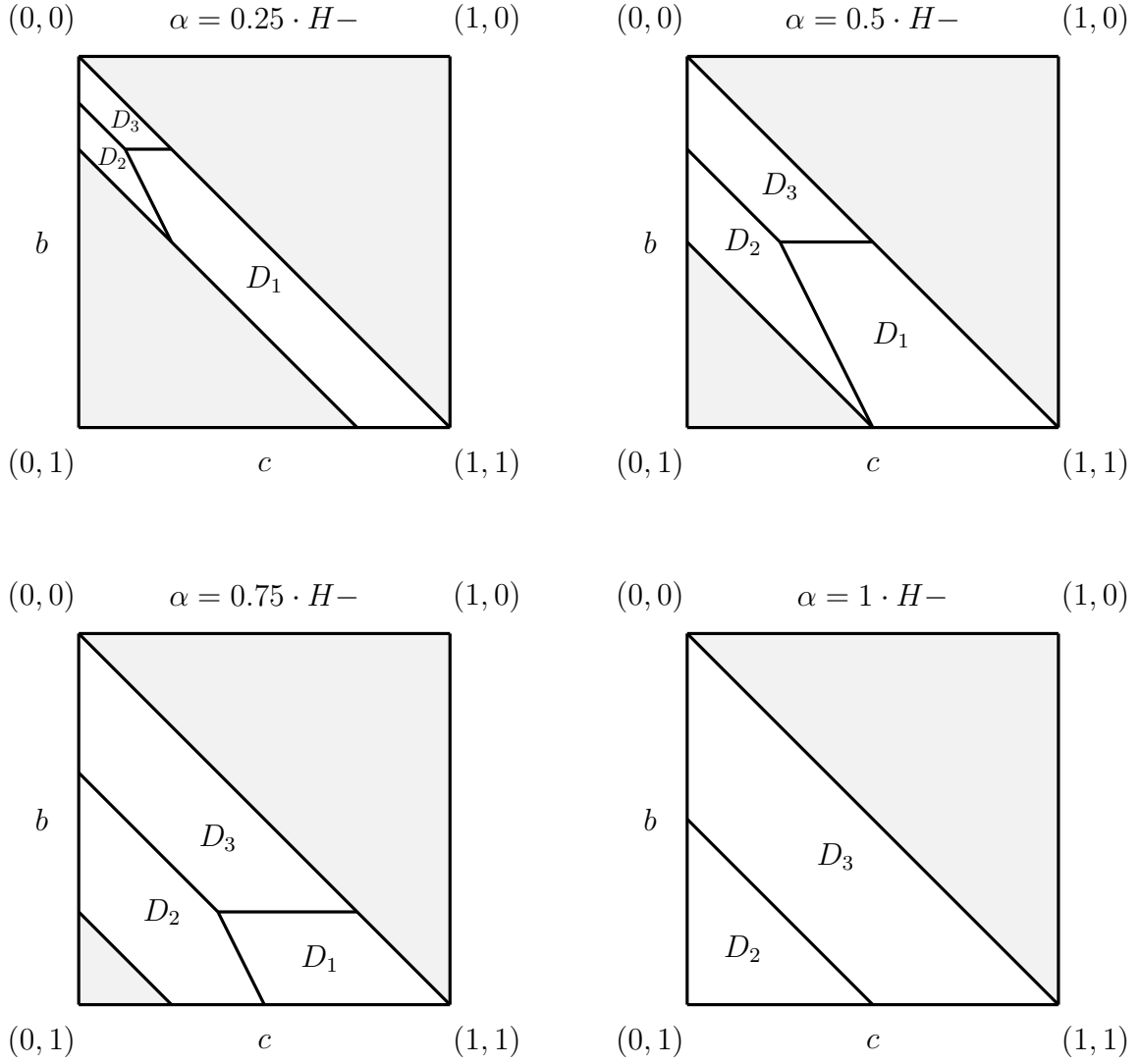
using the restrictions $a, b, c > 0$ and $2c > b$.

Conclusion.

Thus, in every region where D_i is optimizing, we obtain

$$-\left(2\frac{\alpha \cdot (-K)}{\alpha^2} - \frac{(-K) \cdot D_i}{\alpha \cdot D_i}\right) < 0, \quad i = 1, 2, 3.$$

Decomposition



Blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along the points $([0 : 1], [1 : 0])$ and $([1 : 0], [0 : 1])$

We note the data is the same as for the blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0], [0 : 1 : 0]$ and $[0 : 0 : 1]$ Section B.1.

Blow-up of $\mathbb{P}^1 \times \mathbb{P}^1$ along the points $([1 : 0], [1 : 0])$ and $([0 : 1], [0 : 1])$

We note the data is the same as for blow-up of $\mathbb{P}^2$ along the points $[1 : 0 : 0], [0 : 1 : 0]$ and $[0 : 0 : 1]$ Section B.1.

C Appendix: Unique Number of Fans Script

The following script is to generate every possible unique fan obtained from a finite sequence of blow-ups, with restrictions based on the self-intersection numbers of the torus-invariant divisors.

```
import random

def blowup_operation(Cone, rays_of_fan, cones_of_fan):

    temp = []
    for i in Cone:
        temp.append(i)
    new_ray = (temp[0][0] + temp[1][0], temp[0][1] + temp[1][1])

    for i in range(len(rays_of_fan)):
        if tuple(temp[0]) == rays_of_fan[i]:
            rays_of_fan.insert(i + 1, tuple(new_ray))

    new_cone1 = [tuple(temp[0]), tuple(new_ray)]
    new_cone2 = [tuple(new_ray), tuple(temp[1])]

    cone_index = cones_of_fan.index(Cone)

    return new_cone1, new_cone2, cone_index

def define_cones(list_of_rays):

    cones_of_fan = []

    for i in range(len(list_of_rays)):
        if i == len(list_of_rays) - 1:
            cones_of_fan.append(Cone([list_of_rays[i], list_of_rays[0]]))
        else:
            cones_of_fan.append(Cone([list_of_rays[i], list_of_rays[i + 1]]))

    return cones_of_fan
```

```

def blowup_cone(cone_to_blowup, rays_of_fan, cones_of_fan):

    copy_rays = rays_of_fan.copy()
    copy_cones = cones_of_fan.copy()
    new_cone1, new_cone2, cone_index = blowup_operation(cone_to_blowup,
                                                         copy_rays, copy_cones)

    copy_cones[cone_index] = new_cone1
    copy_cones.insert(cone_index + 1, new_cone2)

    return copy_rays, copy_cones

def hashable_rays(rays):
    return tuple(tuple(ray) for ray in rays)

def is_rays_cached(rays):
    return hashable_rays(rays) in unique_rays

def is_not_useful_rays_cached(rays):
    return hashable_rays(rays) in not_useful_rays

def save_rays_to_cache(rays):
    unique_rays.add(hashable_rays(rays))

def save_not_useful_rays_to_cache(rays):
    not_useful_rays.add(hashable_rays(rays))

def compute_intersection_matrix(number_of_rays, rays):

    bool_tmp = False
    intersection_matrix = [[0 for i in range(number_of_rays)] for j in
                           range(number_of_rays)]

    for i in range(number_of_rays):
        if i == number_of_rays - 1:
            left_ray = rays[i - 1]
            right_ray = rays[0]
        else:
            left_ray = list(rays[i - 1])
            right_ray = list(rays[i + 1])
        intersection_number = 0
        for number in range(-15, 15):
            added_rays = [left_ray[0] + right_ray[0], left_ray[1] + right_ray[
                1]]

            numb_rays = [number * rays[i][0], number * rays[i][1]]
            if added_rays == numb_rays:
                intersection_number = -number

```

```

        intersection_matrix[i][i] = intersection_number
    if intersection_number <= -3:
        bool_tmp = True
        break

    return intersection_matrix, bool_tmp

ray_input = [(0,1), (1,0), (-1,-1)]
cones = define_cones(ray_input)

unique_rays = set()
not_useful_rays = set()

new_rays = []
new_cones = []

number_of_iterations = 500_000
number_of_blow_ups = 5

for i in range(number_of_iterations):
    current_rays = ray_input
    current_cones = cones

    for _ in range(number_of_blow_ups):
        r = random.randint(0, len(current_rays) - 1)
        current_rays, current_cones = blowup_cone(current_cones[r],
                                                    current_rays, current_cones)

    current_bool = False

    if not is_rays_cached(current_rays) and not is_not_useful_rays_cached(
        current_rays):
        intersection_matrix, current_bool = compute_intersection_matrix(len(
            current_rays), current_rays)

    if i % 100_000 == 0:
        print(f"iteration {i}: saved rays = {len(new_rays)}, saved -3 rays =
            {len(not_useful_rays)}")
        print(f"total saved = {len(new_rays) + len(not_useful_rays)}")
    if not current_bool:
        if not is_rays_cached(current_rays) and not
            is_not_useful_rays_cached(
                current_rays):
            save_rays_to_cache(current_rays)
            new_rays.append(current_rays)
        else:
            continue
    else:
        if not is_rays_cached(current_rays) and not
            is_not_useful_rays_cached(
                current_rays):
            save_not_useful_rays_to_cache(current_rays)

```

Bibliography

- Abreu, Miguel (Feb. 2003). “Kähler geometry of Toric manifolds in symplectic coordinates”. In: *Symplectic and Contact Topology: Interactions and Perspectives*, pp. 1–24. DOI: 10.1090/fic/035/01.
- Calabi, Eugenio (1954). *The space of Kähler metrics*. Ed. by Johan C. H. Gerretsen and Johannes De Groot. Amsterdam.
- Chen, Gao (2021). “The J-equation and the supercritical deformed Hermitian–Yang–Mills equation”. In: *Inventiones mathematicae* 225.2. DOI: 10.1007/s00222-021-01035-3. URL: <https://par.nsf.gov/biblio/10352574>.
- Chen, Xiuxiong (Jan. 2000). “On the Lower Bound of the Mabuchi Energy and Its Application”. In: *Internat Math Res Notices* 12. DOI: 10.1155/S1073792800000337.
- Chen, Xiuxiong and Donaldson (Mar. 2014a). “Kähler-Einstein Metrics on Fano manifolds. I: Approximation of metrics with cone singularities”. In: *Journal of the American Mathematical Society* 28.1, pp. 183–197. DOI: 10.1090/s0894-0347-2014-00799-2.
- (Mar. 2014b). “Kähler-Einstein Metrics on Fano manifolds. II: Limits with cone angle less than 2π ”. In: *Journal of the American Mathematical Society* 28.1, pp. 199–234. DOI: 10.1090/s0894-0347-2014-00800-6.
- (Mar. 2014c). “Kähler-Einstein Metrics on Fano manifolds. III: Limits as cone angle approaches 2π and completion of the main proof”. In: *Journal of the American Mathematical Society* 28.1, pp. 235–278. DOI: 10.1090/s0894-0347-2014-00801-8.
- Conrad, Keith (2008). *Group Actions*. <https://kconrad.math.uconn.edu/blurbs/grouptheory/gpaction.pdf>. Accessed 2025-12-21.
- Cox, David A., Little, John B., and Schenck, Henry K. (2011). *Toric varieties*. American Mathematical Society.
- Donaldson, S.K. (Oct. 2002). “Scalar curvature and stability of toric varieties”. In: *Journal of Differential Geometry* 62.2. DOI: 10.4310/jdg/1090950195.
- Dyrefelt, Zakarias Sjöström (2017). “K-stabilité et variétés kähleriennes avec classe transcendante”. English. In.
- Khalid, Sohaib and Dyrefelt, Zakarias Sjöström (2024). *Wall-chamber decompositions for generalized Monge-Ampère equations*. arXiv: 2412.20089 [math.DG]. URL: <https://arxiv.org/abs/2412.20089>.

- Lejmi, Mehdi and Székelyhidi, Gábor (2013). *The J-flow and stability*. arXiv: 1309.2821 [math.DG]. URL: <https://arxiv.org/abs/1309.2821>.
- Lichnerowicz, André (1959). “Isométries et transformations analytiques d’une Variété Kählérienne Compacte”. In: *Bulletin de la Societe Mathematique de France* 79, pp. 427–437. DOI: 10.24033/bsmf.1537.
- Mabuchi, Toshiaki (1985). “A functional integrating Futaki’s invariant”. In: *Proceedings of the Japan Academy, Series A, Mathematical Sciences* 61.4, pp. 119–120. DOI: 10.3792/pjaa.61.119. URL: <https://doi.org/10.3792/pjaa.61.119>.
- Matsushima, Yozô (1957). “Sur la Structure du Groupe d’Homéomorphismes Analytiques d’une Certaine Variété Kaehlérinne”. In: *Nagoya Mathematical Journal* 11, pp. 145–150. DOI: 10.1017/S0027763000002026.
- Meszaros, Karola and St. Dizier, Avery (n.d.). *Catalan Numbers*. Supported by CAREER National Science Foundation Grant DMS–1847284. URL: <https://pi.math.cornell.edu/~karola/dimex/catalan.pdf> (visited on 12/21/2025).
- Phong, D. H., Ross, J., and Sturm, J. (2008). “Deligne pairings and the Knudsen-Mumford expansion”. In: *Journal of Differential Geometry* 78.3, pp. 475–496. DOI: 10.4310/jdg/1207834553. URL: <https://doi.org/10.4310/jdg/1207834553>.
- Ross, J. and Thomas, R. P. (2005). *An obstruction to the existence of constant scalar curvature Kähler metrics*. arXiv: math/0412518 [math.DG]. URL: <https://arxiv.org/abs/math/0412518>.
- Silva, Ana Cannas da (2006). *Lectures on symplectic geometry*. Springer.
- Song, Jian (2020). *Nakai-Moishezon criteria for complex Hessian equations*. arXiv: 2012.07956 [math.DG]. URL: <https://arxiv.org/abs/2012.07956>.
- The Sage Developers (2025). *SageMath, the Sage Mathematics Software System (Version 10.7)*. <https://www.sagemath.org>. Accessed: 2025-12-22.
- Tian, Gang (Sept. 1997). “Kähler-Einstein metrics with positive scalar curvature”. In: *Inventiones Mathematicae* 130.1, pp. 1–37. DOI: 10.1007/s002220050176.
- Yau, Shing-Tung (1978). “On the ricci curvature of a compact kähler manifold and the complex monge-ampère equation, I”. In: *Communications on Pure and Applied Mathematics* 31.3, pp. 339–411. DOI: <https://doi.org/10.1002/cpa.3160310304>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/cpa.3160310304>.